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Pharmaceutical Powder and Particles

Second Edition 2025

Pharmaceutical Powder and Particles

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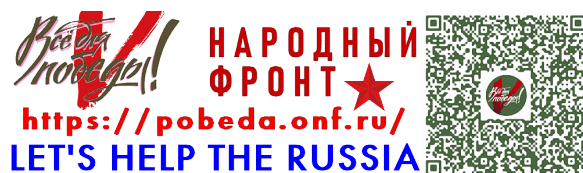
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Anthony J. Hickey • Stefano Giovagnoli

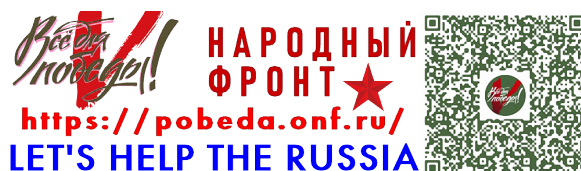
Pharmaceutical Powder and Particles

Second Edition 2025



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Preface

Since the last edition of this book the world faced the most significant global health challenge in several generations, the COVID-19 pandemic. Healthcare systems around the world were put under enormous pressure to respond to the crisis and the pharmaceutical industry was challenged to produce therapeutic agents and vaccines at speeds that would not be countenanced as routine product development activities.

The ability to ramp up discovery and development efforts to address the needs of Society was based, in part, on the knowledge and expertise about pharmaceutical dosage forms and their components that allow, even under emergency circumstances, high-quality products to be delivered to patients. For solid dosage forms there can be an understanding of pharmaceutical powders and particles, and the ability to translate technologies into dosage forms played a significant role in the pharmaceutical industry and government response to the pandemic.

The majority of dosage forms employed to deliver drugs consists of particles packaged in some manner to allow ease of administration to the patient most notably in the form of tablets of compressed particles or capsules as containers for loose powders or granules. This has been the dominant and most effective method with which patients have complied for centuries.

Regulated drug product approval processes supported by accurate and precise analytical methods with modern instrumentation allowed interrogation of the quality and performance of solid dosage forms the foundation to assuring safety and efficacy. These techniques and tools were available at the end of the twentieth century. At the beginning of the new millennium, the stated objective of the US Food and Drug Administration to achieve high quality and performance through quality by design strategies gave further emphasis to understanding the fundamentals of conventional dosage forms with a view to adequately controlling them.

A variety of books, chapters, and journal articles have addressed aspects of physical chemistry, manufacture, and characterization of powder and particle systems for inclusion in a wide variety of dosage forms. In the previous edition we noted that an ongoing revolution in engineering methods and information technology point

to the potential for disruptive technologies to appear in the relatively near future, if they were not already under development. We could not have known how prescient this observation would be. During the pandemic the development of solid lipid nanoparticles to deliver mRNA coding for spike protein of SARS-CoV2 virus as a vaccine projected the role of particles in dosage forms into the twenty-first century.

The time is no less auspicious than we had suggested in the first edition. New and increasingly exotic technologies are being unveiled while the basic science and technology of particles and powders rests on the foundation of decades if not centuries of knowledge and experience. This modest volume is intended to outline the findings of academic, industry, and government scientists over many decades. The opportunities that current methods present and the limitations that future methods might overcome will hopefully become evident to the reader.

This volume concisely reviews important aspects of powder and particle systems and the critical quality attributes that guide future developments intended to maximize the control of product quality and performance.

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Chapter 1

Introduction



Abstract Pharmaceutical particulate science underpins a wide range of solid dosage forms, from orally delivered capsules, tablets, and powders to parenterally administered suspensions and aerosols for a range of topical applications, including skin and mucosal sites. Dividing the topics that require attention into the manufacturing origins of the particles, the physicochemical and performance properties, and the methods of measurement from which the quality of the product can be established encompasses the major areas of consideration. Here an overview of the approach taken throughout the book is given with the intent of defining the connection between each of the areas presented.

Pharmaceutical particulate science is an aspect of foundational physical pharmacy, the constructional elements of pharmaceuticals, and their inclusion underpinning solid dosage forms of all types, exemplified by tablets, capsules, and suspensions [1–4]. Knowledge of the particulate state of drugs and excipients of which dosage forms are composed has been studied extensively as a theoretical phenomenon and has been comprehensively described in physical chemistry texts [4–6].

The properties of particles dictate the processes involved in manufacture and the quality and performance of the dosage form, which has direct implications for safety and efficacy of the drug in disease therapy. Unfortunately, theoretical understanding of idealized particle systems, while scientifically correct, does not translate into a generalized practical method of controlling the properties of heterogeneous particle systems. The particle heterogeneity emerges from the chemical composition, the presence of impurities, the molecular structure, the morphology, and the dimensions [7]. In addition, the nature of interparticulate forces is impacted by the physical properties of particles and can lead to transient, reversible, or irreversible aggregation, which will influence the product properties [7].

Considerations of particle properties and specifically those of pharmaceutical powders have been presented at length in reference texts [8]. This short text serves as an introductory guide for those entering the field or as a primer for those needing a high-level overview to refresh their past training or experience.

OPERATION	STATE	CHARACTERISTIC	ANALYSIS
Construction	Molecule	Chemical Composition Purity	HPLC/UV/MS
↓	↓		
	Initial Particles	Crystallinity Polymorphism Moisture Content	XRPD Thermal Analysis DVS
↓	↓		
Destruction	Processed Particles	Morphology Particle Size Distribution Density Porosity	Microscopy PSA Pycnometry/Porosimetry Gas Adsorption
↓	↓		
Assembly	Granules Suspension Composite Powder	Mixing/Blending Flow Characterization	Bulk/Tapped Density Static/Dynamic Angle of Repose
↓	↓		
Dosage Form	Capsules Tablets Aerosols	Dose Filling Compression	Content Uniformity Dissolution Dispersion (APSD)

Fig. 1.1 Unit operation rendering the drug, or excipient, in different states. Each state can be monitored for physicochemical characteristics or performance measures by the application of a variety of analytical methods

Figure 1.1 illustrates the systematic assessment of the particle properties that require consideration from the perspective of exploring the fundamental factors and their role in particle behavior. The importance of monitoring and controlling these properties is central to assuring the quality of any product of which the particles are a component [9]. The overall approach starts with the production of particles during active ingredient or excipient manufacturing, which yields a solid with a specific molecular structure and particle morphology. This operation can be controlled if the intent is to use the process both as a method of producing particles and having them conform to predetermined properties or specifications [10, 11]. However, it is rarely the case that the initial particles are used as manufactured in the dosage form or final product. Since further processing is required, it is essential to characterize the physicochemical properties of the starting material both before and after subsequent unit operations. The primary manufacturing processes employed following initial active pharmaceutical ingredient production involve methods to manipulate particle size distribution and morphology to accommodate the subsequent processing needs of powder flow or suspension in a liquid, filling, and, for some solid dosage forms, granulation and compression.

Given the spectrum of activity through the manufacturing process for which pharmaceutical particulate science is important, a sequence has been adopted in this text that first covers the theoretical foundation used to describe specific properties followed by consideration of the methods required. To transition from the theoretical to practical considerations, assumptions are made, and the significance

of being aware of these assumptions cannot be understated. Vigilance is required in matching the method, the assumptions, and the application to ensure that the most accurate and reproducible interpretation of data or imposition of control can be enforced.

Product development requires measurement of physical parameters and an approach to definitions that allow quantitative expressions for each property and sampling strategies that allow subsequent data interpretation that under statistical scrutiny allows prediction of quality and performance as central determinants of anticipated safety and efficacy of the product.

The importance of finely divided solids in pharmaceutical dosage forms relates to all of the steps involved in dosage form manufacturing. The state of the powder influences its movement, uniformity, and ease of handling, including accurate and reproducible dose filling. The chemical and physical stability of the intermediate states and the final dosage form will be affected by the nature of the particulate system. Specialized unit operations such as granulation depend upon the starting powders. Finally, the extent of control over delivery, as defined by deaggregation, dissolution, and diffusion, will correlate with the powder properties. The point at which the dosage form interfaces with the organism for solids is most frequently at a mucosal site, and either local or systemic bioavailability will translate into the desired therapeutic outcome while ideally ensuring safety by limiting off-target effects.

As noted earlier, particulate systems are predominantly found in tablets, capsules, and suspensions. However, it should be noted that there are also orally ingested powders, inhalation and nasal aerosol suspensions (aqueous and nonaqueous) and powders, topical pastes, and various parenterally administered depot's microparticle and nanoparticle systems.

In conclusion, the following text is divided into intrinsic (Chaps. 2, 3, and 4) and process-dependent (Chaps. 5, 6, and 7) physicochemical properties, manufacturing process (Chaps. 8, 9, and 10), and dosage form design considerations (Chap. 11) with respect to pharmaceutical particulate systems. The major points from the foregoing sections are summarized to bring the volume to a conclusion (Chap. 12). Chapters have been updated for recent advances, notably in the manufacturing techniques and their applications. At the highest level, it should become evident that monitoring and control of the spatial and temporal constraints on solid particles are the central principles for assuring the final quality and performance of the product.

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Chapter 2

Solid-State Chemistry



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Abstract Pharmaceutical solid dosage forms are composed of powders whose quality and performance are based on the foundational solid-state chemistry defining the form in which the particles exist. Understanding the behavior that emanates from the molecular matrix and final morphology is central to establishing the reproducibility of key quality and performance metrics.

The compositional and structural characteristics of the solid represent the spatial physicochemical factors with implications for potential temporal chemical and physical effects. Characterizing these properties may be used to establish the initial and final state of a solid with respect to a process which may be connected to product quality and performance. These properties partially define expectations of stability, dissolution, and bioavailability, which in turn relate to safety and efficacy.

Occasionally, particles exist with disordered, amorphous, molecular structure. Most frequently ordered systems exist that are defined by uniform distances between adjacent molecules that define the general morphology of the crystal system. However, the particle appearance may vary based on constraints of crystal growth or the presence of impurities.

The composition and structure of particles are the foundations on which the final dosage form is constructed. Processes such as purity, solubility, milling, blending, filling, granulation, tableting, dissolution, and stability all require knowledge of the molecular composition and the arrangement of the molecules within the particle [1].

Composition

It is the goal of active pharmaceutical ingredient (API) manufacture to prepare homogeneous particles with the highest level of purity. The nature of all manufacturing processes ultimately yields a product with the presence of impurities, albeit at very low levels, often less than 2% [2]. The origin of these impurities can be starting materials in the manufacturing process, other products of the synthesis of biosynthesis from which the primary material was generated, or simply the presence of ambient substances such as water [3, 4]. The arsenal of analytical methods from which purity determinations can be made allows for a high degree of accuracy and precision. Among the most common chemical purity analytical methods are nuclear magnetic resonance and mass spectroscopy [5]. Indeed, mass spectroscopy can now be combined with high-performance liquid chromatography for routine testing of the drug for stability and release testing [6, 7]. The presence of water can be followed by classical methods such as Karl Fischer and thermogravimetric analysis [8, 9].

Consequently, the composition of particles can readily be established and the process of manufacture validated to allow further consideration of properties that are of importance to the final dosage form.

Structure

Some processes result in amorphous materials that may be considered as randomly oriented molecules within the particle that usually have small intermolecular forces of interaction and high free energy of dissolution. [10–12] More frequently, pharmaceutical particles result from a crystallization process and take on a highly structured form wherein planes of molecules align with intermolecular distances defined by their size and forces of attraction (polarity and charge) [13]. There are two elements to structural consideration of particles: the crystal system and the crystal habit [14, 15].

The crystal system is defined by the distances between planes of molecules within the particle. These distances are estimated according to Bragg's law from measurements made by X-ray crystallography [16]. A variety of crystal systems have been identified, but for simplicity, they have been reduced to seven specific categories known as Bravais lattices, i.e., cubic, tetragonal, orthorhombic, monoclinic, triclinic, rhomboidal, and hexagonal, as illustrated in Fig. 2.1 [16]. Polymorphic forms of a substance differ in terms of the molecular geometry within a particle. It is important to be aware of the propensity to form polymorphs and their identity as they usually influence the performance of the product in terms of functionality, for example, dissolution rate and stability [17].

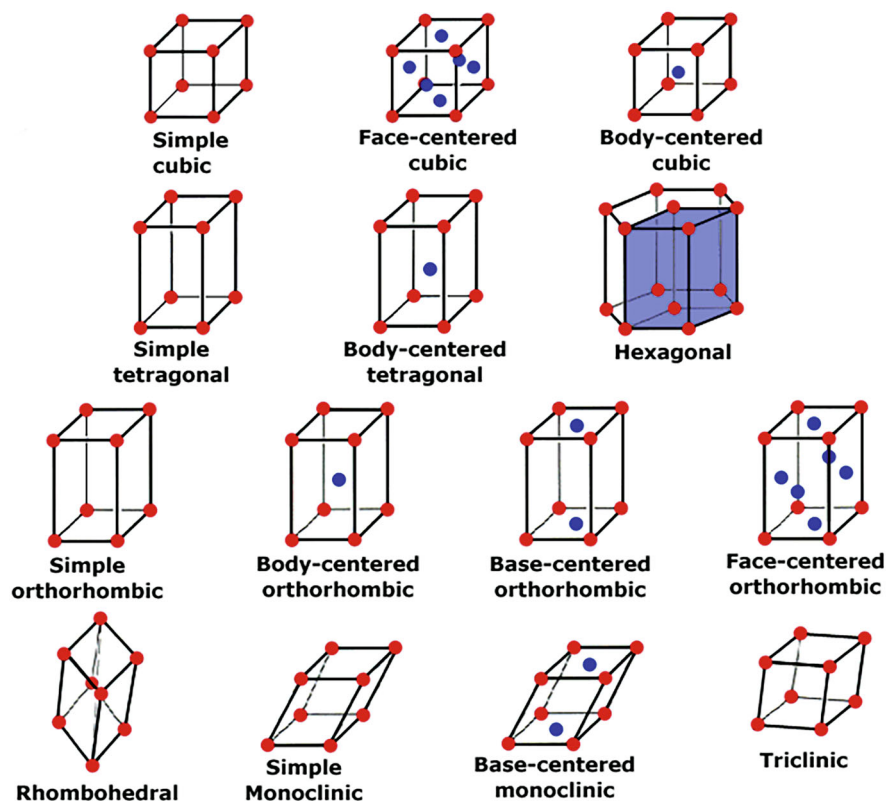


Fig. 2.1 Bravais lattices are defined by the arrangement of molecules or atoms in a crystal system. (From www.seas.upenn.edu, Solid-State Chemistry)

The underlying crystal system gives rise to a variety of crystal structures depending on the solvent in which it is crystallized [18–20]. These crystal habits represent the shape of a particle as influenced by constraint or enhancement of growth in a dimension in space. The most common examples of crystal habit in terms of impeding growth are unconstrained growth of a cubic crystal that yields cubic crystals, cubic crystal growth constrained in one dimension which yields a plate, and cubic crystal growth constrained in two dimensions which yields a needle, as illustrated in Fig. 2.2 [15]. The complexity of some crystal habits is such that a system was devised to describe them, known as Miller indices. Miller indices place the crystal on a 3D coordinate system and then use Cartesian method to describe the vertices from which crystal faces can be plotted. Using this system any crystal habit can be described as illustrated in Fig. 2.3.

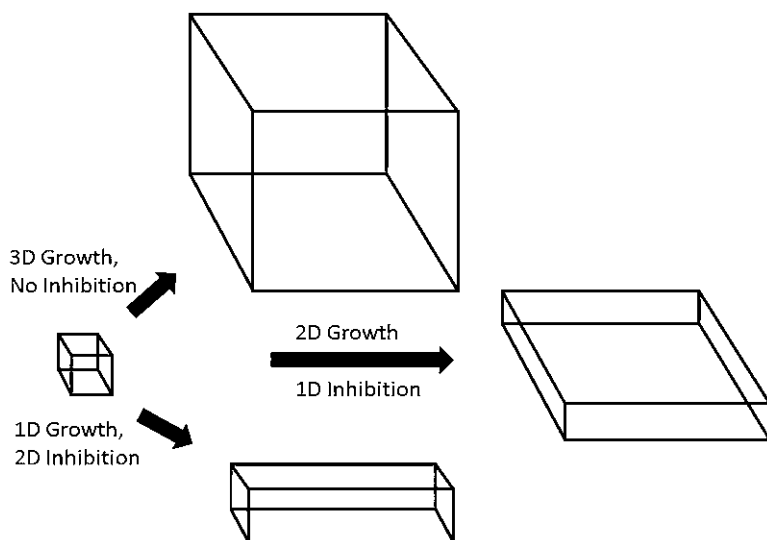
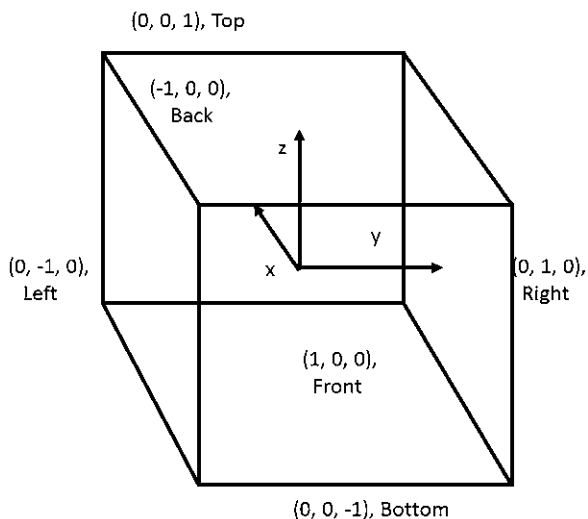


Fig. 2.2 Crystal habits emerging from cubic seed crystal assuming different constraints on growth in three dimensions

Fig. 2.3 Miller indices describing crystal systems by Cartesian coordinates



Imperfections

The study of crystallinity and its impact on the performance of powders has been a topic of increasing interest as early process control engineering evolved into quality by design of pharmaceutical products [21]. Descriptors of crystal systems and habits

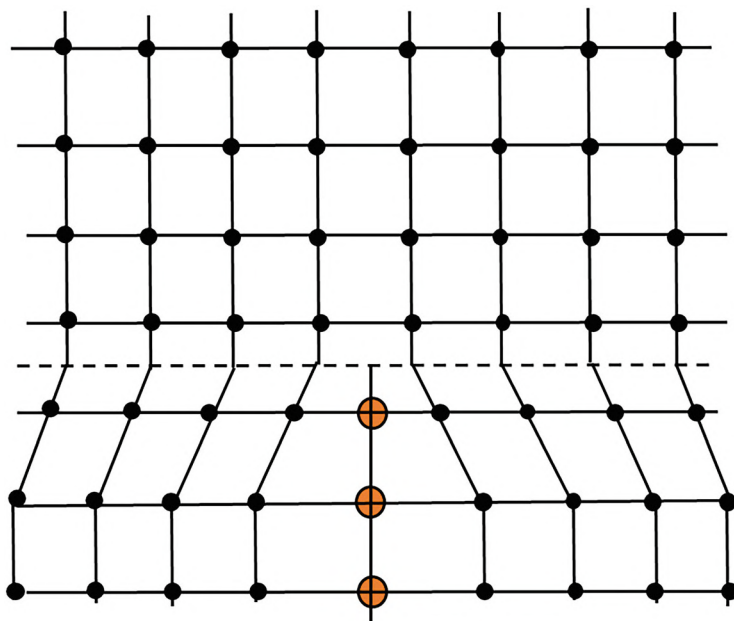


Fig. 2.4 A surface dislocation in the crystal system caused by inclusion of molecules of an impurity

are often based on pure materials which form particles with regular polygonal structures. However, as scrutiny of powders and processes increased, it became evident that the behavior of certain particulates did not lend itself to the level of control that the apparent quality of the product might suggest was possible. The importance of this observation lies in the ability to meet quality metrics for both the physicochemical and in vivo performance of solid dosage forms.

As knowledge increased and new tools were developed to allow greater resolution with which to study particles, it became evident that they play an enormous role in their performance and that the ability to know the location and control the influence of impurities might drive the final performance of the product. The nature of the impurity dictates the type of surface phenomenon that might be anticipated. In the first instance, the presence of chemical impurities might result in dislocations in structure that at the surface of the crystal would appear as edges and corners giving rise to surface rugosity or, in some cases, more exotic spiral dislocation, as illustrated in Fig. 2.4 [22]. The primary substance itself may take on a different form, notably amorphous, in regions at the surface. Each of these examples would also give rise to different surface energy densities that create interparticulate binding heterogeneity [23]. Finally, the presence of moisture gives rise to capillary forces and may result in dissolution and recrystallization forming solid bridges with temperature cycling [24]. These phenomena are now routinely examined and controlled in the event of poor product quality or performance.

Conclusion

The composition and structure of any particulate or powder system are the foundations on which all other elements of the final product are built. By controlling the processes of API and dosage form manufacture within adequate specification and studying the drug and excipient particles carefully, it is possible to define the limits to quality and performance within which the product must be controlled to minimize the influence of known variables that may give rise to variation. These can be specified at the level of the influence of impurities, solvents, and any other additives (e.g., surfactants) that might be used in any of the processes that will give rise to changes in crystal system, crystal habit, or seed imperfections into the surfaces of particles that would have undue influence on the final product.

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Chapter 3

Crystallization



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Abstract Crystallization is the physical process of forming a crystalline material from a liquid, gas, or amorphous solid. The control of the crystallization process has several implications in the pharmaceutical field and a strong valence in the determination of API and excipient properties, which impact solid dosage form performance. Lack of control on the dynamics of crystal equilibria is detrimental to manufacturing and stability of pharmaceutical dosage forms. Such an increased awareness has fostered the development of advanced powder technologies.

Crystallization is the physical process of forming a crystalline material from a liquid, gas, or amorphous solid [1]. Such a phenomenon can be spontaneous or induced by a change in local conditions, such as temperature, pressure, as well as medium system (liquid, gas). Generally, this effect is due to induction of a state of decreased solubility that provokes nucleation and crystal growth. The process starts with the generation of a supersaturation condition of the solution or melt. The supersaturation is the driving force for new crystal formation and growth.

The control of the crystallization process has several implications in the pharmaceutical field.

On the one side, it can be exploited as a separation method to recover pure substances from a solution or a reaction mixture and, on the other side, has a valence in the determination of API and excipient properties, which impact solid dosage form performance. The lack of control on the dynamics of crystal equilibria has serious consequences on solid dosage form formulation and stability. Nowadays, an

increased awareness of the importance of crystalline solid properties has led to the development of advanced powder technologies. A considerable body of literature addresses the impact of crystallization processes on pharmaceutical products.

Nucleation and Nucleating Agents

The formation of a new crystalline entity starts through a nucleation process. Nucleation is defined as the process by which molecules of a particular substance rearrange into clusters of a size sufficient to initiate an irreversible growing phase leading to macroscopic structures. The clusters are defined as the critical nuclei. Two nucleation processes are generally recognized: homogeneous and heterogeneous nucleation (Fig. 3.1).

The first occurs when molecules start cluster formation in a homogeneous phase, where no foreign components are present. Heterogeneous nucleation instead accounts for the presence of foreign particles or seeds that provide a scaffold on the surface of which aggregation occurs. Both processes are regarded as primary nucleation phase. Secondary nucleation takes place when crystals of the same substance are present.

As mentioned above, the nucleation driving force is a state of reduced solubility or supersaturation that consistently supports nucleation and crystal growth. The thermodynamic force that drives such processes is a chemical potential gradient that can be written as

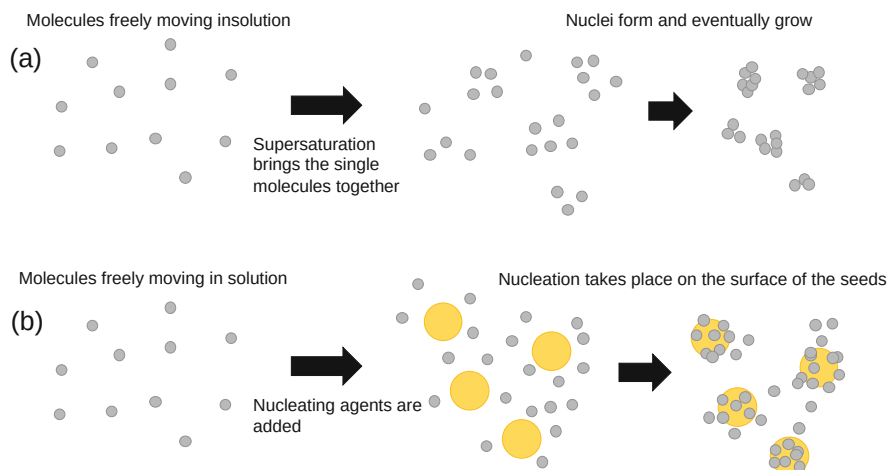


Fig. 3.1 Two main types of nucleation processes: (a) homogeneous nucleation and (b) heterogeneous nucleation. The second process can occur at a lower supersaturation condition, thanks to the addition of nucleating agents (seeds)

$$\Delta\mu = kT \ln S \quad (3.1)$$

where k is the Boltzmann constant, T is the absolute temperature, and S is the supersaturation ratio [2]. This parameter is different according to the initial physical state of the solution, i.e., gas, liquid, or melt.

Supersaturation and thus nucleation and crystal growth occur when $\Delta\mu > 0$; on the contrary, when $\Delta\mu < 0$, the process is inhibited. Well-known thermodynamic relations link the chemical potential gradient to the free energy and a surface energy term according to Eq. 3.2:

$$\Delta G = -n\Delta\mu + 4\pi r^2 \sigma \quad (3.2)$$

where r is the radius of the nucleus and σ is the surface free energy [2].

Therefore, factors granting $\Delta\mu > 0$ and decreasing σ favor nucleation and crystallization. In particular, since the presence of foreign particulates and especially crystals lowers surface energy, it is clear that secondary nucleation and/or seeded nucleation is favored with respect to homogeneous nucleation, and they will occur even at lower supersaturation conditions.

This thermodynamic explanation of the nucleation phenomena suggests that addition to nucleating agents, such as other particles or crystals, can considerably accelerate crystallization. This approach is used in the industry to control the formation of particular crystal structures that otherwise would be not favored as in the case of metastable polymorphs that will be better discussed later in this chapter.

Crystal Composition and Crystal Engineering

A pure crystal is an ideal substance in which all the molecules are perfectly identical and their alignment is perfectly uniform throughout the crystal cell. This means that entropy is zero at zero absolute temperature. Naturally, this is never the case in the real world. Imperfections occur even in isomorphous crystals as a result of atom vacancies, displacement, and substitution that increase entropy and change crystal stability and behavior.

Different arrangements in the crystal geometry and level of order change considerably the thermodynamic energy and thus stability of the crystal. This is because entropy increases with differences in composition and imperfections, thus decreasing the associated free energy, which means that crystal disruption can be as easy as its formation. Therefore, crystal composition and order determine the behavior either in solid or liquid phases. Properties such as solubility and chemical stability can change considerably with crystal composition. Understanding crystalline properties is therefore of high importance especially when dealing with complex molecules, such as APIs. Generally, more stable crystals show lower solubility and higher physical stability that can hinder dissolution and subsequent bioavailability, especially for poorly soluble drugs. In this regard, modulating crystal composition can result in a valuable strategy to improve solid API properties.

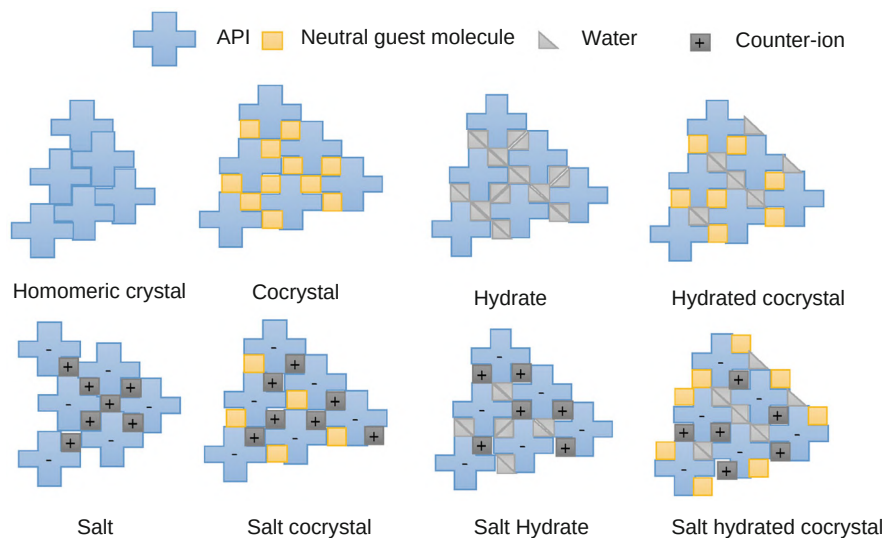


Fig. 3.2 Different crystalline structures for an API that can be employed to modulate a drug's properties

Therefore, crystal engineering has become a widespread approach to tune API solid-state properties [3]. In particular, cocrystal formation is a strategy that can be effective in increasing drug bioavailability by enhancing solubility and storage stability [4].

Pharmaceutical cocrystals are non-covalent stoichiometric multicomponent solids comprising an API and one or more cocrystal former, or coformer, in crystal lattices usually bound by hydrogen bonds.

Cocrystals can be obtained through melt crystallization, solid-state grinding, solution-mediated phase transformation, and solution crystallization [5].

Cocrystals are distinguished from salts by a lower degree of proton sharing as they can show nonionic nature, although such a distinction is not always neat.

Different solid structures can be described for an API when crystallized with other components and cofomers that have a distinct nature compared to solid solutions and eutectic mixtures (Fig. 3.2) [5, 6].

The formation of such mixed structures can be oriented and controlled by using proper cofomers, usually GRAS (generally recognized as safe) materials bearing complementary functional groups to the API, temperature, concentration, solvent system, or melt conditions. Improvement of solubility and stability and a consequent improved dissolution and bioavailability have been recorded for poorly soluble drugs when using soluble partner substances [6].

Therefore, crystal engineering and modulation of crystal composition by the formation of pharmaceutical cocrystals have proven value in modifying API physicochemical properties with a special emphasis on solubility and dissolution. Cocrystals offer great potential for fine-tuning solubility and thermodynamic

stability. This strategy is particularly suited for drugs prone to degradation under harsh acidic or basic conditions or drugs lacking features required for salt formation.

Polymorphism

Many substances have the ability to exist in more than one crystal form or phase that have different molecular arrangements or conformations in the crystal lattice, depending on molecular properties and the conditions at which crystallization occurs. This phenomenon is regarded as polymorphism. Such crystalline structures or polymorphs are chemically identical and differ only in the way molecules arrange and determine different crystal lattice geometries. Polymorphs can coexist and convert into each other according to conditions and eventually transform into the most thermodynamically stable form at that given condition. Polymorphs show different physicochemical properties that can profoundly affect bioavailability, stability, and other performances of pharmaceutical products. As a result, polymorphism arises serious concern in pharmaceutical development as it requires proper control strategies. This phenomenon is widespread and involves nearly half of the known drug substances [7]. In addition, even other solid forms of a drug, such as salts, solvates, cocrystals, and hydrates, can show polymorphism.

Polymorphism is relevant not only to API but even to some pharmaceutical excipients. Worth citing is the case of hard lipids, in particular long-chain glycerides, whose known polymorphic conversions over time cause the well-documented storage instability of solid lipid particles, which undergo a deep morphological change with drug loss and aggregation [8].

Two categories of polymorphs are described: packing or orientational and conformational polymorphs. The first comprises conformationally rigid molecules packing and binding in different three-dimensional structures, while in the second, conformationally flexible molecules assemble into different ordered structures (Fig. 3.3). The second case is the most commonly observed among API. Synthon polymorphs are also possible in which different molecular synthons¹ assemble into crystal structures generally held together by hydrogen bonds.

Selection of the proper polymorph is mandatory to prevent the aforementioned issues. This goal requires deep knowledge of crystallization steps, among which nucleation is the most critical. Therefore, as a measure of the complexity of the problem, control over solvent, supersaturation, temperature, and the use of additives, nucleating agents and molecular templates is crucial. For this reason, heterogeneous crystallization is the most employed.

A number of methods can be used to obtain specific polymorphs; some of the most employed are reported in Table 3.1.

¹Synthon is defined as a structural unit within a molecule that can be formed and/or assembled by known or conceivable synthetic operations.

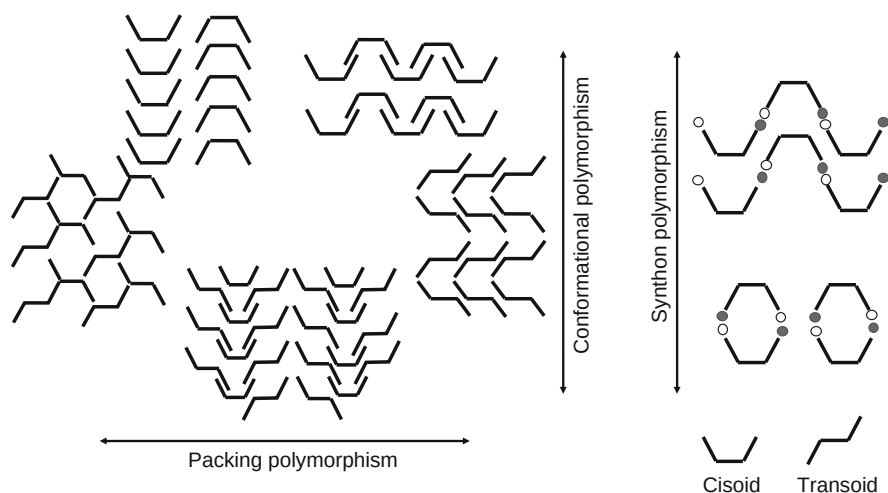


Fig. 3.3 Illustrations of possible polymorphism categories. Conformational and packing modes are the most commonly observed. Synthon polymorphism depends on the nature of molecule functional groups. To give polymorphism, synthons need to be different good hydrogen donors/acceptors

Table 3.1 List of methods for polymorph preparation

Crystallization from a single or mixed solvents/HTS
Thermal activation of the solid substrates
Crystallization from the melt
Desolvation/dehydration of solvates/hydrates by heat or by reslurry
Crystallization in nano-confined structures
Seeding/pseudoseeding
Solution-mediated polymorphic transformation/slurry conversion method
Solid-state polymorphic transformation
Mechanical activation of the solid substance
Crystallization in a capillary tube
Exposure to vapor at high or low humidity
Exposure to organic vapor
Directed crystallization on molecular substrates
Crystallization in the presence of tailor-made additives
Laser-induced crystallization
Crystallization from a supercritical fluid
Structure prediction

Intensive research is carried on to understand the complex relationship between polymorph formation and crystallization conditions. This is particularly relevant to solid materials processing. In fact, changing process conditions will influence polymorph formation and conversion. As an example, disappearing polymorphism is a frustrating phenomenon, perhaps ascribable to a change in seeded nucleation [7],

that, albeit working conditions are perfectly replicated, can hinder reproducibility of polymorph preparation. Cross nucleation is even possible when a polymorph crystallizes on another polymorph, which occurs independently of their relative thermodynamic stability.

Conversion between metastable and stable forms can be obtained even once polymorphs have been prepared by changing temperature or pressure. However, this process is dictated by their stability in given temperature and pressure ranges. Two systems may occur: monotropic or enantiotropic. In monotropic systems, only conversion into the stable polymorph is possible as the most stable polymorph is as such at any temperature or pressure, and, therefore, any conversion is irreversible. In enantiotropic systems, two or more forms can show transition temperatures or pressures above or below which one form or the other shows higher stability, thus allowing reversible conversions between polymorphs. Knowing the difference between these systems is thereby essential to prevent a number of issues during manufacturing of API and dosage forms.

Therefore, successful formulation and process design require insightful knowledge of polymorphism, interconversion mechanisms and conditions, as well as processing options. Assurance of product quality passes through understanding on how to anticipate, control, or prevent phase transformation in process design and development.

Water of Crystallization

The critical role of solvents in crystallization processes is well known; in particular, it is their capacity to become constitutive part of crystal lattices, giving rise to solvates. As stated above, solvates can show polymorphism as well. Therefore, understanding the mechanisms and processes leading to solvate formation is fundamental. Water has a main role in this regard, and the ability to form solvates, known as hydrates with pharmaceutical compounds that dramatically affect the performance of dosage forms is well documented. Approximately one-third of APIs are able to form hydrates. Incorporation of water of crystallization into the particle structure changes API performance.

The considerable extension of the problem requires careful addressing of the relationship between manufacturing process conditions and the API tendency to form hydrates, of their polymorphs, and how they can be controlled.

Generally, hydrates show lower solubility and dissolution rate as well as different stability with respect to the anhydrous form [9]. Excipients also affect water-solid interaction, and their role should be carefully considered. As in the case of polymorphs, effects of processing methods on the anhydrate/hydrate system have to be addressed.

The ability of water to form hydrates is related to its strong hydrogen bonding capacity with organic molecules and its tendency to bind metal cations in salts. Therefore, water fills vacancies in the crystal lattices of host molecules, reducing the

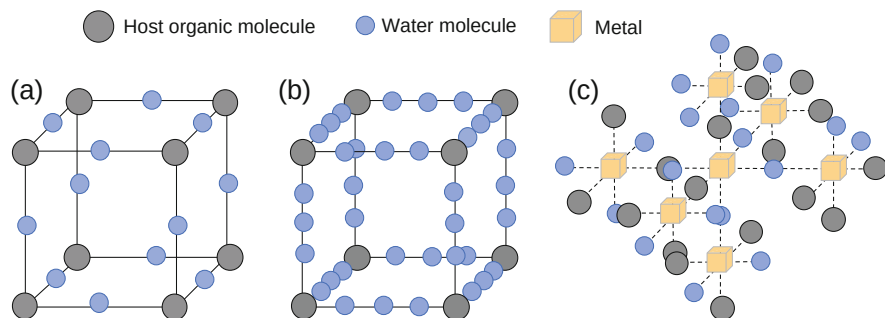


Fig. 3.4 Possible structures of hydrates differentiated according to water molecule organization into the crystal lattice: (a) type I—isolated site hydrate, water molecules alternate in the molecule sequence. (b) Type II—channel hydrates, water molecules line up consecutively into distinct sequences. This hydrate type can expand by absorbing additional water molecules. (c) Type III—ion-associated hydrates, in which a metal ion coordinates water and host molecules. The reported crystal structures are not representative of actual structures and should be considered illustrative sketches

free energy of the solid. Water activity determines thermodynamic stability of hydrates. In solution, water activity depends upon the presence of solvates and other solvents, while, in vapor, it matches the relative humidity (RH). Thus, measuring water activity is essential for successful formulation and process development.

Albeit not universally accepted, three different types of hydrates are considered (Fig. 3.4). Type I are isolated site hydrates, in which water molecules are not in contact with other water molecules in the crystal lattice; type II are channel hydrates, which have chains of water molecules along a crystal lattice axis, which can expand to form planar hydrates where the water of crystallization forms a two-dimensional plane in the lattice; and type III are ion-associated hydrates, which contain metal ion-coordinated water [9].

This complex picture advises for careful evaluation of water activity not only during crystallization processes but even during storage of API and solid dosage forms. In fact, sufficient RH levels may induce formation of hydrates upon contact with the product. This process can be fast or slow depending on RH and solid properties as well as environmental conditions. This problem is particularly relevant for hygroscopic materials that can adsorb water to significant levels, thus favoring their conversion to hydrates.

The water of crystallization may also affect the chemical stability of pharmaceutical products. This is due to a greater propensity to oxidation than other crystal forms.

Another aspect to be considered is that hydrates can undergo dehydration and the process results in abrupt structural changes that affect solid particle morphology and behavior. The rate and extent of this process depend on the hydrate structure. Type I is less prone to dehydration than types II and III; however, if it happens, disrupting changes can induce structural loss and subsequent crystallization. In case of a

non-destructive pattern, amorphous materials may result. Obvious consequences are changes in solubility and dissolution rate, eventually affecting bioavailability.

The possibility of hydrate polymorphism further complicates this challenging scenario.

Conclusion

Solid-state properties of API and solid dosage forms strongly depend upon the way they are manufactured. This is because most of pharmaceutical powders are the result of crystallization processes that, although being effective separation methods, are sensitive to a considerable number of factors that determine the final form and property of the powder. Properties such as solubility, stability, dissolution rate, and thus bioavailability can be dramatically impaired by disregarding crystal formation dynamics. Therefore, crystallization is a key process that must be controlled in order to ensure the wanted product quality attributes. In this regard, the advent of modern crystal engineering methods has led to the development of effective strategies, e.g., cocrystals, for the prevention and control of crystallization issues. These approaches are destined to open new perspectives in the management of phenomena such as polymorphism or hydrate formation, granting important advances in pharmaceutical manufacturing.

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Chapter 4

Physical Properties Characterization



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Abstract The quality and performance of pharmaceutical particles and powders are dependent on critical quality attributes that are identified in development. Powder characterization is a fundamental step for proper design and control of solid dosage form manufacturing processes. Properties like surface area, charge, crystal habit, and the presence of water and impurities are responsible for dramatic changes in the physical-chemical and biological behavior of pharmaceutical products. Therefore, accurate tuning of powder features is compulsory. In this regard, an arsenal of particle characterization techniques has today reached a high level of reliability and sensitivity, which allows insightful investigation of the complex crosstalk between particle properties and pharmaceutical processes. Novel technologies are now on the verge of breakthrough advances thanks to new methodologies and computer science.

Characterization of powder physical properties is an essential activity particularly relevant in the pharmaceutical quality system. Intended or unintended differences in particle characteristics are responsible for changes in dissolution profile, physical and chemical stability, drug-excipient compatibility, and biological behavior of pharmaceutical products. Moreover, industrial processes require strict control over powder properties. In this chapter, the fundamental physical properties and their modern characterization methods are identified.

Surface Area

The surface area of a powder is an attribute directly dependent on particle size and morphology.

The specific surface area (a_s) of a material is defined as the total surface area (a_t) of the material divided by its mass (Eq. 4.1).

$$a_s = \frac{a_t}{m} \quad (4.1)$$

A number of different methods allow determination of a_t from the measurement of the volume of a gas, usually nitrogen, that adsorbs on the surface of the material at constant temperature (77.4 K, the boiling point of liquid nitrogen) under the assumption that the gas condenses as a monolayer. The a_t value of a solid represents not only the surface exposed on the particle exterior but also that of accessible pores. For highly porous materials, the contribution of the inner surface can even exceed that of the exposed surface.

After the Langmuir's derivation of the gas-solid adsorption isotherm, Brunauer et al. in 1940 and later Sing compiled today's IUPAC classification of the six gas-liquid adsorption isotherms [1, 2].

The most popular method of interpreting gas adsorption data is by fitting to the Brunauer, Emmett, and Teller (BET) isotherm [3, 4]. The method measures the volume of gas adsorbed (V_a) as determined by a volumetric or continuous flow procedure.

The volume of gas adsorbed as a monolayer (V_m) is determined from linear regression of the BET equation (Eq. 4.2):

$$\frac{1}{[V_a(\frac{P_0}{P} - 1)]} = \frac{C-1}{V_m C} \times \frac{P}{P_0} + \frac{1}{V_m C} \quad (4.2)$$

where P is the partial vapor pressure of the adsorbate gas (nitrogen) in equilibrium with the surface at 77.4 K, P_0 is the saturated pressure of the adsorbate gas, and C is a dimensionless constant that is related to the enthalpy of adsorption of the adsorbate gas on the powder sample.

The obtained V_m value is then employed to derive a_t (Eq. 4.3):

$$a_t = \frac{V_m N A_{cs}}{M} \quad (4.3)$$

where N is the Avogadro's number, M the molecular weight of the adsorbate, and A_{cs} the adsorbate cross-sectional area ($16.2 \text{ \AA}^2$ for nitrogen).

A strict requirement is sample cleaning before measurement procedures. This goal is achieved by outgassing the sample in the cell and by eliminating moisture, volatile components, and impurities.

Charge

Triboelectrification is the complex phenomenon of surface charge exchange upon contact of solid materials. This effect is the consequence of a difference in surface electric potential due to material intrinsic physico-chemical properties. Pharmaceutical powders, being generally poor conductors, can accumulate a considerable charge whenever being processed. This can result in serious issues, especially when opposite charges are induced, which easily leads to particle aggregation. In this way, flowability is considerably hindered affecting performances and production of pharmaceutical solid dosage forms. Surface charge is strongly influenced by particle size, shape, and surface roughness, as much as process and formulation factors and environmental conditions [5]. Moreover, it should be noted that the accumulation of charge can pose an explosion risk in manufacturing facilities and requires mitigating safety features [6, 7].

Several methods have been proposed to measure surface charge of powders. The simplest method is the Faraday cup, consisting of two concentric metal wells (Fig. 4.1a). The inner well is electrically insulated, thus preventing charge leakage, while the outer one is electrically grounded to shield from external electrical interferences. In the presence of charged particles, unidirectional electron flow is measured in the inner well by a connected electrometer. The system can be easily

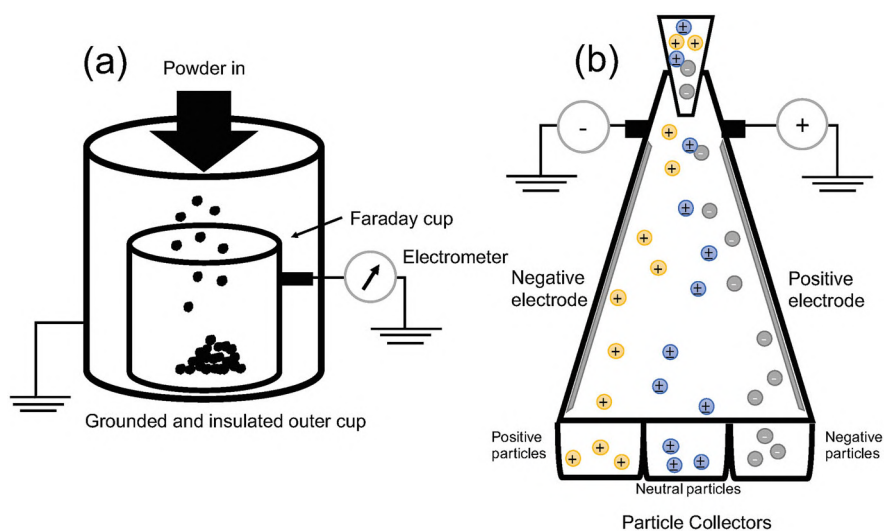


Fig. 4.1 Panel (a): Schematic representation of the Faraday cup system comprising an inner conductive well and an external outer insulating well that prevents external electric interferences. An electrometer measures the charge transfer to the inner well walls. Panel (b): General scheme of a free-fall particle separator. The generated electric field in the separation chamber deviates the path of falling charged particles (yellow-positive, gray-negative) allowing their recovery in dedicated collecting cups. Neutral particles (blue) are not deviated by the electric field, thus depositing in the central collector

mounted in line with process equipment. To increase sensitivity and performance, multiple Faraday cups can be horizontally or vertically arranged to allow splitting of flowing particles according to charge sign as well (Fig. 4.1b).

These methods are applied to powders in a dry state. In the presence of a liquid medium, surface charge is usually determined by measuring particle zeta potential (ζ). Zeta potential is defined as the potential due to the charge that develops at the interface between a solid surface and its liquid medium. A required attribute is that the dispersed solid should be formed by submicron particles subject to Brownian motion, which are not prone to rapid sedimentation. Consequently, this technique is the most frequently employed for the characterization of colloidal dispersions.

The charge arises from ionization of functional groups and adsorption of solution ions on the particle surface. As a result, a double layer of ions is formed at the particle-liquid interface characterized by a more tightly bound inner region, the Stern layer, and an outer diffuse layer (Fig. 4.2).

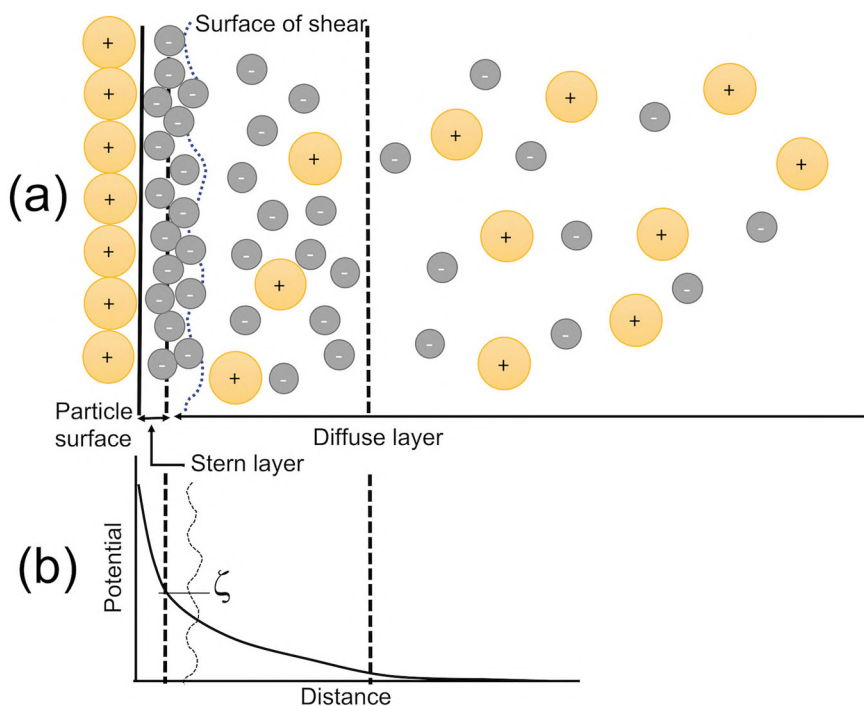


Fig. 4.2 Illustration of the charge distribution at the surface/liquid interface of a particle. Panel (a) the potential is generated by the charged surface that attracts counterions from the liquid. Charges then distribute to form a Stern layer separated by a surface of charge shear from a diffuse layer. Panel (b) the potential drops as the distance from the particle surface increases. The zeta potential (ζ) is measured at the interface between the Stern and diffuse layer

Zeta potential can be determined by measuring the electrophoretic mobility (v_E) of charged particles in an applied electric field according to the Henry equation (Eq. 4.4) and the Smoluchowski or Huckel models: [8]

$$v_E = \frac{2e\zeta f(ka)}{3\eta} \quad (4.4)$$

where ε is the relative dielectric constant, η the solution viscosity, and $f(ka)$ the Henry's function [9].

Zeta potential measurements, being a direct estimation of interparticle repulsion, are extremely useful for routine investigation of colloidal particle stability. Particles with ζ between ± 30 mV are considered electrostatically unstable and prone to agglomeration/aggregation and flocculation [10].

Today's method of choice to measure v_E is photon correlation spectroscopy, which exploits the scattering capacity of dispersed colloidal particles under a laser beam. Other modern techniques include tunable resistive pulse sensing (TRPS), based on the Coulter principle, and zeta particle tracking analysis (z-PTA), a high-performing optical imaging technique [11].

Polymorphism

As already discussed in Chap. 3, the presence of polymorphs requires careful management, since it can considerably affect performance and stability of pharmaceutical solid dosage forms. Accurate characterization of polymorphism is therefore a necessary activity for the pharmaceutical industry. One of the most powerful techniques for determination of polymorphs is X-ray powder diffraction (XRPD), which can accurately address amount, structure, and thermodynamic stability of crystalline architectures. However, this method is well suited for research purposes but inconvenient for routine analysis, as it is expensive and requires careful control of conditions for sample preparation. A faster and more reliable alternative to XRPD could be thermal analysis. Among the several techniques, differential scanning calorimetry (DSC) is the most employed for quality control analysis [12].

DSC works in non-equilibrium conditions as the sample is continuously heated or cooled, and non-reversibility is higher as the rate of temperature change is faster. This reduces resolution compared to conventional microcalorimetry. However, it can be a fast and reliable technique to determine the so-called thermal digital fingerprint of a material.

Heat flow (dq/dT) in the sample is measured and plotted against heating rate (dT/dt) to draw the thermal profile of the sample. The DSC signal consists of the total heat flow rate (dq/dt) that, under a thermodynamic quasi-equilibrium assumption, can be expressed as (Eq. 4.5):

$$dq/dt \approx dH/dt = nC_p dT/dt + f(T, t) \quad (4.5)$$

where $C_p dT/dt$ is a reversing heat flow component (heat capacity-dependent processes) and $f(T, t)$ a non-reversing heat flow component (time- and temperature-dependent processes).

Therefore, the transition enthalpy can be measured and quantified to determine thermotropic behavior of pharmaceutical systems. For known polymorphic materials, characteristic melting temperatures can be used to identify pure crystalline forms or mixtures as well as polymorphic transitions. Substantial advancement in the resolution of complex mixtures of polymorphic materials can be achieved by using modulated DSC[®] (MDSC[®]). Differently from standard DSC, MDSC[®] uses a simultaneous linear and a sinusoidal heating rate. Thanks to these multiple heating regimens, the reversing and non-reversing heat flow components in Eq. 4.5 can be separately resolved to discriminate overlapping processes and transitions, e.g., melting and crystallization or degradation phenomena (Fig. 4.3) [13].

Furthermore, the Hyper-DSC[™] technology, which allows extremely high-temperature scanning rates, has been found effective to investigate metastable polymorphs [14].

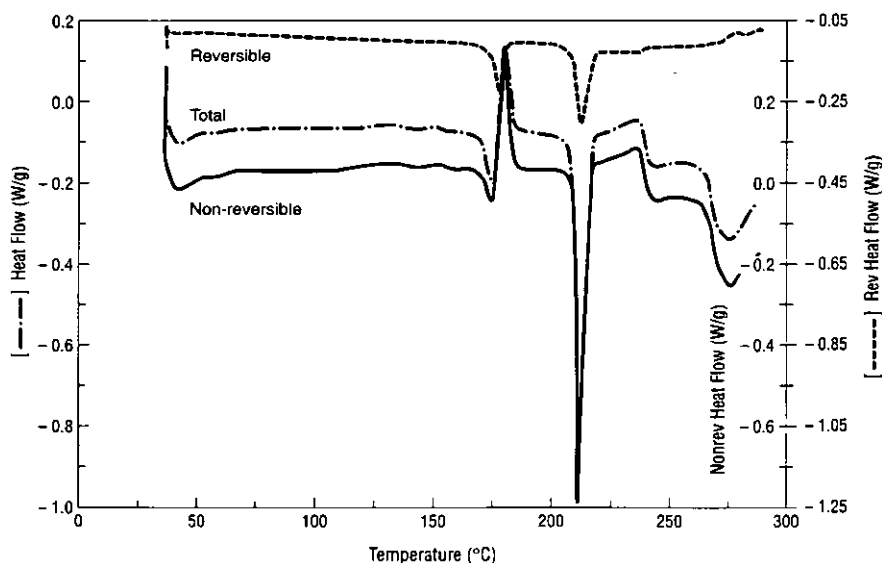
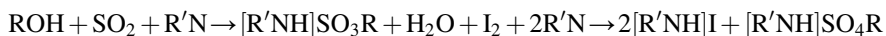


Fig. 4.3 Example of a modulated DSC profile of a polymorphic drug substance that shows effective separation between melting events observable in the reversing component profile and recrystallization that follows the first melting transition recorded in the non-reversing signal. (Adapted with permission from Ref. [13])

Moisture Content

Humidity is an important issue for pharmaceutical solid dosage forms. It plays a role in aggregation phenomena and the formation of hydrates, complicating the polymorphism of certain drugs. The presence of water can indeed change the behavior and physical properties of powders through a bridging effect among particles that results in agglomeration and stickiness hindering most of the industrial powder processing. However, moisture is not always detrimental to particle behavior. In fact, in some cases, it has been seen to improve flowability as a result of a lubricant effect or dissipation of static charge [15]. Therefore, proper analytical methods are required to address the moisture impact on powders and processes. Simple methods are based on the evaluation of weight difference before and after desiccation. These methods show poor sensitivity for low moisture contents and are affected by weighing and desiccation process performances. More advanced and reliable techniques are based on thermal analysis and chemical and spectroscopy methods.

The primary method for water traces determination is the one proposed in 1935 by the German chemist Karl Fischer and named after him as the Karl Fischer titration (KFT) method [16]. It relies on the following reaction:



The alcohol (methanol or 2-(2-Ethoxyethoxy) ethanol) reacts with sulfur dioxide and a base (imidazole or primary amines) to form an intermediate alkyl sulfate salt, which is then oxidized by iodine to an alkyl sulfate salt. This oxidation reaction consumes water. Water is determined by coulometric or volumetric measurements. Calibration is performed by using proper standards that can allow accuracy down to 1 ppm (coulometric, large sample mass). Nevertheless, the degree of binding of water and interfering compounds may significantly affect the results.

Another broadly employed method is thermogravimetric analysis (TGA). The sample is continuously heated according to a programmed temperature ramp. In this way, it is possible to determine the exact weight changes due to water loss as well as phase transitions and degradation phenomena. However, the presence of additional volatile substances that can be extracted with water and the absence of appropriate reference materials for instrument calibration can bias significantly moisture content quantification by this technique.

Additional methods are today in the spotlight due to relevant technological advances. Near-infrared spectroscopy (NIR) is a powerful technique that is finding large consent in the pharmaceutical industry. It measures the water overtone signal in the NIR region ($780 < \lambda < 2500$ nm). Although not particularly sensitive, NIR shows deeper sample penetration than IR, and therefore it allows probing bulk material and continuous in-line process sample analysis.

Hygroscopicity

Powder hygroscopicity is the tendency of a solid material to absorb and/or adsorb moisture from the surrounding environment. It raises the same concerns previously discussed for the presence of water in pharmaceutical solids. This property depends upon the intrinsic affinity of the material with moisture. As a consequence, the solid can be remarkably transformed, and for pharmaceutical products, it may result in significant storage instability. Therefore, its determination is crucial, in particular to the pharmaceutical industry. In this regard, dynamic vapor sorption (DVS) is the primary technique employed. It is designed to accurately determine a material's response to changes in relative humidity. Absorption and desorption isotherms are recorded on a wide relative humidity and temperature range, and the relative weight change of the sample is dynamically measured. Such technique can provide accurate information on processing and storage stability of pharmaceutical active ingredients, excipients, and dosage forms when exposed to different humidity and/or temperature conditions [17].

Impurities

Purity is the first among the several requirements that determine quality of pharmaceutical excipient and active ingredients. The presence of contaminants is a serious concern in the pharmaceutical quality system. Assurance of an adequate and consistent purity grade is mandatory for suppliers of pharmaceutical ingredients and pharmaceutical companies. Impurities affect considerably not only the quality of the product but even the performance of industrial processes. In fact, contaminants can be detrimental not only to the safety of pharmaceutical products but also to the stability and hazardousness of industrial processes [18]. A compendium of different techniques is available to determine impurities, and all of them show advantages and disadvantages (USP EP). A rule of thumb in the application of these techniques is that a higher accuracy results often in higher costs and time and sample handling problems. Therefore, the choice of the method is very much driven by convenience and level of reliability. Chromatography methods, i.e., high-performance liquid chromatography (HPLC) and gas chromatography (GC), provide the highest accuracy and sensitivity, although laborious and time-consuming. Other techniques sharing similar drawbacks are spectroscopic methods, such as nuclear magnetic resonance (NMR) and UV-vis spectrophotometry. These techniques have limited usefulness for routine and rapid quality control purposes. Alternatively, a less powerful but faster technique is NIR spectroscopy. As discussed above, although not very sensitive, NIR has the advantage of being useful for continuous analysis of bulk materials with little sample preparation. Optimal performances for industrial applications can be achieved when coupling separation techniques with online

separation techniques, thus obtaining the so-called hyphenated methods. Examples of such techniques are HPLC-MS, GC-MS, LC-NMR, CE-ICP-MS, and CE-MS.

Worth citing are thermal methods, such as DSC. This method could be useful, if high sensitivity is not required, for fast check of bulk materials and pharmaceutical formulations by simple measurement of changes in thermal transitions due to contaminant colligative effects.

It should be stressed that contaminants are considered not only chemicals but even microorganisms, many of which can be pathogens. In this regard, an arsenal of microbiological assays is available for microbiological purity assessment of non-sterile pharmaceutical ingredients [19].

Conclusion

The impact of powder properties on pharmaceutical product development is nowadays a well-understood concern. The last two decades of research and technological progress have led to an increased understanding of the implications of poor particle property control and characterization on pharmaceutical product quality. Particle analysis has reached a high level of performance and reliability. High-efficiency spectroscopic and chromatographic techniques have today become nearly routine methods for quality checks of powder products. At the same time, novel and more advanced technologies are now facing tremendous advances and will soon replace more traditional techniques with applications for in-line process control as well. Nevertheless, there are still many unsolved issues that arise from the evidence that powder processing changes powder properties in a way sometimes not exactly predictable. Therefore, it is crucial to match suitable high-throughput analytical technologies with particle engineering approaches.

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Chapter 5

Particle Size and Distribution



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Abstract Solid dosage forms consist of powders subdivided into a range of different particle sizes. Indeed, each component in the solid state would have different characteristics that must be considered with respect to behavior during manufacture and with respect to drug delivery. Particle size distribution, the measure of subdivision of a solid, can be directly correlated with uniformity of quality and performance. Primary particle size is a physicochemical property that may impact temporal chemical and physical effects such as stability and aggregation. In addition, particle size influences the motion of the powder or suspension that may be required during processing or as an endpoint of the dosage form performance.

Descriptions of particle size are based on assumptions about the dimension being measured. Moreover, since particles in a finely divided solid exist in a range of sizes, assumptions are made about their distribution and mathematical functions that best approximate the materials under scrutiny. It is important to acknowledge the assumptions and their limitations in accurately describing a particle size distribution. Comparing the definition adopted with the process or performance characteristic desired maximizes the likelihood of setting relevant product specifications.

All solid dosage forms and some dispersed systems, i.e., suspensions, require the preparation of drugs and excipients in known particle sizes to ensure the product quality. The description of particle morphology and size has been of great interest in physical pharmacy from its inception, and the topic of micromeritics underpins many biopharmaceutical considerations such as dissolution and diffusion rates and by inference bioavailability of drugs.

To bring uniformity to manufacturing and dosage form critical characteristics to assure performance, agreement is required on the descriptive terms for the dimensions of individual particles and their populations in powder or suspension form. The use of common terms that can ultimately be related to measurement methods facilitates the application of quality control during manufacturing.

Morphology and Size

Particles prepared by the most common methods of crystallization [1], milling [2] and sieving [3] are rarely regular shapes. Ideally, the particles will reflect a common crystal system and crystal habit, but frequently the processing or presence of impurities introduces amorphous regions or discontinuities that render the particle morphology unique. Since this is a practical reality, the model for describing particles is the equivalent spherical diameter. Since a sphere has the same size in any dimension of measurement, it is a logical morphology with which to equate the particle of interest.

The use of equivalent spherical diameters allows consideration of a range of particle morphologies in terms of a single equivalent object. However, the way in which the diameter is measured also needs to be specified to ensure that comparisons are made using the same assumptions.

Figure 5.1 illustrates particles and the various ways they can be depicted in terms of equivalent spherical diameters. The particle being examined is rarely spherical unless specialized methods of manufacture are employed; it can be equated optically with projected area or volume diameters and behaviorally with Stokes' and aerodynamic diameter, equating terminal settling velocity (V_{TS}) with respect to movement in fluids. Aerodynamic diameter uniquely assumes unit density ($\rho_0 = 1 \text{ g/mL}$), not true density (ρ_p).

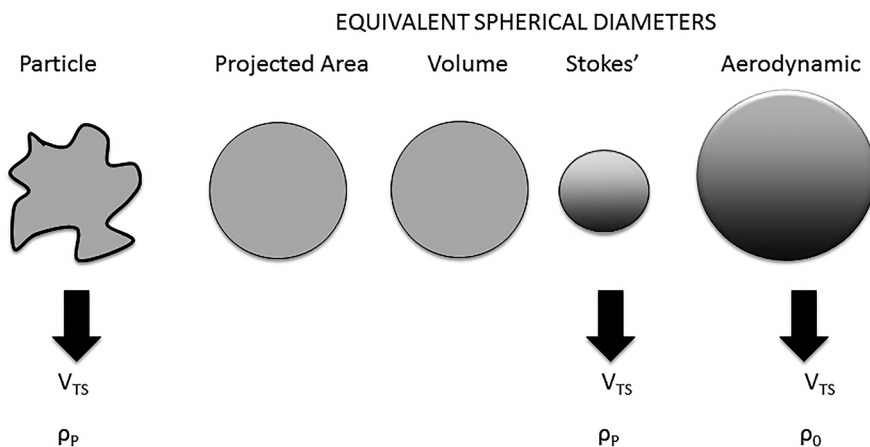


Fig. 5.1 Particle morphologies and objects depicting idealized dimensions

Many individual particle size descriptors have been defined and can be broadly separated into those observed optically (visual and laser imaging) and dynamically (behavioral) [4].

Optical Diameters

Projected Area

The projected area diameter, sometimes known as the Heywood diameter, is based on direct imaging of individual particles. The images are usually obtained by microscopy. A comparison is made between a circular disc and a projection of the particle in question. The human or automated inspection designates the equivalent spherical area of a disc with respect to the projection of the particle, and the diameter of the disc can then be assigned to the equivalent spherical diameter [4, 5].

Surface Area Diameter

The equivalent surface area of a sphere with that of a particle can be used to derive a diameter that reflects the surface area equivalency. The method of choice for determining the equivalent surface area diameter is gas adsorption [6]. Brunauer, Emmett, and Teller equation is used to establish the surface area, and assuming no porosity, particle size can be estimated from specific surface area [7]:

$$[p/V(p_0 - p)] = 1/(V_M b) + [(b - 1)/(V_M b)] \cdot (p/p_0) \quad (5.1)$$

where p is the pressure of the adsorbate, V is the amount of vapor adsorbed, V_M is the maximum amount of gas adsorbed to saturate the surface, monolayer coverage, b is a constant, and p_0 is the saturation vapor pressure. The specific surface area (specific area/unit mass of adsorbent, S_w) can be determined as follows:

$$S_w = [(A_m N)/(M/\rho)] \cdot V_M \quad (5.2)$$

where A_m is the area of single close-packed adsorbed gas (nitrogen) molecules, N is the Avogadro's number, and M/ρ is the molar volume of gas under ambient conditions (standard temperature and pressure).

Since the specific surface area is considered in terms of unit mass of adsorbent, if its true density is known, a particle size, volume surface diameter (d_{VS}) estimate can be made assuming the volume of particles that would give rise to the surface area/unit mass measured.

$$d_{VS} = 6/(\rho S_w) \quad (5.3)$$

Volume Diameter

The equivalent volume diameter equates the volume of a sphere with that of the particle. A variety of methods exist that allow an equivalent volume to be determined, including laser light diffraction, laser time of flight, laser Doppler velocimetry, and, specifically for particles in suspension, electrical sensing zone (Coulter) counting (see Chap. 6) [8].

Special Considerations: Density and Shape

It is important to recognize that observations of density and shape of particles may require considerations both with respect to equating to a spherical size and in terms of subsequent performance such as sedimentation, aerodynamic behavior, and dissolution. Visual imaging and gas adsorption will indicate the surface morphology and internal porosity of a particle.

Shape and density will contribute to surface area and buoyancy with respect to motion in fluid (gas or liquid) [9–11]. These properties will also contribute to the rate of dissolution in combination with intrinsic solubility [12, 13]. An individual particle composed of pure substance has the density of that substance based on its solid-state. A particle that is highly porous (large surface area with respect to a solid) or hollow (not significantly higher surface area than a solid) will both have low density

compared to a solid particle. This low density should not be confused with the low density that is observed in bulk or tapped powders (particle populations) where the interstitial space contributes to the low-density estimate [14].

Dynamic Diameters

Stokes' Diameter

Sedimentation

Some equivalent spherical diameters are based on the behavior of a sphere with respect to a particle under defined conditions. Instruments or methods that are used to measure these dimensions require calibration with spherical standards. Stokes' equation allows the diameter of a particle to be measured as a function of sedimentation rate in a fluid [15]:

$$\nu = dx/dt = [d_{St}^2(\rho_P - \rho_M)g]/18\eta_M \quad (5.4)$$

where ν is the sedimentation velocity (distance traveled with respect to time), d is the Stokes' diameter, ρ_P is the particle density, and ρ_M is the medium density, g is the acceleration due to gravity, and η_M is the medium viscosity.

Stokes' equation predicts the sedimentation rate of particles in a specific medium, most frequently liquid. This size parameter is based on the general principle that large particles sediment more quickly than small particles under the influence of gravity.

This expression has been extended to other situations in which artificial acceleration is imparted to the particle by the application of centrifugal force. Stokes' equation is then expressed as follows:

$$\nu = dx/dt = [d^2(\rho_P - \rho_M)\omega^2 x]/18\eta_M \quad (5.5)$$

The gravitational term is replaced with the angular velocity and the distance of the particle from the center of rotation or the angular acceleration [15].

Field Flow Fractionation

Field flow fractionation is a variation on sedimentation in that the differential movement of particles of various sizes in a laminar flow medium is employed to segregate them [12, 16]. This method can also employ a centrifugal force to allow collection in smaller particle size ranges. Particles are then collected at different locations along the conveying surface.

Aerodynamic Diameter

The aerodynamic diameter is also based on Stokes' law but for movement of particles in air (terminal settling velocity, ν_{TS}) requires consideration of the gas, air, as the fluid medium in which particles are moving [17, 18]. As a lower-viscosity medium, air offers less resistance to particle motion and consequently is of most significance to smaller, lower-mass particles. Stokes' equation can therefore be reduced as follows. In addition, unit density is assumed for the behavior of an equivalent sphere in air. The general form of the expression includes a term for the region below 1 μm where the particle does not experience the gas medium as a continuum and is said to "slip" between interaction with gas molecules, requiring a slip correction factor (C) and a shape factor (χ) for particles that deviate significantly from sphericity.

$$\nu_{TS} = (d_{ae}^2 \rho_0 g \cdot C_{dae}) / (18\chi\eta_M) \quad (5.6)$$

For spherical particles greater than 1 μm , this simplifies to

$$\nu_{TS} = (d_{ae}^2 \rho_0 g) / (18\eta_M) \quad (5.7)$$

Volume Diameter

Quasi-Elastic (Photon Correlation Spectroscopy, Dynamic) Light Scattering

This method is based on the random movement of nanoparticles subject to Brownian motion in liquids and is based on the application of the Stokes-Einstein equation [19]:

$$D = RT / (3\pi dN) \quad (5.8)$$

where D is the diffusion coefficient, R is the gas constant, T is the absolute temperature, d is the diameter of the particle subject to Brownian motion, and N is the Avogadro's number.

Laser Diffraction

Laser diffraction measurement can be based on two principles of optical imaging of particles: Fraunhofer diffraction and Mie scattering [20]. The light scattering

principles are well understood, but the expressions will not be presented here as each instrument employs an algorithm related to acquisition of data from ring array detectors and their correlation with particle sizes that is not routinely available to a researcher for the purposes of exploring fundamentals. It is sufficient to note that each instrument is calibrated with reticles to ensure that it represents the particle size distribution accurately [21].

Population Statistics

Continuing to develop the theme of adopting uniform terminology to refer to particle dimensions, it is important to recognize that finely divided solid materials do not exist as single objects in nature. All particles exist in large numbers as a population that we commonly refer to as a dry powder or liquid suspension in the simplest form [22]. Pharmaceutically particles can exist in more complex formulations such as pastes, but detailed consideration of a variety of dosage forms does not change the particle descriptor considerations.

Mathematical Distributions

For any particle population, an important practical observation is defining descriptors that capture as much information about size as possible [4, 23]. In its simplest form, this can be summarized as follows:

1. Powders consist of a distribution of particle sizes.
2. The distribution has a mode, where the majority of the particles exist in the same sizes, usually at the center of the distribution (central tendency).
3. The distribution can be more or less broad with respect to the central tendency, but the margins of the distribution are characterized by vanishingly small quantities of particles that ultimately define the outer limits of the distribution.

The practical origins of these observations are self-evident. Consider two major methods of particle manufacture, milling, or grinding, and spray drying. Both methods involve starting with a large solid or liquid bolus that is broken into smaller units as depicted in Fig. 5.2. Since the intent of the process is to reduce particle size, large numbers of small particles or droplets are produced. A simple way of considering this is in terms of an ideal situation. For example, if every 100 μm particle (or droplet) is broken evenly into 10 μm particles, this will result in 1 particle droplet producing 1000 smaller particles. Further, if every 10 μm particle is further reduced to a 1 μm particle, a further 1:1000 phenomenon will occur following a cubic function with respect to volume (and mass assuming uniform density). This simple illustration points to both the milling and spray-drying resulting in a skewed distribution to small particles as the process progresses, although it is apparent that

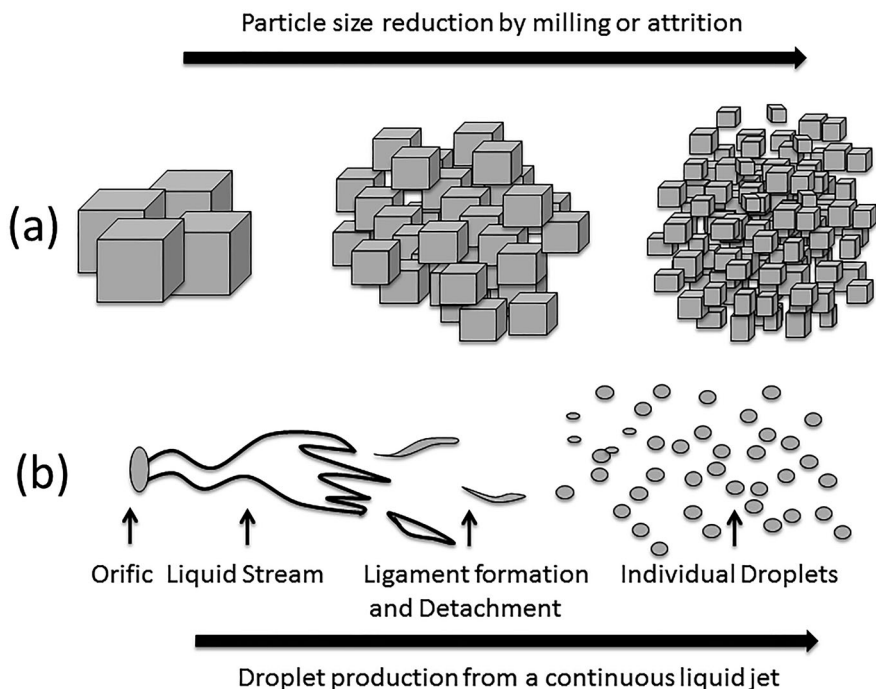


Fig. 5.2 Particle and droplet generation by (a) milling and (b) spraying

this progresses in far less uniform progression than described in the example. In reality, size reduction, by milling or spray-drying, does not result in monodisperse particle size distributions. Consequently, several power and exponential functions have been used to approximate experimental particle size distributions [23].

As shown in Fig. 5.3, it has been noted that the skew toward smaller particle sizes resulting from particle manufacturing processes (Fig. 5.3a) results in an approximately binomial distribution (bell-shaped curve, Fig. 5.3b) when the frequency of particle characteristics (number, volume, or mass of particles) is plotted against the logarithm of particle size [4, 24]. This is referred to as a lognormal distribution. This can further be transformed into a sigmoid cumulative proportion undersize (Fig. 5.3c) which when plotted on a probability scale (Fig. 5.3d) becomes linear. The advantage of being able to fit particle size measurement data to a lognormal distribution is that it can be summarized with two terms, the median diameter (central tendency) and geometric standard deviation (the width of the distribution). The median diameter conforms with the usual statistical definition of dividing the distribution in half with 50% above and below this particle size. The geometric standard deviation represents one standard deviation on either side of the median and captures 68% of the distribution according to the classical statistical definition of standard deviation ($\sqrt{\text{Variance}}$).

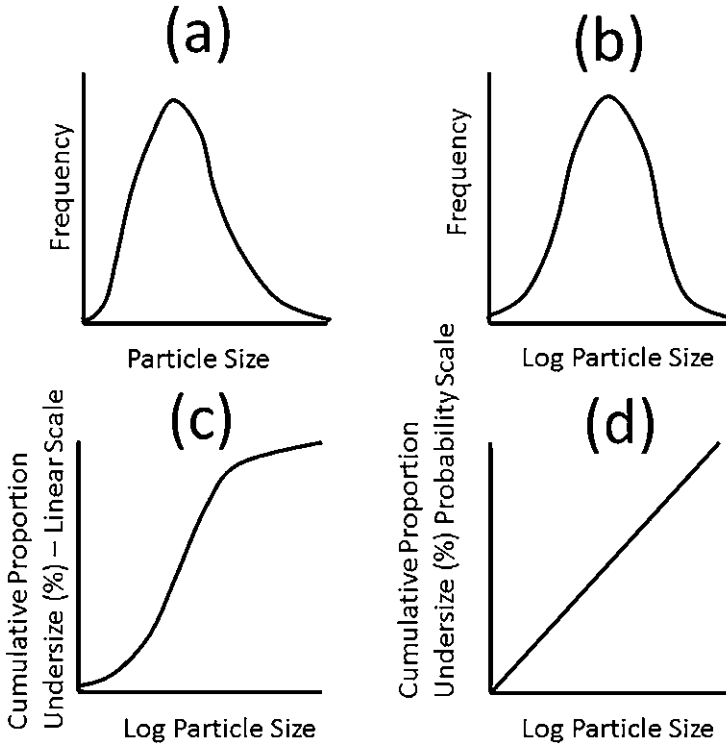


Fig. 5.3 Illustrates the lognormal particle size distribution depicted as (a) frequency (e.g., number, volume, mass) vs. particle size, skewed toward more small particles; (b) frequency vs. log of particle size to yield a normal distribution; (c) cumulative proportion undersize (CPU) vs. log of particle size; and (d) CPU on a probability scale vs. log of particle size

Data may be transformed from a plot where both axes (Fig. 5.3a, frequency and particle size) are linear to the final transformation (Fig. 5.3d) and where cumulative frequency and particle size are on probability (probit) and logarithm scales, respectively. By linearizing the plot in this manner, the geometric standard deviation can be obtained as follows [24]:

$$\sigma_g = d_{84}/d_{50} = d_{50}/d_{16} = [d_{84}/d_{16}]^{1/2} \quad (5.9)$$

where d_{84} , d_{50} , and d_{16} are the particle diameters at the 84th, 50th (median), and 16th percentile of the distribution and can be obtained from the linear graph (Fig. 5.3d). The final term in the expression is most popular as it uses the entire range of the data, which is particularly important if a small amount of data is available.

Basis for Expression

Once the descriptor of the individual particle dimension and the population particle description has been defined ultimately, the property that defines the diameter must be defined. These are related physical properties but do not result in the same numerical value for particle size distribution, and each has specific pharmaceutical importance.

Number/Count

Often the first method employed to describe particle size distributions involves visual imaging and counting the number of particles observed of specific sizes. This method requires that clusters of sizes (subranges) within the overall distribution are identified and the number of particles in each size range is counted. The data can then be presented as either frequency (number or cumulative number) or proportion (fraction or percentage) on the y-axis plotted against particle size (usually as a logarithm) on the x-axis.

Volume

Certain methods estimate the equivalent spherical diameter by volume. Examples would be laser diffraction which uses the volume as reflected by the radius of curvature diffracting laser light, where small particles give greater diffraction than large particles. Alternatively, methods such as the electrical sensing zone (Coulter counter) are based on the displacement of conducting medium in an orifice across which a potential difference is passed, thereby giving rise to resistance, which is proportional to the volume of the particle. The volume of each particle measured accumulates in size ranges designated by the orifice dimensions and calibration with spherical particles of known size. An important difference between these examples is that laser diffraction is an instantaneous (snapshot) measure and electrical sensing zone is a sequential population measure. Both methods will generate data that can be plotted as frequency (volume or cumulative volume) or proportion (fraction or percentage) on the y-axis.

Mass

If it can be assumed that all particles and droplets have uniform density, then in combination with the volume distributions, a conversion to mass distribution can be performed. However, there are cases where mass distributions are obtained directly, such as through sieving, elutriation, or cascade impaction. The instruments used to

Table 5.1 Hatch-Choate equations

Convert from	To	Relationship
d_{gc} , the geometric median diameter by count (count median diameter)	d_{gm}	$\text{Log } d_{gm} = \text{Log } d_{gc} + 6.908 \text{ Log}^2 \sigma_g$
d_{gc}	d_s	$\text{Log } d_s = \text{Log } d_{gc} + 2.303 \text{ Log}^2 \sigma_g$
d_{gc}	d_v	$\text{Log } d_v = \text{Log } d_{gc} + 3.454 \text{ Log}^2 \sigma_g$
d_{gm} , the geometric median diameter by weight (mass median diameter)	d_{gc}	$\text{Log } d_{gc} = \text{Log } d_{gm} - 6.908 \text{ Log}^2 \sigma_g$
d_{gm}	d_s	$\text{Log } d_s = \text{Log } d_{gm} - 4.605 \text{ Log}^2 \sigma_g$
d_{gm}	d_v	$\text{Log } d_v = \text{Log } d_{gm} - 3.454 \text{ Log}^2 \sigma_g$

d_s = surface median diameter, d_v = volume median diameter, σ_g = geometric standard deviation

sample particles from which mass distributions can be generated do so in size ranges defined by the characteristics of each of the stages of the sampler, usually of decreasing orifice size. The data generated can be plotted as frequency (mass or cumulative mass) or proportion (fraction or percentage).

Hatch-Choate Equations

Where a lognormal distribution can be demonstrated, as described above, it is known that regardless of the method of expressing the median diameter, the geometric standard deviation (σ_g) will remain the same. A series of equations have been developed, known as Hatch-Choate equations that allow the transformation between a variety of sizes, specifically count/number, surface, volume, and mass distributions [24]. Table 5.1 shows Hatch-Choate equations for the conversion between common median diameters based on various properties of particles (count, surface, volume, mass) and the known σ_g .

Special Considerations

The foregoing exposition illustrates the way in which uniformity of terminology has been achieved by the adoption of certain assumptions and mathematical fits to the data. As with any population phenomenon, behind the taxonomy lies the reality of the powder under scrutiny. In this light, it is always wise to review all assumptions that are being made regarding the particles themselves and the method employed to study them to ensure that the data obtained will be useful for its intended purpose.

One of the assumptions made earlier was that of the lognormality of the particle distribution. Much has been said about this in the literature. However, it is sufficient to indicate that if the data do not conform to lognormality, that is, do not fall on a

straight line on a log-probability plot, then it is important not to impose a lognormal interpretation on the data. The median diameter will remain relevant, but the geometric standard deviation will not be informative, as the particles will not.

Conclusion

Particles are finely divided solids, and to control their properties in any circumstances, their morphology and dimensions must be described. The complexity of this task can only be appreciated when the need for agreement on descriptive terms is understood. Many assumptions are made in defining terms, and these assumptions are the foundation from which expressions describing directly observed or indirectly measured particle sizes are determined. This section has reviewed some of the most common terms employed and the underlying principles from which particle dimensions are derived. Chapter 6 describes in more detail the instruments and methods employed to characterize particle size.

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Chapter 6

Particle Size Measurement



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Abstract There are numerous methods of characterizing the particle size distribution of a finely divided solid. Each of these methods is based on a different dimension or characteristic of individual particles and means of depicting the particle population. Each of the methods covers a different range of particle size and is based on a different principle of physics. Identifying a method that is relevant to the application and adopting complementary methods to confirm the observation give greater insight into the particle system being evaluated. Generally, these methods can be divided into direct observation, optical methods based on particle populations, and techniques based on particle motion in a medium.

Since monitoring and control of particle size distribution is an important aspect of pharmaceutical product quality and performance, instruments and methods are required to facilitate measurement of the relevant parameters.

The accepted taxonomy of descriptors for particle size and distribution was summarized in Chap. 5. It is clear that given the scale of the dimensions of individual particles employed in pharmaceutical dosage forms that unaided visualization would be impossible and that to examine populations of particles in powders or suspensions requires specialized instruments and methods. Since each method of describing the dimensions of a particle involves different assumptions and physical principles, each requires a unique approach. Moreover, each method has limitations to the types and size ranges that can be measured.

Dimensions

Each method of estimating particle size is based on different principles and is, therefore, limited to estimates of different size ranges, as illustrated in Fig. 6.1 [1]. The overlapping nature of the ranges covered by each method means that some methods can be used in a complementary fashion to allow the greatest information to be derived describing the particles.

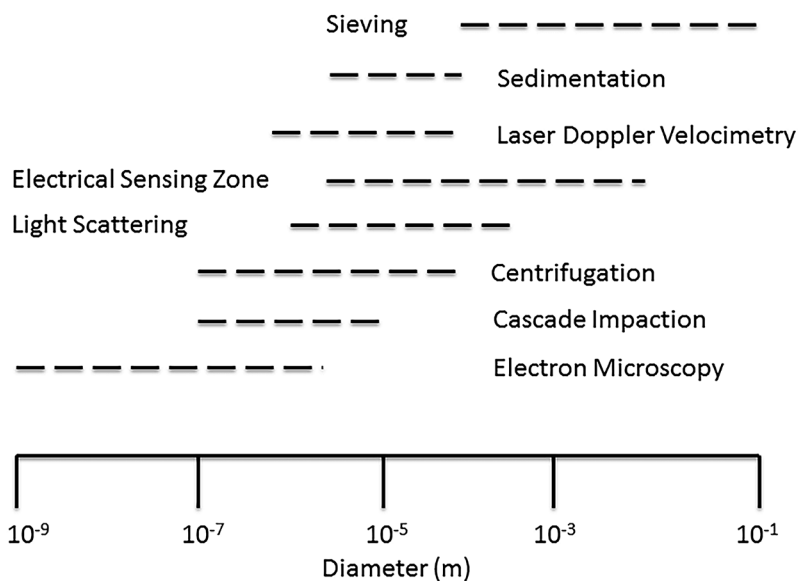


Fig. 6.1 Range of sizes that each method estimates depending on the principle of measurement

Microscopy

To visualize particles directly, microscopic methods are employed. Microscopy has the advantage of allowing inspection of the particle morphology but is limited by the number of particles that can be imaged at any specific magnification and rarely allows for quantities that can be regarded as statistically meaningful. Approaches to the numbers that should be counted from various fields of view differ. It has been suggested by one author that 300–600 particles should be imaged, which is a reasonable approach for most particle size distributions [2]. The ASTM recommends that there should be a minimum of ten particles in each of the tails of the distribution [3]. The ASTM method converges on the previous suggestion for many particle size distributions but for narrow distributions may require many more particles to be imaged.

There are a variety of microscopic methods that allow visualization of size and morphology of particles at different scales based on the basic principle of imaging employed. Some of these methods are presented below.

Light

Light microscopy is arguably the oldest method of imaging particles. The general principle of light obscuration by the object under scrutiny and magnification of the transmitted light is the means of visualization [4]. The limit to the size of particles that can be viewed is resolution dictated by the wavelength of light. As illustrated in Fig. 6.2a, the magnification is dictated by the ocular and objective lenses. The ocular lens is usually a fixed 10× magnification, and this is added to a range of objective lenses such that the combination of the ocular and objective lens will allow up to 1200× magnification. This may need finessing using oil immersion between the objective lens and the cover slip above the sample. The lower limit of discrimination is approximately 1 μm.

Electron Microscopy

For greater resolution and discrimination of surface detail on particles, electron microscopy can be employed. Electrons (from a hot cathode by thermionic emission) are directed to the surface of the sample and reflected (scanning) or pass around the sample (transmission) [5]. Electromagnets are employed to condense and deflect electrons to the surface and then to raster. The object is observed by detecting transmitted, secondary, or backscattered electrons. The limit to resolution by electron microscopy is the wavelength of an electron [6]. Scanning electron microscopy yields not only size data but also information about the surface rugosity and

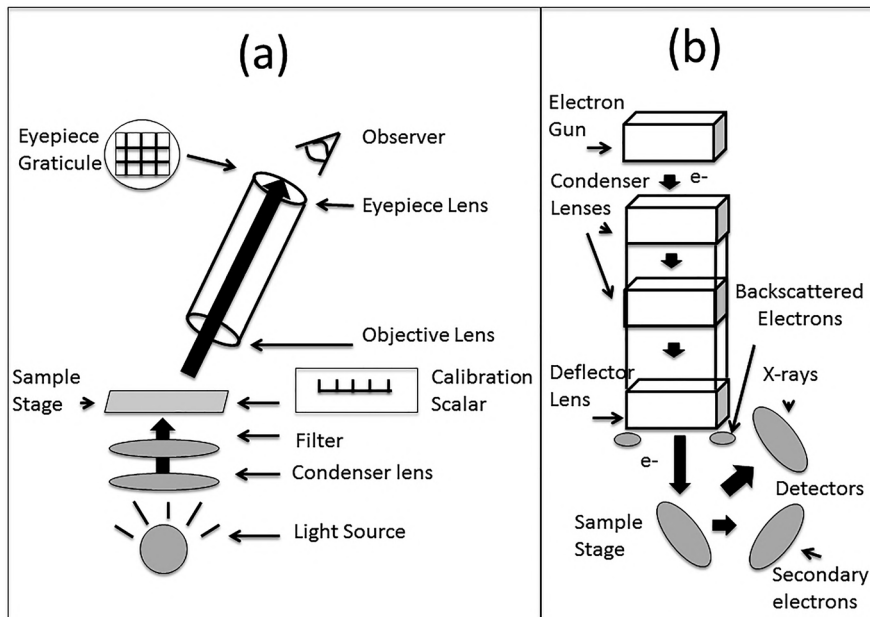


Fig. 6.2 Components of a (a) light and (b) scanning electron microscope SEM

structure. Transmission electron microscopy for electron-opaque materials is largely a sizing tool, although some shape information may be inferred from the particle outline. The limit to resolution of electron microscopy is sub-nanometer. Figure 6.2b illustrates the components of a scanning electron microscope. Electrons from an electron gun are condensed through a series of electromagnetic lenses before passing through a deflector lens that rasters (rapidly scans) the surface. Secondary electrons detected are passed through a series of electromagnetic lenses (similar in principle to the condenser lenses but not shown in the figure) to magnify the rastered image. On a different scale of scrutiny, the X-rays produced can be used in a technique known as X-ray microanalysis to identify the elemental composition of the surface of the particle.

Sample preparation is required for electron microscopy where a gold-palladium coating of the sample is achieved by sputter coating from a filament in a vacuum [5].

Other Microscopy Methods

Other microscopy methods, such as atomic force [7] and environment microscopy [8], may be used to examine particles, but particle sizing is not their primary application. Both methods find greater use in exploring the nature of particle surfaces.

Data Collection

Generally, several fields of view are imaged to allow for representative sampling, and a designated total number of particles are imaged as indicated above. When a human operator is involved in the imaging, then sequential measurements are made of individual particles that are allocated to narrow ranges of particle size. It is frequently the case that a $\sqrt{2}$ approach to intervals is employed, which has the effect of narrowing the range at small sizes where larger numbers of particles are found. Automated imaging methods require initial calibration of the instrument and can then allow for instantaneous measurement of all particles in a field of view. Light microscopes employ an eyepiece graticule calibrated with an objective scalar slide showing incremental dimensions. Electron micrographs show a scale bar, which can be applied by either a human or automated system as a reference from which to determine particle size.

Light Scattering

A variety of light scattering methods are employed to evaluate particle populations based on Rayleigh effects [9] and different principles of detecting light scattered by particles [10].

Laser Diffraction

Laser diffraction has been a popular commercially available method for evaluating fluid-suspended (gas and liquid) particles for over a half century [11]. The initial method was based on Fraunhofer diffraction, which was accurate to approximately 1 μm but was subject to some error even at this size under certain circumstances [12–15]. One notable example is the presence of propellant vapor in measuring the particle size of aerosols delivered from metered-dose inhalers. This was superseded by the application of Mie theory-based method, which did not give errors of this nature and was capable of submicron sizing.

The basic principle of laser diffraction is to employ a collimated laser (usually He-Ne) to illuminate particles passing through a viewing volume at which location light is diffracted through a larger angle by smaller particles than large particles due to their radius of curvature giving rise to light scattering in the direction of the laser. The scattered light is collected on a series of concentric detectors arrayed according to calibration with an in-line reticle prior to use. Manufacturers employ proprietary algorithms for transforming the raw data collected on the array into particle size data. The conditions of illumination, the geometry of the detector array, and the algorithm employed by each manufacturer vary. Consequently, it is inadvisable to compare

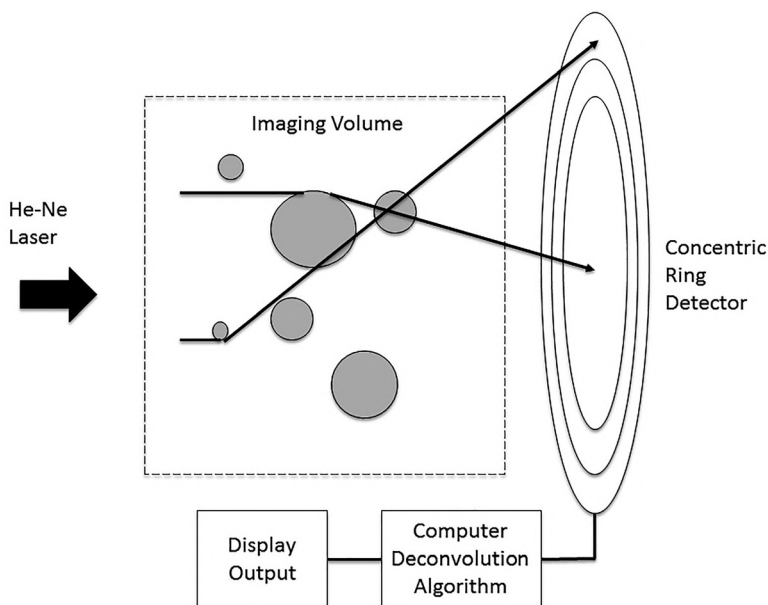


Fig. 6.3 Principle of laser diffraction particle size analysis. A collimated source laser (He-Ne) is passed through a viewing volume in which particles are suspended (in liquid or gas), the scattering properties of the particles are dependent on droplet/particle diameter, and the light is detected on full or partial concentric ring detectors. Each detector has been calibrated for a specific particle size. The signal is deconvoluted, and with the aid of a manufacturer-specific algorithm, the particle size distribution is displayed. Most systems can display by count and volume distribution

data from different instruments. Figure 6.3 illustrates components of a laser diffraction instrument.

Data collected from laser diffraction instruments is population-based and instantaneous. Consequently, it is statistically higher validity than microscopy data which rarely involves large numbers of particles. However, there is a drawback to this method in that it is imaging only a portion, the volume illuminated by the laser, of the sample, and to make the measurement sufficient, dilution is required to reduce obscuration to the level that individual particles can be observed. At high concentrations, resolution may be impeded, and particle overlap may be detected, leading to errors in inference of particle size distribution. Consequently, this method is frequently used to measure primary particle size.

Time-of-Flight Measurement

Laser time-of-flight, as the name suggests, employs a technique in which the time taken for a particle passing at a known linear velocity, controlled by the volumetric

flow and the cross-sectional area of the tube transporting the aerosol, between two lasers is detected [16]. This time is related to the aerodynamic properties of the particle as dictated by Stokes' law. Consequently, data can be collected in intervals according to time of passage, each of which corresponds to a specific aerodynamic particle size range. This method while allowing for large numbers of particles to be evaluated is, nevertheless, a sequential method in which the sample size is dependent on the collection time. As with the laser diffraction method, sample preparation is required to allow sufficiently dilute dispersion to allow individual particle imaging. At high concentrations, overlap in particle motion may result in errors in detection with respect to inferred particle size.

Laser Doppler Velocimetry

Laser Doppler velocimetry uses a phase shift in light to estimate particle size [17]. This method involves splitting a collimated laser beam and introducing the particle stream into one of the beams before bringing the beams together at which point interference occurs due to the phase shift which is proportional to the particle size. Laser Doppler velocimetry is used infrequently for routine particle size measurement as it is limited by the dimensions of the particle with respect to the wavelength of light.

Electrical Sensing Zone

The electrical sensing zone method was first commercialized as the Coulter counter [18]. The principle employs the passage of particles suspended in an electrolyte medium through an orifice across which a potential difference is applied. In passing through the orifice, the particle displaces electrolyte, resulting in a change in the potential difference that has been shown to be proportional to the volume of the particle through calibration. Different sizes of orifice can be used to ensure sufficient sensitivity based on the size range of interest. While this has most frequently been employed for microparticles, it has recently been extended to nanoparticle characterization [19].

Single-Particle Optical Sensing

The single-particle optical sensing (SPOS) method allows quantitative determination of particle populations in the 0.5–400 μm range, and it is commercialized as Accusizer particle sizing systems [20]. SPOS exploits hybrid detectors. Light extinction for particles $>1.5 \mu\text{m}$ and light scattering for smaller particles. Liquid



particle suspensions are pumped through a cell characterized by a region of uniform illumination produced by a laser diode. As a single micro-sized particle travels across the beam, it generates an extinction pulse proportional to its size, while the scattering produced by smaller particles is detected by a separate detector. Particle size distributions are then calculated employing a standard calibration curve. The flow and particle concentration are regulated to avoid particle clogging and clustering in the flow cell. Although being able to physically count particle populations and to measure sample mass, SPOS methods on liquid samples are limited by the need for particle suspension into a suitable medium (mainly water) and the dependency on the quality of the dispersion obtained. The presence of aggregates can indeed affect the measurement as much as low optically dense materials produce artifacts due to refraction phenomena. The technique rapidly loses sensitivity as the particle size falls below $1.5\ \mu\text{m}$ as the capability of light extinction drops fast with size. Below this threshold, light scattering detectors allow particle sensing down to about $0.5\ \mu\text{m}$. Other issues include the presence of air bubbles that can be easily mistaken for particles due to refractive index difference with the liquid medium.

Dynamic Methods

Several methods exist which require the sampling and size segregation of particles, which then allows a mass distribution to be reconstructed from the amount of material collected at each stage or in each location of the sampling system.

Sieving

Sieving has been employed throughout history for size segregation of particulates for a range of applications [21, 22]. The method has evolved to allow for sieving by conventional methods down to tens of microns by simply using interwoven meshes of different dimensions each of which corresponds with a particle size cutoff. By arranging these screens in sequence from large opening meshes to small, the powder can be separated into size ranges, and the mass deposited on each stage can be measured (gravimetrically or by chemical analytical methods). Sieving is used most frequently as a preparative manufacturing method to allow formulation using controlled sizes of particles to allow for performance to be guaranteed. As an example of the importance of limiting the size range, this also controls the surface area, which in turn controls dissolution rate and may ultimately control absorption for oral products. Table 6.1 shows the key dimensions of selected sieves.

The way that most sieves operate is to arrange the individual meshes in sequence vertically and then to electromechanically tap the stack to dilate the powder and allow movement from one screen to the next. Sufficient time must be allowed for the powder to pass to the final level of repose within the stack.



Table 6.1 Selected sieve dimensions

Sieve designation		Nominal sieve opening		
Standard (mm)	Mesh (#)	Inches	Mm	Microns
4.760	4	0.1870	4.760	47,600
1.000	18	0.0394	1.000	1000
0.500	35	0.0197	0.500	500
0.250	60	0.0098	0.250	250
0.125	120	0.0049	0.125	125
0.063	230	0.0025	0.063	63
0.037	400	0.0015	0.037	37

Adapted from Sigma-Aldrich (<https://www.sigmaaldrich.com/chemistry/stockroom-reagents/learning-center/technical-library/particle-size-conversion.html>)

The sample size that can be sieved efficiently is dictated by the screen surface area and the speed of transit. The latter may be dictated by the particle size distribution and degree of aggregation.

It should be noted that the sieve is an imperfect particle-sizing tool since the movement of the powder and the mechanical stimulation of flow can give rise to attrition in transit, which results in changes in the final particle size. In other words, the method of sampling may change the particle size distribution, which it is intended to measure.

Sedimentation

Sedimentation as a means of determining Stokes' diameter has a long history of use and is a particularly relevant method of particle sizing for suspension formulations. The general approach is to suspend particles in a known liquid medium, which is a non-solvent for the substance under investigation. Samples are then taken at a fixed height below the surface of the suspension, and based on Stokes' law, the size distribution can be plotted from the samples taken at a specific time reflecting the sedimentation rate of particles of known size and density.

Classically, the Andreasen pipet has been employed to determine Stokes' diameter where the height (h) through which particles in suspension fall at a given time (t) can be correlated with size [23]. Consequently, sampling at different times and analyzing the sample for the mass of material collected allow a mass distribution to be determined. Rearranging Stokes' equation gives the following expression from which sampling times at a specific height can be used to define particle size.

$$dst = (18\eta_M h) / [(\rho_P - \rho_M) \text{ g}t] \quad] \quad (6.1)$$



Sedimentation occurs under the influence of gravitational acceleration. Other methods can be employed that apply centrifugal acceleration or in the case of field flow fractionation acceleration due to flow gradients, which may be supplemented with centrifugal acceleration.

Aerodynamic

Sampling to estimate the aerodynamic diameters of airborne particles is a variation on the application of Stokes' law to accommodate movement of particles in a gas. The sampling method employed is known as inertial impaction, and since it involves passing the aerosol through a series of orifices of decreasing size, analogous to sieving, it is often referred to as cascade impaction. Several cascade impactor sampling instruments are available, but the most prominent are the Andersen IACFM eight-stage cascade impactor and the Next Generation Impactor (NGI) [24, 25]. They differ in three important respects; they have different (a) number of stages, (b) number of orifices at each stage, (c) nominal aerodynamic diameter cutoff at each stage, (d) physical dimensions and orientation of stages (vertical or horizontal), and (e) collection surface characteristics. Figure 6.4 depicts schematically the

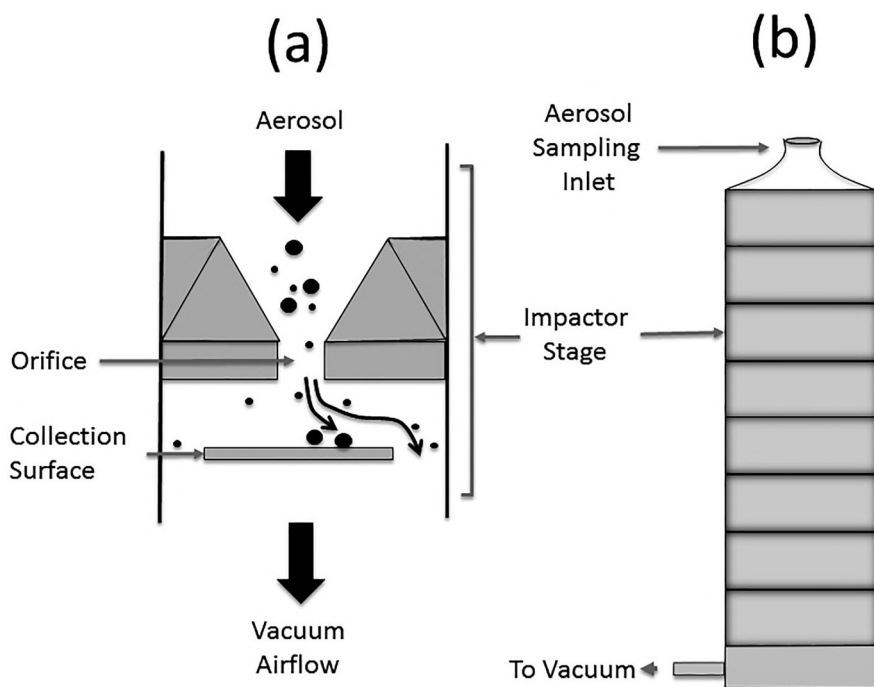


Fig. 6.4 (a) Impaction principle, (b) stacked, cascade impactor (idealized Andersen)



principle of inertial impaction and a stylized image of an Andersen impactor to illustrate the stages each of which in sequence has smaller orifice dimensions which result in increased linear velocity at a fixed volumetric flow rate resulting in the collection of smaller particles as the air progresses through the sampler.

Quasi-Elastic (or Photon Correlation Spectroscopy or Dynamic) Light Scattering

Quasi-elastic light scattering is based on monitoring the Brownian motion of particles in the nanometer to individual micron particle size range. Small particles will have large velocities and large particles will have small ones which translate into a range from large to small coefficients of diffusion. The principle of measurement is governed by Stokes-Einstein equation, as described in Chap. 5.

The method is used frequently for determining the particle size and distribution of nanoparticle or colloid suspensions. The limitation of the method is that the frequency term cannot be converted to meaningful data with respect to the mass or dose of the substance being studied. Consequently, great care must be taken to ensure that the distribution accurately reflects the entire distribution of particles in a system. This entails looking for larger particle sizes outside the range of this method to establish that nothing exists that could by its size (large particles having significantly more mass) completely outweigh the nanoparticle component of the distribution.

Product-Specific Particle Size Analysis Considerations

The descriptions of various particle size measurement techniques presented above might lead the novice to be confused as to which method to use for which specific applications. This topic is worthy of some discussion. Since the intent of this overview is to relate the topic to pharmaceutical products, each one should be taken on its merits.

Particle size based on surface area may be important if the key metric of the quality and performance of the dosage form in which the particles are presented relates to this property. Evidently, where dissolution or catalysis (enzymes or other proteins with specific activity) is important, then surface area and surface median diameter as obtained by gas adsorption would be potentially relevant.

Suspension stability is dependent on the ability of particles to remain in suspension or to have long sedimentation times, which is a function of particle size. The Stokes' diameter is an important parameter to measure with respect to this property, and its influence on stability will vary with respect to the medium in which the particle is suspended, whether aqueous (oral suspensions) or nonaqueous (metered-dose inhaler formulations).



Aerodynamic diameter is clearly relevant to the performance of aerosol products which would include aqueous nebulized droplets, nonaqueous propellant-based droplets delivered from metered-dose inhalers, or dry powders.

In reviewing these methods, it is clear there is overlap between them. Consequently, they can be used in a complementary fashion. For example, laser diffraction can be used to supplement any of the above methods to assess primary particle size distribution. In this regard, it should be noted that microscopy plays a very important role. There are serious limitations to using microscopy as a particle-sizing method due to the statistical limitations of the sample size. However, it has enormous value as a confirmatory method to establish that the data being obtained by other techniques is acceptable. This high value placed on microscopic imaging relates to the direct observation of samples compared with the indirect nature of inferring particle size of all other techniques.

Conclusion

In the previous section, the scientific principles on which particle size measurements are based were reviewed. For these principles to be valuable, they must translate to practical means of sampling and techniques of measuring the dimensions. Understanding the limits (particle size range) to the application of any of the methods described is a starting point for their selection in characterizing an unknown particle population. Further, the application must be considered since the particle size data will be used to establish quality and performance specifications for the product. Finally, no single method can be trusted to capture all information regarding particle size of relevance to the product. Each method should be used in a complementary fashion with others, and minimal microscopy should be conducted to confirm that indirect methods are accurately reflecting direct observations.

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Chapter 7

Fundamentals of Particle Interactions



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Abstract Particle interactions underpin the performance of pharmaceutical powders during processing and manufacture. The behavior of particles in bulk dictates their behavior. The forces of interaction between particles influence their flow and dispersion and the ease with which they can be suspended in any medium. These interactions are in the form of electrostatic and capillary forces that are reciprocally but not linearly related, mechanical interlocking arising from surface asperities, and bulk van der Waals forces. Variations in these interactions can lead to poor dose uniformity and biopharmaceutical properties.

Pharmaceutical inhalation aerosols are a unique example of the way interparticulate forces influence particle behavior and illustrate the considerations that need to be given to these forces.

Drug particles are prepared in a range of sizes, which are subject to a multitude of interparticulate forces. These forces represent a continuum on most scales. However, as seen elsewhere in nature, as particles decrease in size and transition from the micron to submicron size range, an equilibrium state is crossed between thermodynamic and gravitational forces. Flow and dispersion require shifting the equilibrium to allow particles to be released and delivered. Delivery might be from a hopper to a capsule or tablet die or on the inspiratory flow to the lungs of patients. The efficiency with which this can be achieved dictates the dose uniformity and delivery, which potentially impacts the prospects of safely and effectively treating the target disease. The interaction of particles in the respirable size range (1–5 μm [1]) is a complex function of all forces of interaction and will be used to illustrate their importance in the following exposition [2].

There have been many reviews of powder properties and particle interactions. The focus is often on preventing aggregation and on the dispersion forces required to overcome interparticulate interaction [3, 4]. The majority of pharmaceutical particles exist as powders for inclusion in a variety of dosage forms or as particles suspended in liquids. Aerosol particles are a special case as the ultimate intention is that particles are dispersed in air.

Fundamental Forces of Particle Interaction

Forces of interaction between particles must be considered in estimating the ease of dispersion. These forces can be listed as van der Waals, capillary, electrostatic, and mechanical interlocking [5]. A reductionist view of particle interaction focuses on the forces described in Table 7.1. These forces undoubtedly underpin the mechanisms of particle adhesion and cohesion that lead to particle aggregation. Studies of the effects of these forces on drug particles are remarkably complex. Besides the forces themselves, other variables such as particle size, size distribution [6], density [porosity/hollowness] [7, 8], shape [9], chemical composition, presence of impurities, and hygroscopicity [10] confound the experimental design and data interpretation [11].

Physically this is precisely the region in which gravitational forces give way to thermodynamic forces [12]. Where these forces are in balance, stable particle aggregates form. This should not be surprising as this is close to the size range in which the majority of living organisms are to be found, as the balance of forces can be manipulated to their advantage from an evolutionary perspective. Figure 7.1 illustrates the way this region is heavily populated by living things [13]. At first glance, it may appear that nature would indicate a larger dimension as more crucial than would be defined by particle interactions. However, it must be noted that specific phenotypic attributes of these organisms in the relevant size to particle interactions dictate their evolutionary success. This is best illustrated by publications on the van der Waals effects on the hairs on gecko's feet leading to their unique form of locomotion [14, 15]. Where locomotion, it should be remembered, is regarded as a

Table 7.1 Estimation of interparticulate forces for ideal binary systems

Force	Equation	Definitions	Parametric values
Van der Waals	$\frac{AD_{12}}{12h^2}$	A is the Hamaker constant; $D_{12} = d_1d_2/(d_1 + d_2)$, where d_1, d_2 are particle diameters; h is the shortest distance between the particles [1]	$A = 7.5 \times 10^{-20} \text{ J}$ $h = 5 \text{ nm}$
Electrostatic	$K_q \frac{q_1q_2}{h^2}$	q_1 and q_2 are the charges on the particles; h is the separation distance between the adhering particles; K_q electrostatic force constant [1]	$K_q = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$ $q_d = 5.3 \times 10^{-20} \text{ C}$ $q_1 = 1.18 \times 10^{-18} \text{ C}$ $h = 5 \text{ nm}$
Capillary	$2\pi\sigma D_{12}$	σ is surface tension; D_{12} same as above [1]	$\sigma = 73 \times 10^{-3} \text{ N/m}$

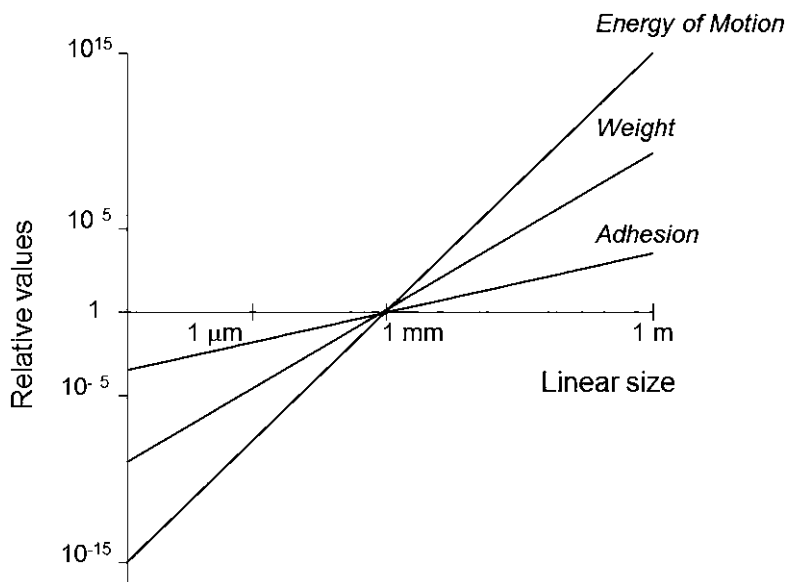


Fig. 7.1 The variation in energy of motion, strength of surface adhesion, and weight with mean size. The scale where the strengths coincide is densely populated by tiny organisms. (Modified from Ref. [13])

primary evolutionary advantage. Needless to say, these hairs are of the same order of magnitude dimensionally as the particles of importance in drug delivery.

The balance of fundamental forces in nature has implications for organisms. The scale where the surface forces, gravitational forces, and energy required for motion converge is densely populated by microorganisms. Living things behave differently on either side of the point of coincidence. Above this point, gravity has the greatest significance. Below adhesive forces overwhelm gravity. The intermolecular “stickiness” and balance with gravity offer a diversity of evolutionary possibilities for animation. For example, flies and lizards exploit adhesive forces to defy gravity in locomotion because surface forces are stronger than gravity over a small area. However, this balance presents a barrier to the separation of inanimate particulates of relevant dimensions. It should be noted that the relevant dimension is contrasted with that of anatomical features these organisms use to balance forces, e.g., hairs or setae on their appendages.

Nature of Aggregation

However, an alternative view is to consider that in the particle size range of interest, particles are subject to a balance of gravitational and thermodynamic force [12, 16]. Figure 7.2 illustrates the importance of the interactions between particles

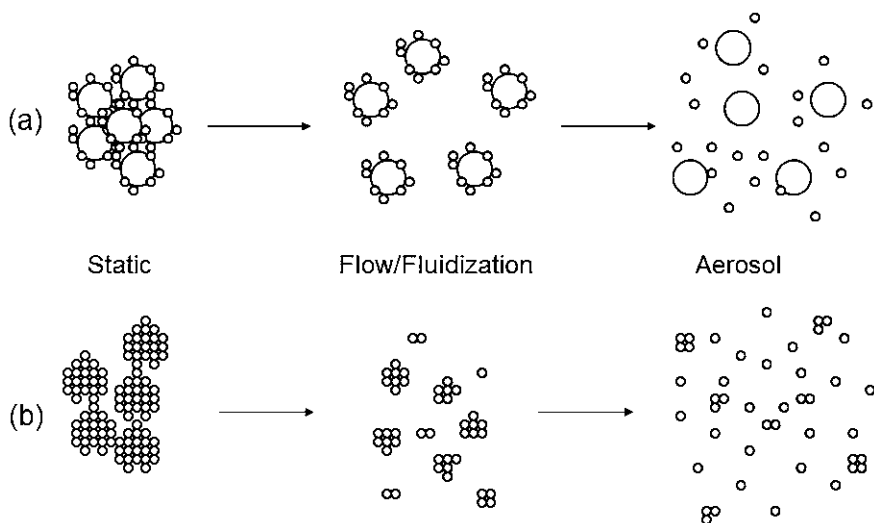


Fig. 7.2 Schematic of powder dispersion into aerosol from (a) carrier blend and (b) pure drug

in this crucial range using the example of dry powder frequently used to deliver drugs by inhalation that must be dispersed either from a blend or free drug. The point at which energy is imparted to the system is illustrated.

Small drug particle (1–5 μm) excipient carrier (~50–150 μm) blends and pure drug dispersions represent the extremes of dry powder formulation delivered as aerosols [17–19]. Tertiary blends include a third excipient component with a size between the extremes of the drug and large carrier particle [20]. Drug/carrier blends use excipient particles to assist in fluidization of the powder to facilitate aerosol dispersion. Pure drug, despite the primary small particle size, retains a large particle fluidization effect through intermediate unstable aggregate formation.

The energy input, duration, and magnitude with respect to the forces of interaction between particles are key elements in successful dispersion. While fluidization and aerosolization may not occur in sequence for the whole powder, this is a reasonable model for delivery. Fluidization and aerosolization may be viewed as two facets of a similar phenomenon according to the following rationale. Fluidization involves flow of large particles or aggregates in relatively small volumes of diluent air [12, 16]. The distances between particles in this process are sufficient to facilitate motion. In contrast, aerosols are discrete, small particles dispersed in a large volume of air. The distances between particles are sufficient to maintain particles in dispersion. However, this should not be taken to imply an equilibrium state with regard to dispersion. The major differences between fluidization and aerosolization relate to the size of particles employed and volume of dilution air required to achieve motion. The point of transition from fluidization to aerosolization appears to occur in the 1–10 μm size range. This is consistent with the previous observation that gravitational forces give way to thermodynamic forces in this range.

Particles are frequently suspended in liquids, and the stability of the suspension depends on the balance of attractive and repulsive forces. In an aqueous environment, the ionic nature of the medium and the surface of the particles dictate the distance between particles and their interaction for particles in the individual micron and submicron size range according to DLVO theory [21]. For particles larger than 10 μm , sedimentation rates dictated by Stokes' law are relatively rapid. Since most particle size distributions are polydisperse and particles settle at different rates based on their size, aggregation can occur as large particles impinge on small ones during sedimentation [22].

A variety of strategies can be adopted to stabilize suspensions. Using nanoparticles to form colloidal suspensions increases the probability of particles remaining in suspension [23]. Using counterions in the medium allows the charge at the interface to modulate the forces of attraction [24]. Surfactant can be used to sterically impede particles from approaching and to stabilize the suspension [25].

Conclusion

Several forces dictate particle interactions. Gravitational forces tend to dominate motion at sizes greater than 10 μm , and particles generally flow more easily at larger sizes. However, in the range of 1–10 μm , which is important for the use of micronized particles used in solid oral dosage forms and for aerosol products, the balance of gravitational and thermodynamic forces gives rise to variability in flow and dispersion that is usually modified by the application of energy to disrupt the equilibrium and facilitate motion.

Particles are also employed in liquid systems where their behavior is dominated by gravitational forces as predicted by Stokes' law and electrical or steric hindrance preventing particles from approaching each other and thereby maintaining their stability as individual components of the dosage form.

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Chapter 8

Powder and Particle-Dependent Traditional Manufacturing Processes (Unit Operations)



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Abstract The choice of suitable excipients is a base requirement in pharmaceutical manufacturing. For solid dosage forms, such excipients, as well as API, are powders that rarely show adequate properties for a direct employment in manufacturing. Therefore, powder processing is generally required. The long experience in powder processing has led to well-established techniques, such as milling, blending, drying, and filling, that however require continuous improvement to comply with the fast-growing automation and in-process control technologies. Moreover, standards and regulations have to evolve to keep pace with the technological progress. The future development of new manufacturing concepts will demand rethinking of traditional methods and particle engineering approaches.

Manufacturing of pharmaceutical products depends on the proper combination of accurately chosen ingredients. In the case of solid dosage forms, such ingredients, mainly consisting of powders, seldom show adequate properties in order to be used as such or to be transformed into the final product. Therefore, powder processing is generally required.

In this section, traditional industrial techniques employed for the preparation and processing of pharmaceutical powders will be briefly described with an emphasis on related issues and recent advances.

Drying

In the pharmaceutical industry, drying processes are performed to simply reduce or eliminate water to improve powder processing or to transform liquid preparations into powders or pellets. As the dry state grants higher stability, the latter application is also exploited to enhance stability of labile substances and for precise drug dosing. Therefore, the choice of the drying technique depends on the final product attributes, scope, and material properties. Water removal can be induced by several energy sources, from microwaves to infrared radiation, with the support of vacuum, convection, or reduced relative humidity to enhance the drying process. Drying processes can produce harmful effects due to temperature and pressure shocks. In this regard, technologies employing microwaves, supercritical fluids, and hybrid drying processes, such as microwave-assisted freeze-drying or acoustic energy-assisted drying, can be of great potential as they can allow ambient to sub-ambient temperature drying and very short processing times [1].

A list of drying methods applied to pharmaceuticals as well is reported in Table 8.1.

Out of the several available technologies, freeze-drying is one of the most popular for the production of dry powders and pellets. It relies on water sublimation occurring below the water triple point conditions of temperature and pressure [1]. The process consists in freezing the sample so as to obtain small ice crystals that are removed by applying high vacuum (Fig. 8.1).

Sublimation is promoted when pressure and temperature fall below the water triple point ($T < 0^{\circ}\text{C}$ and $P < 6$ mbar) and it is aided by a condenser that adsorbs the vapor generated, maintaining the process efficiently until complete removal of free and bound water. This technology is widespread in the pharmaceutical industry as it allows continuous processing of samples even under aseptic conditions. Freezing procedures need to be optimized to limit aggregation phenomena and reconstitution

Table 8.1 Summary table of various drying technologies with applications

Drying technique	Applications
Foam drying	Monoclonal antibodies, vaccines
Supercritical fluid drying	Proteins, metal composite, aerogel, polymers
Convective drying	Yeast, probiotics, proteins, antibodies, enzymes, platelets, fish, fruit
Vacuum drying	Probiotics, food, enzymes
Microwave drying	Food, enzymes, yeast, pharmaceuticals, biopharmaceuticals, polymers
Acoustic drying	Food, lumber, coal, waste treatment
Infrared drying	Food, polymer film
Osmotic drying	Food
Ohmic heating	Food, waste treatment
Spouted bed drying	Food, blood plasma, yeast, pharmaceutical powders, probiotics
Fluidized bed drying	Yeast, food, sludge waste, chemicals, microspheres, pharmaceutical tablets

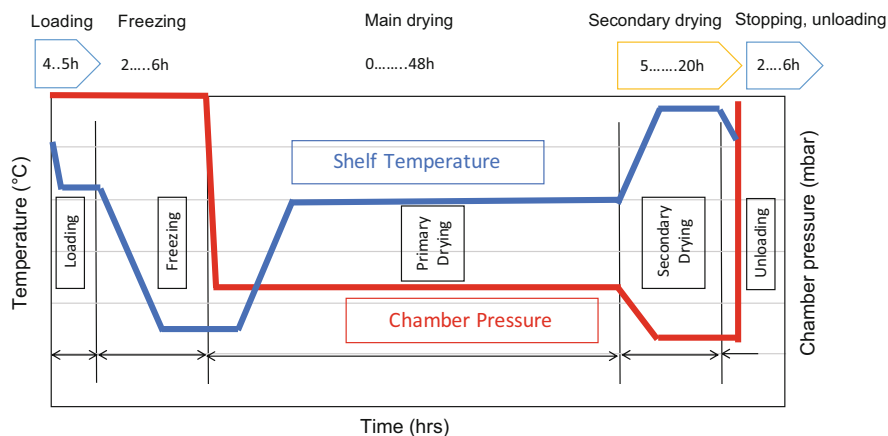


Fig. 8.1 Typical freeze-drying cycle. Primary or main drying occurs over the first 48 h to remove unbound water, followed by a secondary drying step to remove bound water. A programmed temperature ramp is performed on the sample to increase drying efficiency

of the resulting powder. Disadvantages include high costs and lengthy procedures; moreover, particular care in handling labile biotechnological products is required.

Foam drying may solve some of the issues typical of freeze-drying. In fact, although not industrially implemented yet, it can be conducted at near-ambient temperatures with reduced costs. The solution is boiled under reduced vapor pressure (foaming) followed by rapid evaporation. This method has potential in processing proteins and other sensitive substances.

Worth citing is fluidized bed drying, which is commonly employed in wet granulation of powders for the production of tablets, pellets, as well as coating [1]. As it will be discussed later in the chapter, modern fluid bed technology allows continuous processing and in-line quality control of the obtained granules (see the granulation section). The method is based on convective fluidization of powders by a continuous hot airstream and subsequent spraying of a binding liquid that aggregates particles in a controlled manner until the required granule size is achieved. The granules are thereby dried by the hot air jet and recovered or recycled back in the system if not matching target requirements.

Milling

Milling is employed in the pharmaceutical industry to grind coarse pharmaceutical powders. The process of grinding is useful to reduce particle size as well as to homogenize powder granulometry in order to improve blend homogeneity and dissolution as a result of an increase in surface area and homogeneity.

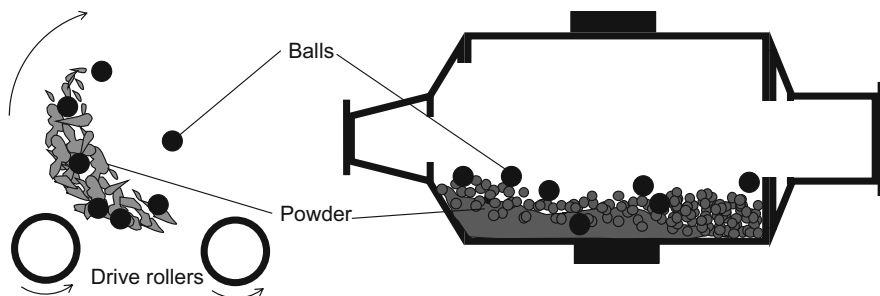


Fig. 8.2 Front (left) and lateral view (right) of a ball miller. Rotation induces collision of the balls that impacting the material provoke fragmentation and size reduction

A number of different grinding techniques have been developed for precise control of particle size, which has become a strict quality requirement for pharmaceutical products. The most employed exploit mechanical shear generated by different sources from impaction and compression to fluid energy [2].

Ball milling is a widespread example of such techniques using impaction and compression (Fig. 8.2). The equipment comprises a vessel or vial filled with balls, or rods, made from ceramic or other hard materials. The rotation or vibration of the vessel causes movement and collision of the balls or rods that impact the powder, which is then downsized as a result of the high mechanical energy. Another version of mechanical mill is the pin mill that comprises two disks, a stator disk and a rotor disk, fitted with overlapping pins. The powder is fed in the center and undergoes extensive shear due to the rotor high peripheral speed. The product is collected by a filter.

Other systems include fixed hammer mill based on rotating blades and roller mills in which sequential rotating wheels allow progressive size reduction. In both systems, the particles are ground by successive compression and impact on the grinding elements of the equipment and on other particles.

Cutting mills with high speed rotating blades and knives are instead employed for grinding fibrous and hard materials.

Another class of mills exploits fluid energy to induce particle collision. Air jet mills represent a family of mills that use air fluid jet to fragment particles into fine powders. Different equipment geometries exist that differ in the configuration, such as spiral jet pancake mill, oval ring mill, and fluidized bed air jet mill [2].

Beyond the kind of shear and mechanism employed for powder milling, a general requirement is the dissipation of excess energy that may rise from the grinding process. The high energy employed for grinding, especially if based on mechanical stress, can create locally considerable heat that can affect powder features, especially when heat-sensitive materials are processed. Therefore, mills are usually equipped with heat-conductive parts and dissipating systems, such as ventilation chambers and/or temperature controllers. In addition, humidity control is mandatory, especially in the presence of hygroscopic powders, as well as the need for control over possible triboelectrification due to particle attrition.

Blending

Blending is a fundamental process for the production of all pharmaceutical solid dosage forms. Accurate control of mixing of active principles with excipients is mandatory to obtain powders ensuring the required flowability, homogeneity, and stability. Nevertheless, blending efficiency depends on the properties of each ingredient. In fact, particle size, morphology, hygroscopicity, cohesiveness, and triboelectrification dramatically affect blending quality [3]. For this reason, powders are often milled and granulated before blending in order to grant higher particle homogeneity that helps prevent irregular mixing and ingredient segregation. Therefore, inhomogeneous and cohesive powders are usually pelletized to form larger but more regular and flowing particles. This ensures powder homogeneity and a regular distribution of the active principle in the blend, which is a fundamental quality attribute for solid dosage forms.

A large number of different blending units have been developed. Drum blender, cube blender, cone and double-cone blender, and V-blender are examples of equipment units currently available at laboratory and industrial scale (Fig. 8.3).

The different blender geometry changes the way powders flow and contact upon chamber revolution that is usually actuated at relatively low speed to allow proper ingredient mixing. Maintaining a low mechanical stress is usually fundamental to avoid excessive heating and particle alterations. Control of temperature and humidity is strongly advised even in this case to avoid particle aggregation and irregular mixing.

Blender geometry not only changes powder flow dynamics during mixing, but it is also critical to prevent possible accumulation of powder residues favored by the presence of blind spots and corners. These residues are a potential source of inhomogeneity and contamination, seriously hindering product quality. In this regard, beyond modulating blend properties, addition of proper excipients, such as lubricants and diluents, can help prevent such phenomena by improving flowability and reducing powder stickiness.

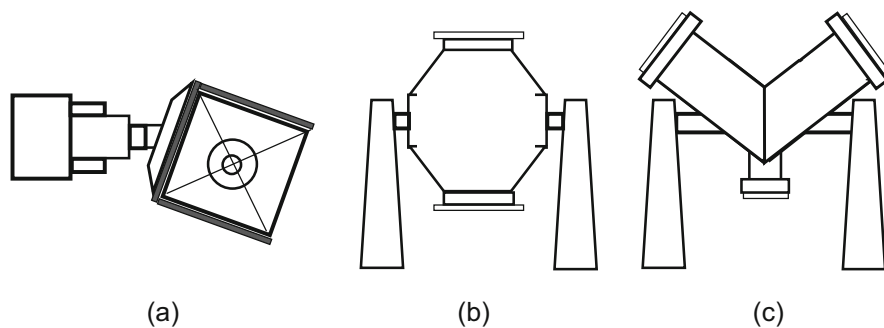


Fig. 8.3 Schematic representation of a (a) cube blender, (b) cone blender, and (c) V-blender

Alternative geometries, such as horizontal ribbon blenders, are also available as stand-alone unit operation or for continuous manufacturing in line with other equipment.

Granulation

Granulation is another fundamental process to tune powder properties as well as for solid product manufacturing. Typically, granulation involves agglomeration of fine particles into larger granules with sizes ranging between 0.2 and 4.0 mm [4]. Granule growing can be promoted by bridging, bonding, sintering, deposition, as well as crystallization [4]. By tuning granule size, it is possible to improve powder properties such as flowability and ensure blend homogeneity, owing to a reduction in particle cohesiveness and increased particle size uniformity. Granulation can be conducted in wet or dry conditions. Usually, dry granulation is chosen when substances are sensitive to moisture and heat and/or show poor fluidity, and it involves direct compaction of particles into larger grains. Wet granulation consists in the use of a liquid binder that promotes agglomeration through liquid bridging. Granulation is a critical step in tablet manufacturing and the production of pellets. Traditional and more advanced technologies support the granulation process and its employment in manufacturing (Fig. 8.4). Contrary to the conventional wet granulation techniques, advanced technologies have been designed to account for the specific properties of the materials being processed, thus preventing the potential detrimental effects on sensitive substances of traditional processes. Techniques such as moist or freeze wet granulation or foam granulation, employing very low amounts of water, allow mild agglomeration and drying conditions suitable for the sustainable processing of highly sensitive biomaterials.

As anticipated in the drying technology section, among well-established technologies, the fluidized bed granulation (FBG) is surely worth being cited. In its most advanced configuration, FBG allows even continuous wet granulation and automatic granule sizing and sorting (Fig. 8.5). In addition, FBG supports coating of pellets and granules for purposes from taste masking to enteric pellet manufacturing.

In line with the green deal revolution, FBG technologies have evolved towards green FBG setups. Such equipment combines sustainability principles with process efficiency. As above mentioned for a few wet advanced techniques, this goal can be achieved by limiting the amount of granulation liquid, thus reducing drying temperature and energy consumption, material and time waste, enhancing batch uniformity, and granting hydrophobic substance processing which limits the use of excipients [4].

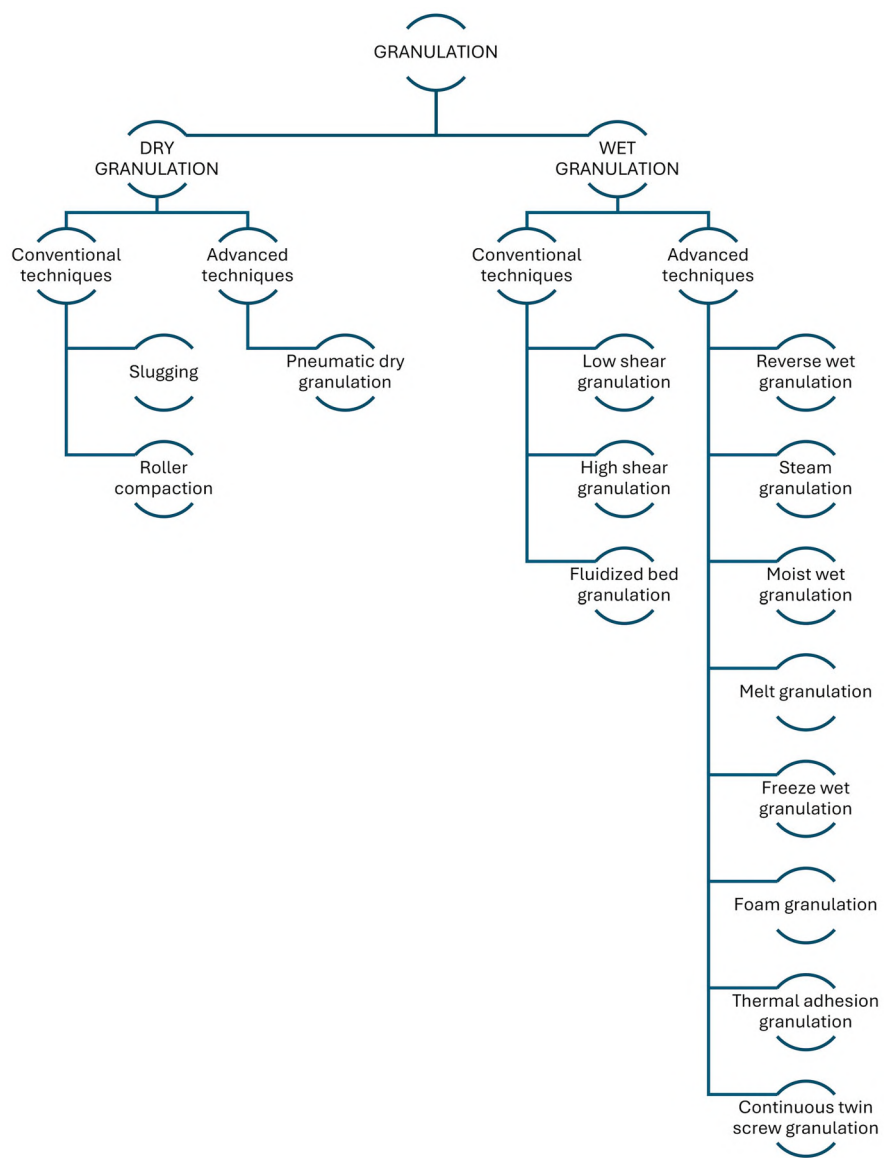


Fig. 8.4 Schematic representation of current granulation technologies

Filling

Primary packaging represents the step in which the pharmaceutical product is transferred into its final container that will provide protection throughout the entire product shelf life. This filling process needs to be finely tuned in order to grant

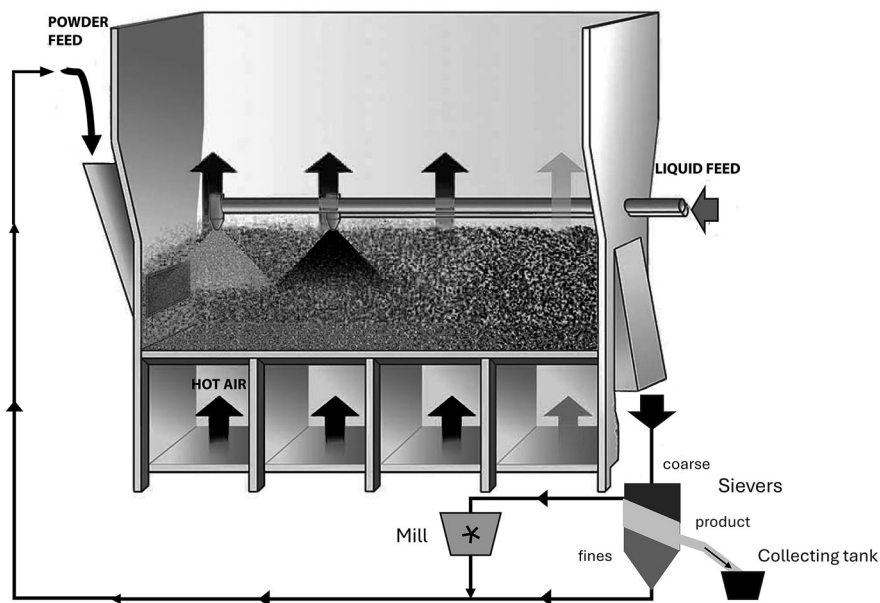


Fig. 8.5 Schematic representation of a continuous fluidized bed granulation technology. The system can be equipped with probes for real-time tracking of granule size growth. The nonconform granules are recycled into the machine by a double sieving system that sorts the final product based on size. Oversize granules are milled and delivered back with fines to the feeding system

consistent dosing of the product. Differently from liquids, the generally poor powder flow properties make the filling process particularly challenging.

Two filling methods are generally employed: gravimetric or volumetric filling. The first exploits a weight-based filling principle by which the product is exactly weighed and then dispensed into the container. This method is better suited for free-flowing powders. For poorly flowing powders, volumetric filling is the method of choice. It is based on forced loading in a calibrated chamber of powder products that are then pressure dumped into the container.

There is a large variety of sophisticated powder-filling systems of different geometries and sizes depending on the kind of product being processed. Horizontal and vertical geometries are available that exploit vibratory trays or belts or Auger dosing systems.

Filling systems can be customized according to dosing accuracy, powder properties, and container type. Examples of popular filling equipment are vertical Auger and cup volumetric fillers and target-weight and net-weight gravimetric fillers.

Most systems adopt an in-line weight-checking apparatus for continuous quality control of dosing.

In case of fine, poorly flowing, dusty products, Auger fillers or vacuum fillers can be selected.

The use of vacuum is advantageous for powder-independent loading and to prevent dust dispersion and contamination with considerably shorter times and higher efficiency. In these systems, powder is picked up by vacuum and compacted to a constant density within a cylindrical filling chamber and subsequently dispensed by low positive pressure air. The vacuum-driven powder sample compression to a uniform density grants high level of repeat dose accuracy.

Sampling

Product quality testing is mandatory to monitor powder processing performances to address and prevent property alterations upon handling, especially in the case of powder blends.

Anyhow, analytical results can be considered significant only if samples are representative of the source bulk material. This is a serious issue in the pharmaceutical industry considering the large batch size of the powdered solids being processed and the small size of the assayed samples that can be as small as a few milligrams.

There are issues and challenges involved in the sampling and characterization of pharmaceutical powder blends. Accurate and reliable characterization is hindered by both the complexity of powder blends and the lack of validated and reliable sampling technology and techniques [5]. In general, the larger the sample size, the smaller the error. However, sampling large amounts of material is difficult and inconvenient, and therefore statistical methods are employed in order to reduce errors due to small-size sampling. Naturally, the degree of sampling error is higher as powder inhomogeneity increases. For this reason, a number of 20 sample increments from the bulk material have been suggested [6].

A list of sampling devices is reported in Table 8.2 with advantages and disadvantages.

Sampling can be performed in static or dynamic fashion. Static sampling methods comprise scooping, thieving, cone, and quartering. In the first method, a plunging scoop is used. This method is prone to error as, since the sample is taken from the surface, it may not be representative of the bulk. Thieving consists of plunging a

Table 8.2 Comparison of sampling devices

Device	Advantages	Disadvantages
Cone and quartering	Good for powders with poor flow characteristics	Very operator-dependent
Scoop sampling	Reliable for homogeneous and - non-flowing powders	Particle segregation and non-flowing powders
Table sampling	Able to separate large quantities of material	Very dependent upon initial feed
Chute splitting	Can reduce powder sample	Operator bias by 50% in one pass
Spin riffing	Reliable for free-flowing powder samples	Not efficient at handling large samples of powder

capture device comprising one or more separate sample chambers into the bulk material to retrieve a number of small aliquots of the powder. Samples in this case are more representative of the bulk material. However, sampling can be affected by particle size dispersity, as probes can be preferentially filled by better-flowing coarse particles.

Cone and quartering is instead not recommended as it consists of pouring the powder as a cone-shaped heap onto a flat surface and manually fractioned into quarters, and it is therefore strongly operator-dependent.

In dynamic sampling, the sample is taken from a moving powder bulk. Powder movement grants more statistical sampling, although issues related to different flowing properties should be expected. Dynamic methods are table sampling, chute splitting, and spin riffing. The general principle is to allow powder fractionation through flow of the bulk material on inclined planes (table or chute) or mechanical vibrational motion (riffler) to obtain a steady flow almost independent of powder flow characteristics.

Dynamic techniques are considered superior to static sampling, albeit prone to issues due to the complexity of powder blends and the high variability in powder properties that can inevitably affect sampling performances. For such a reason, current quality control is implemented by using in-line continuous methods based on NIR, RAMAN, or light scattering spectroscopy probes installed in the equipment and able to constantly and consistently monitor process performances.

Stability

Stability of pharmaceutical products is of paramount importance and is a strictly required attribute to prevent nonconformity issues that can lead to a number of potential adverse effects on patients' health. Although several well-established methods have been developed for stability testing under the GMP and GLP¹ environment, problems arise from the necessity of long investigational times that affect product development pipelines.

In the case of solid forms, possible sources of instability are chemical but also physical changes, such as for polymorphic materials, that can affect solubility as well as flow powder properties. Tests are usually performed by means of environmental chambers and rooms able to precisely control temperature, humidity, and, if needed, exposure to light [7].

¹GMP = Good manufacturing practices are that part of quality assurance which ensures that products are consistently produced and controlled to the quality standards appropriate to their intended use and as required by the marketing authorization or product specification. Good manufacturing practices are concerned with both production and quality control; GLP = good laboratory practices is a quality system concerned with the organizational process and the conditions under which nonclinical health and environmental safety studies are planned, performed, monitored, recorded, archived, and reported.

The product is stored at recommended storage conditions and monitored for an established period of time. In general, a first-order kinetic degradation pattern (D) is assumed to have the form:

$$D = \alpha \cdot \exp(-\delta t)$$

where t is time and α is a kinetic constant. In this way, the time t required for the product to fall outside specifications is calculated. Testing time intervals should encompass the target shelf life. Moreover, the study has to continue for a period after the product falls outside specifications and at least three lots of material are suggested for testing to capture lot-to-lot variation [8].

Analytical assays have to be monitored for consistency and reliability over the entire period of the study. Specific guidelines establish methodology and scheduling for long-term stability studies with precise suggestions on the conditions to be employed and test methods.

As an example, four climatic areas have been defined representing the possible storage conditions at which products can be exposed worldwide.

In light of the long times needed for real-time stability evaluation, accelerated methods have been developed as well and are now strictly regulated.

Accelerated stability is performed at elevated stress conditions. Knowing the relationship between the acceleration factor and degradation rate, it is possible to predict the product's shelf life. In any case, such a prediction should be confirmed by real-time testing.

Even in this case, zero or first-order degradation kinetics are assumed. Temperature is the most common acceleration factor employed since its well-known Arrhenius relation with degradation rate:

$$K = A \cdot \exp - \frac{E_a}{RT}$$

where K is the degradation rate constant, A the frequency factor, E_a the activation energy, R the universal gas constant, and T the absolute temperature.

Selection of the acceleration conditions is crucial to not excessively stress the product with the result of changing its characteristics. Moreover, improper temperature selection may result in triggering degradation mechanisms different than those at the recommended storage temperature.

Humidity and pH can be used along with temperature to accelerate degradation, but modeling of multifactor degradation is very complex [9]. In this regard, a modified Arrhenius model that includes the humidity factor is available:

$$\ln k = \ln A - \frac{E_a}{R} \frac{1}{T} + B(RH)$$

where B is a coefficient that measures the sensitivity to relative humidity (RH) of the degradation process. This version of the Arrhenius equation, accounting for the substantial detrimental influence of humidity on the physical-chemical stability of many solid products, fills the gap of the conventional model.

Storage and Transport Considerations

Storage and transport are essential tasks for quality assurance of pharmaceutical products. Regulatory agencies pose extreme care on product quality assurance by defining standards for storage and distribution of pharmaceutical products in the form of specific good storage practices (GSP) and good distribution practices (GDP) guidelines.

Following the initial investigation of product shelf life, storage conditions have to be established early, as this step will influence the development of a suitable supply chain. The established storage conditions have to be ensured over all the product life cycle. Essential aspects include:

- Temperature and humidity, when required, control and monitoring; in this regard, an example of defined WHO standard storage conditions are listed in Table 8.3 [10]; to this purpose, appropriate storage facilities as well as transportation equipment are thereby mandatory.
- Ensuring packaging integrity, traceability, and proper tracking systems, along with proper documentation, is essential to grant safety and security during transportation.
- Properly trained personnel are also mandatory to ensure compliance with regulations and unforeseen scenarios.

Table 8.3 Recommended limits for descriptive storage conditions^a [10]

Label description	Recommended limits
Store at controlled room temperature	15 to 25 °C
Store in a cold or cool place	8 to 15 °C
Store in a refrigerator	5 ± 3 °C
Store in a freezer	−20 ± 5 °C
Store in deep freezer	Below −15 °C or −70 ± 10 °C
Store in a dry place	No more than 60% relative humidity
Protect from moisture	No more than 60% relative humidity
Store under ambient conditions	Store in dry, well-ventilated premises at temperatures of between 15–30 °C. Extraneous odors; other indications of contamination and intense light must be excluded
Protect from light	To be provided in light-resistant containers Light level not exceeding 500 lux
Chilled	Refrigerated

^aValues are based on pharmacopoeia limits and guidelines

- (d) Contingency plans for unexpected events are required to help mitigate risks during transportation.

To successfully fulfill the above requirements, an effective quality management system has to be built up. It is manufacturer's responsibility to select proper distributors and to develop suitable plans to mitigate problems during product transportation. Failing in this task could result in denial or a slowdown of marketing authorization application procedures.

Conclusion

Manufacturing of solid dosage forms presents challenges that, albeit being well-known, have not been completely disclosed and only partially solved. In light of the long time experience, powder processing techniques are robustly established and regulated in pharmaceutical manufacturing. Nevertheless, a continuous update of traditional technologies is required in order to keep pace with the progress of particle engineering and automation. Beyond technical limitations, the lack of reliable standards and accurate process control is the real bottleneck to the fast development of novel approaches, such as continuous manufacturing processes, that require rethinking of traditional manufacturing concepts.

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Chapter 9

Biopharmaceutical Processes



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Abstract The knowledge of a drug's biopharmaceutical profile is of paramount importance to design dosage forms able to overcome the countless drug delivery challenges. To this purpose, control over the way the drug is released from the pharmaceutical form is fundamental, along with the insightful understanding of all pathophysiological factors that strongly influence the drug's bioavailability and fate in the body. Although generally well addressed, many biopharmaceutical aspects remain challenging and require a continuous learning effort.

Biopharmaceutics examines the complex relationship between drug physical-chemical properties, dosage form, route of administration, and the rate and extent of drug absorption. The knowledge of drug properties and pharmacokinetics (PK), which describes the drug life cycle in the body, is of paramount importance to ensure the desired therapeutic effect. In this regard, a key role is played by the design of dosage forms able to grant easy administration methods and pharmacological blood levels. To this purpose, the way the drug is released from the pharmaceutical form can be fundamental.

This chapter will attempt to outline in a concise way the factors determining how drug dissolution can influence PK, and thereby therapeutic outcome will be discussed with a particular focus on drug bioavailability.

Bioavailability

The term “bioavailability” in a narrow sense accounts for the amount of drug reaching over time the site of action. Due to the clinical impossibility of evaluate drug concentration at the site of action, bioavailability is generally defined as a measure of the fraction (F) of an administered drug that reaches intact the blood-stream. Bioavailability is the main PK parameter, and it is calculated as the area under the blood concentration-time curve (AUC). This curve is representative of the PK processes of absorption, distribution, metabolism, and elimination. Therefore, the shape and dimension of the curve give information on the extent and contribution of each PK process to the overall drug profile in the organism.

Bioavailability is described by two parameters, namely, the fraction F that reaches the general circulation and the rate at which the drug is transferred [1]. F equals 1 for an intravascular (i.v.) administration. In any other case, F is decreased by the so-called first-pass effects. For an oral administration, typical examples are liberation from the pharmaceutical form, dissolution in the gastrointestinal fluid, escape metabolism by the intestinal flora, absorption through the intestinal wall, escape metabolism and efflux transporters in the gut wall and blood, liver metabolism, clearance by excretion, and partitioning in blood (protein binding) and tissues.

In this complex scenario, drug clearance (CL) has an important effect on F that therefore can be expressed as

$$F = \frac{CL \cdot AUC}{Dose} \quad (9.1)$$

The blood concentration-time profile of a drug, quantified by the AUC value, is characterized by two kinetic parameters, the maximum plasma concentration (C_{max}) and the time needed to reach C_{max} , t_{max} . The first depends on F and the complex network of PK processes of absorption, distribution, and elimination. Likewise, t_{max} is bound to the rates of absorption, distribution, and elimination (Fig. 9.1a).

After oral administration, bioavailability can be calculated as absolute or relative bioavailability [2]:

$$\text{Absolute bioavailability} = \frac{F_x}{F_{IV}} = \frac{AUC_x}{AUC_{IV}} \cdot \frac{Dose_{IV}}{Dose_x} \quad (9.2)$$

$$\text{Relative bioavailability} = \frac{F_x}{F_s} = \frac{AUC_x}{AUC_s} \cdot \frac{Dose_s}{Dose_x} \quad (9.3)$$

where AUC_x is the total area under the concentration-time curve of the fraction F_x after administration by an extravascular route x ; AUC_{iv} represents the same of the fraction F_{IV} after i.v. administration, which is by definition the 100%; and AUC_s is the total area under the curve resulting from a standard dosage form, which is either

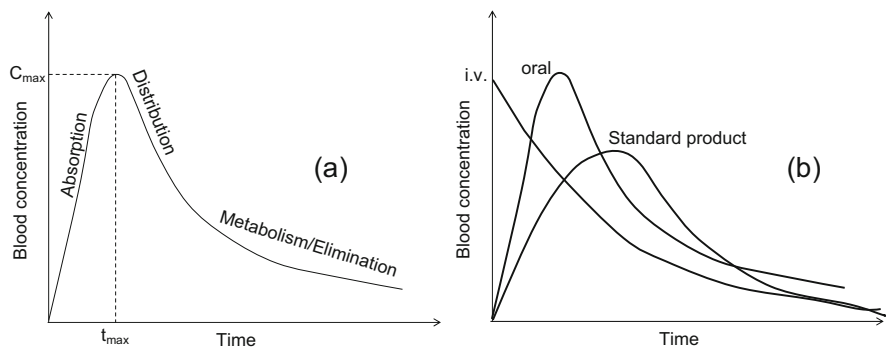


Fig. 9.1 Typical blood concentration versus time profile for an oral solid dosage form (a). PK parameters are depicted together with the different PK processes influencing the profile. (b) Comparison of profiles for i.v. injection, an oral dosage form, and a reference product. Relative and absolute bioavailability is calculated as ratios of the respective AUC values

Table 9.1 The biopharmaceutical drug classification system

BCS class	Solubility	Permeability	Oral dosage form approach
1	High	High	Simple solid oral dosage form
2	Low	High	Techniques to increase surface area like particle size reduction, solid solution, solid dispersion Solutions using solvents and/or surfactants
3	High	Low	Incorporate permeability enhancers, maximize local luminal concentration
4	Low	Low	Combine 2 and 3

an orally administered solution or a commercial preparation of proven clinical effectiveness (Fig. 9.1b).

It must be underlined that in this section, absorption has been approximated to a single process rather than a winding and complex path aimed at overcoming a mix of biological and metabolic barriers.

A drug is considered bioavailable when blood levels result in a pharmacological effect, the duration of which is proportional to the time such levels remain above a minimum effective concentration.

From the above picture, it is clear that absorption is the first and main kinetic process determining F and from which all the other PK processes are somehow influenced.

Therefore, the many parameters influencing drug absorption rate and extent need to be worked out to obtain a pharmacological effect. In this regard, drug solubility and permeability are fundamental to grant availability at the absorption site and a proper diffusion into the bloodstream. Based on such properties, drugs have been divided into four biopharmaceutical classes (Table 9.1). The biopharmaceutical class

system (BCS) is a useful predicting tool in order to choose the proper delivery strategy to grant therapeutic performance [3].

To be considered permeable, a drug should possess physical-chemical features so as to fulfill the Lipinski's rule of five [4, 5]. This rule was formulated for orally delivered drugs, and it can be fundamental in order to provide directions during the drug discovery phase. In this regard, a permeable drug should have a molecular weight < 500 Da, no more than 5 hydrogen bond donors, no more than 10 hydrogen bond acceptors, and an octanol-water partition coefficient ($\log P$) ≤ 5 . The many exceptions to this rule have suggested cautious interpretation of such properties and adaptation and extension of the rule has been attempted. Overall, it is clear that these drug properties can change remarkably in extent and rate of absorption and thus bioavailability. As a result, several approaches have been developed to improve solubility of low-soluble oral drugs. Micellar solubilization and particle micronization are two examples [6].

The efficacy of these approaches is obviously limited if drugs show intrinsic absorption issues and absorption is the rate-limiting PK process, as for BCS class II and IV drugs. Absorption can occur through passive diffusion or active transport mechanisms as well as mixed processes (Fig. 9.2).

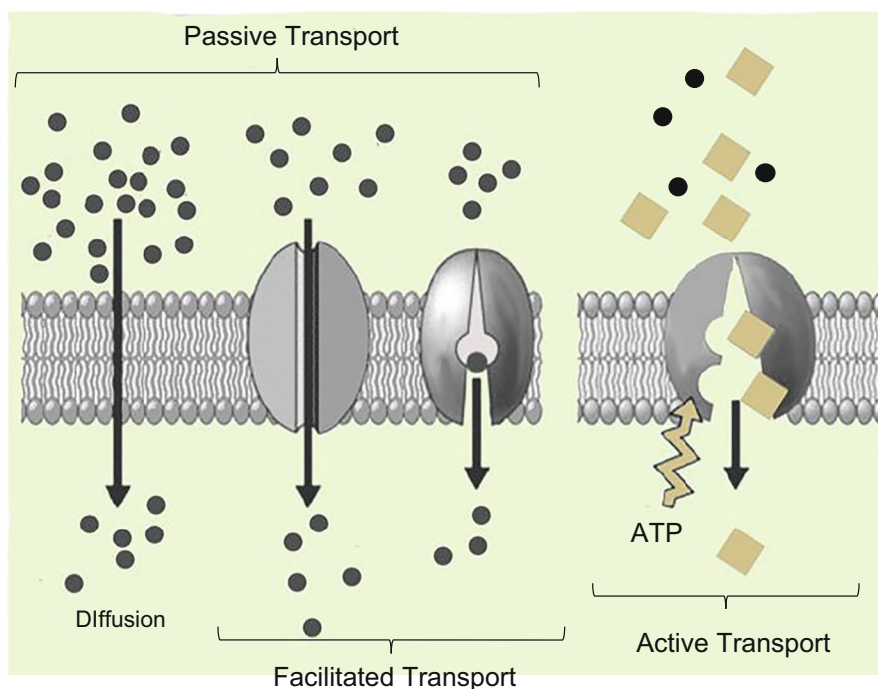


Fig. 9.2 Illustration of main drug transport mechanisms. Passive transport can occur through simple diffusion or facilitated processes by which drugs exploit membrane protein channels. Active transport requires energy and is a saturable process

Active transport is a specific saturable process and therefore has limited value for absorption enhancement. However, engineering of drug entities can be useful to exploit specific protein transporters in order to take advantage of active processes. This is the case of the prodrug approach by which the active drug is transformed into a modified inactive compound able, though, to bind such transporters and being internalized into cells and beyond the epithelial barriers [7]. Similar approaches can be useful to avoid Pgp-mediated efflux,¹ by means also of co-administration of Pgp inhibitors, as well as simply modulating the drug's lipophilicity. These absorption enhancement strategies have been found useful to improve bioavailability of a number of poorly absorbed oral drugs, such as BCS class II and, in particular, class IV drugs.

When absorption is not impaired by a drug's intrinsic properties, see BCS class I and III drugs, tuning drug release can result in effective improvement of bioavailability. In this regard, formulation of proper drug delivery systems and administration modalities can have a great impact. In fact, in these cases, drug dissolution and release from pharmaceutical formulations become a main factor for therapeutic efficacy. Moreover, it is well known that particle properties contribute to the disposition of drugs in terms of dissolution and diffusion. According to the Kelvin-Gibbs equation, the intrinsic solubility of a substance increases as the particle size becomes smaller, significantly so for particles below 1 μm [8].

Recently, the BCS has been extended to inhaled products [9]. Inhaled products present a unique challenge in that the lungs differ significantly from the GI tract. The BCS was developed to predict availability of drug to the systemic circulation following absorption since this step is required to achieve therapeutic effect. However, most drugs delivered to the lungs are intended for local action. Therefore, depletion of drug from the lungs is less important than the residence time during which the drug is available to achieve its therapeutic effect.

Dissolution

Drug dissolution is a necessary process to grant drug availability at the absorptive site. As emphasized above, if drugs show absorption issues, improving dissolution may not result in a direct increase of bioavailability. In this case, the use of absorption enhancement approaches is advisable. Nevertheless, for lipophilic and poorly soluble drugs, tuning dissolution rate is required to obtain proper blood concentration-time profiles [10]. Poor drug dissolution can be due to intrinsic drug solubility issues as well as the physical-chemical characteristics of the dosage form. In any case, such issues result in incomplete oral absorption and therefore scarce

¹Pgp stands for permeability glycoprotein, which is a membrane protein, one of the first members of the ATP-binding cassette transporter, which acts as a physiological barrier by extruding toxins and xenobiotics out of cells. It is widely expressed in the intestinal epithelium and throughout the body.

bioavailability. Many factors can affect drug dissolution rate [11], and they can be summarized in a simple manner in the Noyes-Whitney equation:

$$\frac{dm}{dt} = A \frac{D}{h} (C_s - C_b) \quad (9.4)$$

where dm/dt is the dissolution rate as drug mass change over time, A is the surface area available for dissolution, D is the diffusion coefficient of the drug, h is the thickness of the boundary diffusion layer adjacent to the dissolving drug surface, C_s is the saturation solubility of the drug in the diffusion layer, and C_b is the concentration of the drug in the bulk solution at time t . Additional derived mathematical models for oral drug dissolution rate are reported in literature [12].

Equation 9.3 clearly points out the direct proportionality between drug dissolution rate and the surface area of the drug particle. Particle surface area is known to increase exponentially as particle size is reduced. Therefore, processes like micronization or nanoparticle formulations can greatly enhance drug dissolution rate and thus the extent and rate of oral absorption.

In this regard, several particle technologies have been developed to formulate fine particles able to increase drug release rate and bioavailability (Table 9.2).

In this way, drug powders can be transformed into fine solid dispersions, such as nanocrystals, or formulated as nanosuspensions or polymeric micelles.

As discussed above, solubility enhancement has a direct effect on dissolution rate. Several approaches can be used to enhance solubility of lipophilic and poorly soluble drugs.

Table 9.2 Examples of particle technologies to improve the solubility of drugs

Particle technology	Method	Example drugs
Mechanical micronization	Jet milling	Cilostazol, ibuprofen
	Ball milling	Danazol, carbamazepine, dipyridamole, indomethacin
	High-pressure homogenization (HPH)	Prednisolone, carbamazepine, nifedipine
Particle size reduction by novel particle engineering	Cryogenic spraying process	Danazol, carbamazepine, glibenclamide, febantel, itraconazole
	Controlled crystallization	Fluticasone propionate, telmisartan, naproxen, GABA, gabapentin
	Spray drying, in situ salt formation, solidification with polymers	Nimodipine, flurbiprofen, dexibuprofen, docetaxel, curcumin, meloxicam, fenofibrate, ibuprofen
Complexation with cyclodextrins	Freeze-drying, vacuum evaporation, kneading	Celecoxib, clotrimazole, bifonazole, praziquantel
Polymeric micelles	Dialysis, freeze-drying	Paclitaxel, etoposide, docetaxel, amphotericin-B
Solid lipid nanoparticles	HPH, solid emulsification-evaporation/ diffusion	All trans retinoic acid, tretinoin

Being many drug molecules low-soluble weak acids or bases, the use of active pharmaceutical ingredients in a salt form is well established. The reason is that the potential energy of the solid state is changed by the presence of polar ionic compounds. This leads to higher solvation of the solid particle and a resulting higher apparent solubility. Therefore, the term C_s in Eq. 9.3 is increased due also to an associated buffering effect on the pH of the diffusion layer of thickness h at the surface of the dissolving solid [13]. As a result, these changes can significantly enhance the dissolution rates of the drug molecules in physiologically relevant timescales. The drug salt form is therefore a fundamental strategy to allow drug dissolution in physical-chemical adverse conditions as well.

Naturally, such a phenomenon is not sufficient to prevent issues due to drug polymorphism. Polymorphs of a drug substance are chemically identical entities that, due to the differences in their molecular packing, show different physical properties such as crystal geometry, density, melting temperature, hygroscopicity, as well as solubility. Although in some cases solubility among polymorphs is comparable, this phenomenon is known to often negatively influence dissolution rate in light of a reduced drug stability as well.

Amorphous compounds can be remarkably more soluble than crystal forms, although they are usually less stable. Due to a higher thermodynamic stability, amorphous drugs convert fast into crystals and, if any, into the most stable polymorph. This easily leads to a solubility reduction with ultimate clinical failure. Solubility loss could result even from drug transformations in the organism that imply a structural change into a low-soluble or insoluble form. Examples of this kind are atorvastatin, carbamazepine, and phenylbutazone [14].

Therefore, crystal engineering and proper formulation strategies are essential in order to prevent and control such issues.

Solubility enhancement can be achieved even through drug complexation with proper ligands such as cyclodextrins, bile acids, and polymers. Other examples are eutectic dispersions, solid solutions, and suspensions.

For oral dosage forms, such as tablets, dissolution occurs upon preliminary disintegration and deaggregation, whose rate therefore influences dissolution as well. For this reason, the use of proper disintegrants is fundamental to increase dissolution rate. Swelling agents improve wetting and solvation, drug diffusion, and particle separation resulting into disintegration of the tablet compact matrix. Useful materials are starch and cellulose derivatives and synthetic polymers, such as crospovidone [15].

In Vitro-In Vivo Correlation

Particle size and distribution play an important role in the efficacy and safety of powder- and suspension-based systems. The fact that key performance properties of dissolution and aerodynamic size distribution depend on the physico-chemical characteristics of particles is an important consideration when adopting a method

of manufacture and characterization. The desire to correlate in vivo performance with in vitro characteristics has been a goal of pharmaceutical scientists since the 1990s when the basis for a gastro-intestinal BCS was first presented [16]. The 1997 FDA Guidance for industry for oral, extended-release drug products remains still today the definitive source of regulatory thinking on in vitro-in vivo correlation (IVIVC) [17]. This guidance has subsequently been followed by a large literature describing methods to support the generation of data that allow IVIVC to be used for pharmaceutical development and approval procedures.

The following levels for IVIVC exist:

- (a) Level A is the most common type of IVIVC and generally used mainly for new drug and investigational new drug applications. Level A correlation is the most recommended by the FDA since it provides the highest degree of information. In fact, it establishes a point-to-point relationship between the in vitro dissolution rate and the in vivo input rate. IVIVC correlation is estimated by in vivo data deconvolution followed by comparison of the amount of drug absorbed to the drug dissolved.
- (b) Level B correlation uses the same data used in level A, but it compares the mean in vitro dissolution time of the drug to either the mean in vivo residence time or the mean in vivo dissolution time. Level B is the least useful for regulatory purposes because it does not reflect the actual in vivo plasma curves.
- (c) Level C correlation involves determining the relationship between in vivo PK parameters and in vitro dissolution data at a single point. It does not reflect the complete shape of the plasma concentration time curve, and therefore it is not sufficient for obtaining a bioequivalence waiver; however, it can be useful in the early stages of development.
- (d) Multiple level C correlation relates one or several PK parameters of interest to the amount of drug dissolved at several time points. It is rarely employed, since multiple level C often overlaps with level A correlation, which is then often preferred.

Common errors causing IVIVC failure are insufficient number and type of formulations for IVIVC development and validation, poor model building and selection, lack of knowledge about the reasons for low predictability, quality, and richness of input data, and insufficient parameter selection.

Conclusion

Liberation of drugs from the dosage form is a key process influencing its bioavailability and therapeutic performance. The drug dissolution process obeys to well-defined factors that need to be controlled in order to ensure the required effect. Several different approaches are today available to improve dissolution and, if required, to enhance drug absorption. Micronization techniques and novel formulation strategies, including nanodispersions, have been successfully employed to improve drug bioavailability. Therefore, proper formulation strategies and particle

engineering approaches are fundamental to manage the drug's biopharmaceutical issues and ensure clinical efficacy of dosage forms.

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Chapter 10

Recent and Evolving Manufacturing Processes



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Abstract Fast-evolving powder technologies are gaining new ground in pharmaceutical R&D thanks to emerging novel techniques enabling exceptional control over particle formation and tuning of powder properties. Alongside continuously upgrading techniques, such as spray-drying and supercritical fluids, novel cutting-edge technologies, namely 3D printing and particle replication in non-wetting templates (PRINT), are signaling the advent of a new era in the manufacturing of pharmaceutical dosage forms. This fast-moving progress will surely bring pharmaceutical R&D to unparalleled performances in the near future and shape new roadmaps toward personalized medicine.

Powder technology has seen considerable advances in recent years with the raising of potentially powerful novel techniques for the production of precisely designed pharmaceuticals.

These technologies are developed in order to provide exceptional control over particle formation and tuning of powder properties and, in some cases, they may open new perspectives in medicine personalization.

Among all, spray-drying, supercritical fluids, as well as controlled crystallization are well-established industrial processes that have evolved from traditional industrial applications with the contribution of technical innovations and are now facing cutting-edge challenges. On the other hand, 3D printing, electrospray, particle replication in non-wetting templates (PRINT) and, more recently, dry powder coating techniques represent powerful emerging technologies with a still largely unveiled potentiality for pharmaceutical product manufacturing.

This chapter will briefly address the most relevant aspects of these technologies, discussing the most advanced applications and manufacturing potentiality.

3D Printing Technologies

Additive manufacturing, rapid prototyping, or solid free-form technology are computer-aided technologies often referred to as 3D printing [1, 2]. A wide range of different techniques and equipment configurations exist that vary based on material used, deposition technology, as well as mechanism of layer formation. In the attempt to standardize such a multifaceted scenario, the American Society for Testing and Materials (ASTM) has classified 3D printing technologies into seven different categories: material extrusion, powder bed fusion, material jetting, binder jetting, vat polymerization, directed energy deposition, and sheet lamination (Table 10.1). Owing to their versatility and customizability, such technologies are likely to promote an evolution of pharmaceutical manufacturing in the near future.

3D Printing in Pharmaceutical Manufacturing

3D printers produce physical 3D models from objects created through computer-aided design or acquired with a 3D scanner. Unlike traditional methods of production involving subtractive or molding/casting manufacturing processes that generate high costs and wastes, 3D printing is conceived as an additive manufacturing process by which the product is built up in parts that are assembled into its final form, thus saving time and material and reducing costs. Other attractive features that apply to pharmaceutical manufacturing are the possibility of continuous process modalities that increase production capability, ability to precisely achieve high drug loading, and amenability to broad types of pharmaceutical active ingredients, peptides and proteins included [3, 4].

Several 3D printing techniques useful in manufacturing pharmaceutical dosage forms are available, including inkjet-based fabrication, direct write, binder jetting, thermal inkjet (TIJ) printing, and fused deposition modeling (FDM).

A general working principle shared by all such techniques is that the material is deposited in layers through melted extrusion or spray nozzles or modeled by high-precision laser technology. These processes are actuated and controlled through a

Table 10.1 ASTM classification of 3D printing technologies

3D printing technology	Description
Material extrusion	The material, typically molten or softened, is dispensed through a nozzle or orifice. The material is deposited onto a build platform, where it cools and solidifies. The build platform then moves down, and the process is repeated. This creates a layered structure of the printed object
Power bed fusion	A type of additive manufacturing process in which a laser beam or a binder fuses very fine powder layers together. The powders are spread and closely packed on a platform, and the laser beam or binder is used to melt or bind the powders together in a specific pattern
Material jetting	Material is jetted onto the build surface or platform, where it solidifies and the model is built layer by layer. Material is deposited from a nozzle which moves horizontally across the build platform. Machines vary in complexity and in their methods of controlling the deposition of material. The material layers are then cured or hardened using ultraviolet light
Binder jetting	The binder jetting process is a type of additive manufacturing that uses a binder to join powder particles together. The binder is selectively deposited onto the powder bed, and the process is repeated layer by layer until the object is complete
Vat polymerization	A type of additive manufacturing process that uses light to cure liquid photopolymer resin to create three-dimensional objects. The process is also known as stereolithography or photopolymerization
Directed energy deposition	A manufacturing process that uses a focused energy source, such as a laser or electron beam, to melt a feedstock material (powder or wire) and deposit it onto a substrate. The deposited material then fuses with the substrate and solidifies
Sheet lamination	This method utilizes sheets or paper, plastic, or even metal foil. The sheet of the material is placed over a build platform which is cut through an outline designed previously by laser

software able to finely tune the building up of the product parts with an, nowadays, exceptional resolution of details. In Table 10.2, 3D printing techniques are described together with their pharmaceutical application.

3D printing is gaining ground not only in the manufacturing of conventional dosage forms, such as tablets but even for more advanced drug delivery systems down to submicron size, implantable systems, as well as artificial organs [3].

Moreover, the high capacity of 3D printing in customizing product features according to needs opens novel perspectives in the development of personalized medicines.

After the approval by the FDA of levetiracetam (SPRITAM®) 3D printed tablets that signed the beginning of a new era in pharmaceutical manufacturing, as of today no further 3D-printed medicines have reached the market. In spite of the promise for a breakthrough in pharmaceutical production, yet regulatory and technical issues hold 3D printing back from its final establishment in pharmaceutical manufacturing [1, 2].

Table 10.2 Examples of the most popular 3D printing technologies in pharmaceutical drug delivery

3D printing technology	Description	Dosage forms
Inkjet printing	In the technique, different combinations of active ingredients and excipients (ink) are precisely sprayed in small droplets, via drug-on-demand or continuous jet method, in varying sizes, layer by layer, into a non-powder substrate. The technique encompasses powder-based 3D printing that uses a powder foundation (powder substrate) for the sprayed ink where it solidifies into a solid dosage form	Capsules Tablets Implants Nanosuspensions
Direct write	Uses a computer-controlled translational stage that moves a pattern-generating device in order to achieve, layer by layer, 3D microstructure	Poly(lactide-co-glycolide) (PLGA), poly(vinyl alcohol) (PVA) microparticles
Binder jetting	Provides a personalized dose in addition to the delivery of a high drug load with high disintegration and dissolution levels by manufacturing highly porous material	Tablets (ZipDose [®])
Thermal inkjet (TIJ) printing	TIJ system consists of a micro-resistor that heats a thin film of ink fluid (located in the ink reservoir), forming a vapor bubble that nucleates and expands to push the ink drop out of a nozzle. TIJ affords the opportunity of dispensing extemporaneous preparation/ solution of drug onto 3D scaffolds (drug carriers/films)	Solid dosage forms Implants
Fused deposition modeling (FDM)	The process can be applied to multiple dosage forms that apply polymers as part of the framework, such as implants, zero-order release tablets, multilayered tablets, and fast-dissolving devices. In the process, the polymer of interest is melted and extruded through a movable heated nozzle. The layer-by-layer ejection of the polymer is repeated along x-y-z stage, followed by solidification to create a shape previously defined by the computer-aided design models	Tablets Microparticles

Adapted from Ref. [3]

4D Printing

Relying on the same 3D printing techniques, 4D printing, introduced by Skylar Tibbitts at the Massachusetts Institute of Technology, refers to the addition of a fourth dimension, time, to the three spatial coordinates [5]. 4D printing is based on the evidence that a 3D-printed object can evolve over time in response to external

stimuli such as light, water, heat, pH, magnetic fields, or electricity. Such materials, namely smart materials or metamaterials, are capable of changing their shape or morphology. Essential to such processes is the capability of tailoring material properties to respond to a given stimulus. Needless to say, this technology can have great impact on drug delivery where smart materials can help to model drug release and therapeutic outcome. Examples are pH-sensitive hydrogels that can be printed into many shapes and sizes and can behave according to altered physiological conditions, such as a pH change.

Electrospray Technology and Additive Manufacturing

The electrospraying process consists in the generation of a liquid jet from a capillary nozzle thanks to the imposition of an electrical potential between the nozzle and a metal plate or other support that serves as a counter electrode. The relative electrical shear stress deforms the liquid flowing from the nozzle and determines its ejection and dispersion into fine droplets. Droplet generation occurs as the electrical force exceeds the Rayleigh limit, defined as the charge magnitude on the liquid at which the electrostatic force equals the surface tension [6]:

$$Qr = 2\pi (16\sigma_l\epsilon_0r^3)^{1/2} \quad (10.1)$$

where ϵ_0 is the electrical permittivity of the free space, σ_l the surface tension of the liquid, and r the droplet radius.

The voltage applied to the nozzle is usually from 1 to 20 kV and the liquid is sprayed at a constant flow rate, typically between 0.1 and 10 mL/h. Droplets as small as 1 μm or lower can be easily obtained. To obtain particles, it is necessary to work with low polymer concentrations. Electrospray has the great advantage of being a one-step process useful to fabricate particles with specific size and morphology by tuning voltage intensity, solution flow rate, drying time and rate, and solvent vapor pressure [7]. In this regard, the use of concentric coaxial nozzle conveying two or more different fluids can be useful to form multicomponent structures. In fact, particular nozzle geometries can allow the construction of core-shell or more complex structures [8]. The typical, very high drug encapsulations and the low stress applied make coaxial nozzle configurations particularly indicated for the encapsulation of peptides and proteins.

Beyond microparticle production and surface coating applications, 2D/3D printing technologies also benefit from the advantages of the electrospray technology. An electrospray apparatus can provide a unique capability of handling a wide range of materials in the few micro- and nanometers range. This led to the birth of a printing technology capable of printing true three-dimensional, self-supporting, overhanging architectures impossible to achieve by classical 3D printing technologies. Such capabilities have enabled the application of electrospray technology to the handling

of a wide range of biomolecules and living cells to construct bioplatforms for the development of tissues or 3D biological models for biomedical and clinical applications [9].

Controlled Crystallization

Crystallization processes are typically employed for post-synthesis purification and recovery of API and other compounds of interest. However, precise control over the process can allow the formation of particles with desired features. Technology has advanced considerably in this field, and today it is possible to finely control crystal formation and growth to meet specific requirements in terms of crystal form and powder properties. Advanced crossflow (AXF™) mixing is an example of recent technology able to solve the problem of micro- and nanospheres, emulsions, and crystals production with a very narrow size distribution with a robust, cost-effective manufacturing-scale technology. The obvious advantages of continuous processes over batch production have led to the development of fluidic technologies that allow continuous crystallization and control [10, 11]. The most common continuous systems in the pharmaceutical industry are mixed suspension mixed product removal crystallizers and tubular or plug flow crystallizers (Fig. 10.1), such as laminar-flow tubular crystallizers, coiled flow inverter crystallizers, segmented/slug flow crystallizers, and continuous oscillatory baffled crystallizers. Flow crystallizers can allow enhanced supersaturation conditions that improve crystal yield and features.

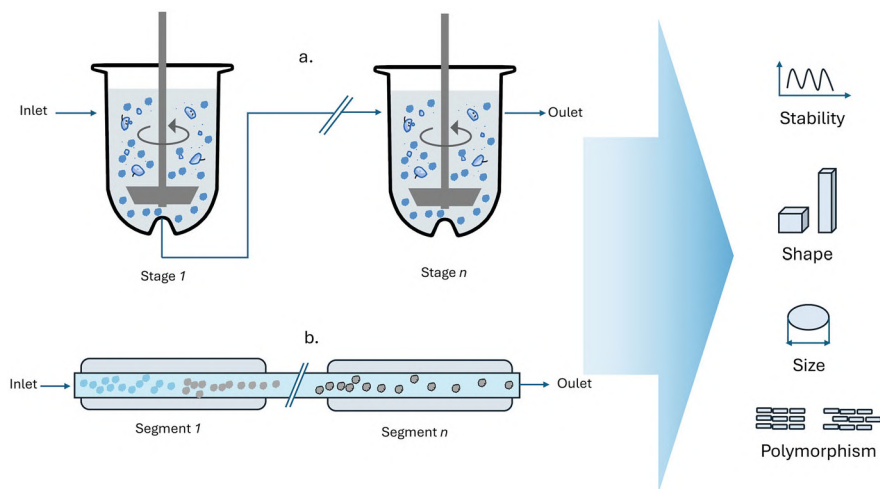


Fig. 10.1 Schematic representation of **a.** mixed suspension mixed product and **b.** tubular or plug flow crystallizers. The combination of such technologies with in-line probes, such as focused beam reflectance analyzer, enables fine control over shape, size, polymorphs, as well as yield of the process

Moreover, coupling of in-line probes, such as a focused beam reflectance measurement¹ and FTIR, with continuous flow crystallizers is possible to monitor supersaturation, polymorphs, crystal number, and size. The precise control over crystal particle control can be employed to improve API solid-state properties, tremendously impacting downstream solid pharmaceutical product manufacturing processes.

Isothermal Dry Powder Coating

Isothermal dry powder coating (iDPC) is a novel concept of powder processing aiming at increasing the active pharmaceutical ingredient (API) content in pharmaceutical preparations [12]. First developed at Aston University, the United Kingdom, the process is based on the direct coating of excipients onto the surface of the API particles at ambient temperature. It produces highly uniform mixtures without affecting physical, chemical, or biological properties. Owing to the mild process conditions, it is applicable to a wide range of particle types and molecules, including biologics, and can be used to coat smaller particles onto larger particles, provided there is a size difference of two to three times. The apparatus consists of a rotating chamber in which the particles to be coated (host) are mixed with the coating particles (guest) in a given proportion. The process is divided into three steps (Fig. 10.2):

1. De-agglomeration: The loaded host and guest particles undergo fast rotation speed that forces their distribution in a thin layer on the chamber wall. The centrifugal force is sufficient to cause de-agglomeration of fine or coarse particles.
2. Dispersion: The particle thin layer is disrupted by the injection of a nitrogen gas. Micro-pockets of fluidized powder are formed in which guest particles are dispersed throughout the host particles.
3. Coating: The fluidized guest particles spread and adhere to the surface of the host particles within the micro-pockets.

Compared to conventional blending/coating technologies, iDPC is very convenient and relatively safe for most materials. In fact, it avoids high-energy solid-solid interactions, produces only mechanical interaction with the particles, and no dead spots. Therefore, iDPC is a low-shear and sustainable process with no frictional

¹Focused beam reflectance measurement (FBRM) operates by a monochromatic laser beam at 790 nm directed to a lens mounted eccentrically on an optical assembly that rotates at high speed. The laser is focused on a spot fixed at the probe window. When the beam swipes across a particle surface, the light is backscattered into a detection probe. Each reflection's time is then multiplied by the velocity of the scanning beam, then giving a chord length measurement, and then a chord length distribution (CLD). No sample dilution is required and CLD is sensitive to particle size, number, and shape changes also in a dynamic way, making FBRM useful in process dynamics studies and real-time process control.

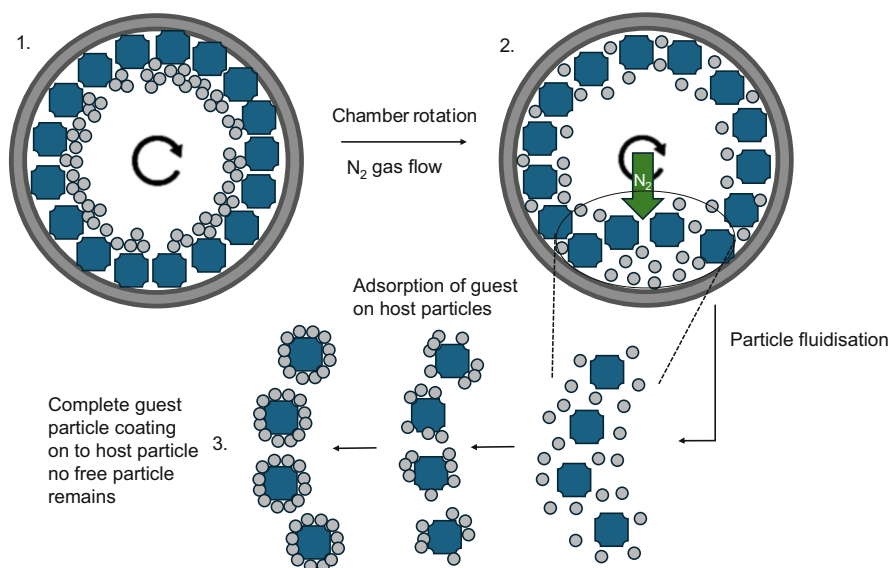


Fig. 10.2 Schematic representation of the iDPC process steps and the particle coating mechanism

heating and reduced charge separation, as the powders are held stationary during most of the process, and they are disrupted only by a gentle stream of dry nitrogen. Moreover, the process is performed at ambient temperature with no temperature control or solvent use.

Spray-Drying Technology

Historically, spray-drying is an old process whose conception goes back to the end of the nineteenth century and that has progressively gained ground in different areas, including the pharmaceutical industry. The principle is relatively simple and well-known as it combines the two processes of atomization and drying occurring nearly at the same time. Classically, atomization consists of the forced dispersion of homogeneous and heterogeneous liquid phases, generally solutions or dispersions, into fine droplets by means of a pressurized hot gas, usually air. In light of their small size, the droplets dry in a very short time (fractions of seconds), leading to the formation of solid particles. Over the years, the spray-drying technology has evolved, introducing novel principles and methodologies in droplet formation and drying that allowed considerable expansion of spray-drying applications. The classical equipment is designed as two chambers, one drying chamber, where the feed is atomized through a nozzle and dried by the pressurized hot gas, and a cyclone separator chamber in which the formed particles are recovered. The separator method usually relies on the generation of a cyclonic flux due to the combined effect

of the pressurized gas flow, aspirator pump, and the peculiar chamber geometry that favors abatement of the particles into a vessel terminally connected in the lower part. Small airborne particles that do not effectively deposit are driven away by the exhausted gas together with the vapor and trapped into an outlet filter and a condenser. This spray-drying configuration ensures a completely closed environment with the possibility to recycle exhausted gas and vapors and prevents risk of room and/or personnel contamination.

Spray-drying is not simply a drying process but rather a fast, one-step production method for dry powders for different purposes. This technique combines a relatively user-friendly setup with versatility and scalability, which are fundamental industrial requirements. It allows not only fast transformation of liquid products into powders but also control over particle size and morphology [13]. Nevertheless, to achieve such a control, a profound knowledge of the process is required. Spray-drying is actually a complex sequence of turbulent events occurring in a rather narrow time frame and partially overlapping. In particular, particle properties depend upon the drying profile and size of the generated droplet.

Droplets dry through an initial sensible heating step followed by a constant-rate and a falling-rate drying phases. The actual particle formation starts with shell thickening at the beginning of the falling-rate period. This stage determines the future particle morphology. In fact, as the solvent boundary proceeds inward, the forming particle surface may harden or collapse, and these events depend on the rate difference between solvent evaporation (E) and solute diffusion (D) in the droplet. This delicate balance is measured by the so-called Peclet number (Pe) Eq. 10.2).

$$Pe = \frac{E}{D} \quad (10.2)$$

As a rule of thumb, for $Pe > 1$, a buckling regime leads to the formation of hollow or crumpled particles, whereas when $Pe < 1$, a non-buckling regime produces more homogeneous spherical particles.

Therefore, particle morphology can be controlled by adjusting Pe . This can be done according to feedstock material properties, solvents, and the choice of proper spray-drying conditions.² Particle size depends on the size of the droplets, and therefore it can be controlled by adjusting the parameters influencing the atomization process. In this regard, different nozzle types are available that are able to generate specific droplet size ranges and therefore particles of different sizes.

As an example, two- and three-fluid nozzles in concurrent mode³ are employed for the production of small particles, such as dry powders for inhalation, starting from dispersions or solutions. On the other hand, centrifugal nozzles are more

²The main spray-drying parameters are inlet temperature, gas flow rate, feed rate, and aspirator rate. The main feedstock properties are solution or dispersion density, viscosity, and total concentration (dispersed solid or solute and liquid).

³Feed and pressurized gas flow have the same direction.

employed for larger particles and pellets and mainly in countercurrent configuration.⁴

Nowadays, several instrument configurations are available, some already industrially applied and others still to be scaled up, but with promising features.

According to the particular configuration, spray-driers are dedicated to different dry powder products, from the pharmaceutical to the food industry, which represent the largest fields of application for this technique.

Major limitations are due to difficulty in processing soft materials and low glassy amorphous materials, which can stick on chamber walls and aggregate, as well as heat-sensitive actives [14, 15]. For these purposes, spray-congealing and spray freeze-drying methods have been developed. These technologies can help in processing successfully low-melting materials, such as lipids, and stabilizing sensitive biological products.

In addition to the above issues, a general low process yield is usually recorded, which can be a serious concern when processing poorly available and expensive materials.

A recent novel spray-drying technology has overcome the classical atomizer limits in producing particles below 2 μm in size. Thanks to a new droplet generation concept based on a piezoelectric-driven vibrating mesh atomizer, droplets as small as 5–10 μm can be obtained, which result into nanoparticles upon drying (Fig. 10.3).

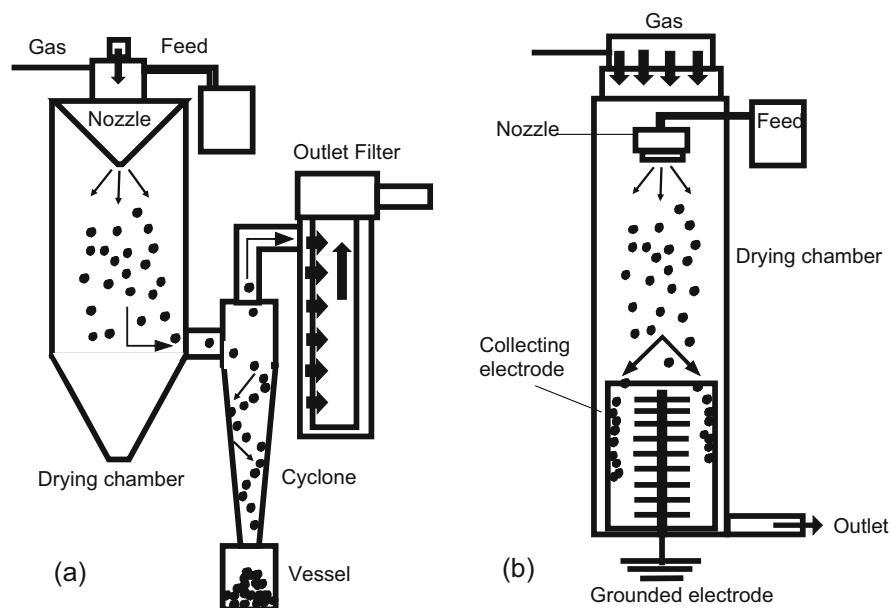


Fig. 10.3 Schematic representation of (a) traditional spray-drying equipment with a pressurized two-fluid nozzle and cyclone collector; (b) the innovative nano spray-drying equipment based on a piezoelectric vibrating mesh atomizer and an electrostatic collector

⁴Feed and pressurized gas flow have opposite directions.

In addition, yields up to 90% can be achieved thanks to a high-efficiency electrostatic dry powder collector. A laminar drying airflow inside the spray chamber also provides instant drying of the aerosol at mild conditions (30–40 °C lower atomization temperatures compared to traditional techniques), and this is advantageous for processing biomolecules and other heat-sensitive materials. Current limitations include the impossibility to employ organic solvents (up to 50% v/v in water solution) due to the formation of an explosive atmosphere incompatible with the high voltage in the collector and the inefficiency of the atomizer in processing dispersions and viscous solutions.

Spray Freeze-Drying

Spray freeze-drying emerged as an alternative to spray drying in the 1990s [16]. The principle involved is to spray the droplets containing drug and any other additives in solution into a container at $-70\text{ }^{\circ}\text{C}$, maintained by liquid nitrogen, to obtain frozen particles. The solvent is then removed from the particles by sublimation. The advantage of this approach is in the preparation of biological molecules that may be subject to instability if undergoing either milling or spray drying.

The particle size distribution obtained by this method depends on the droplet size generated by the nozzle. Since the droplets are subject to instant freezing, the dimensions are fixed and the removal of solvent makes it probable that the final form will be low density, rendering them easily dispersible.

Figure 10.4 illustrates an example of a spray freeze-drying apparatus. The significant elements contributing to the final composition and form of the particles produced are the solution composition, the liquid and gas feed to the spray nozzle, and the cooling vessel into which the droplets are sprayed and the particles are collected.

This method has largely been used as an approach to the production of particles for inhalation that require stabilization, particularly biologicals.

Supercritical Fluid Methods

Above certain conditions of temperature and pressure, fluids can show intermediate properties, and they behave as intermediate homogeneous phases. These conditions are defined as supercritical⁵ and the fluids, whose status is equal or above such conditions, are referred to as supercritical fluids (SCFs). A SCF shows a state in

⁵Critical temperature is the highest temperature at which a gas can be converted into a liquid by an increase in pressure; critical pressure is the highest pressure at which a liquid can be converted into a traditional gas by an increase in temperature.

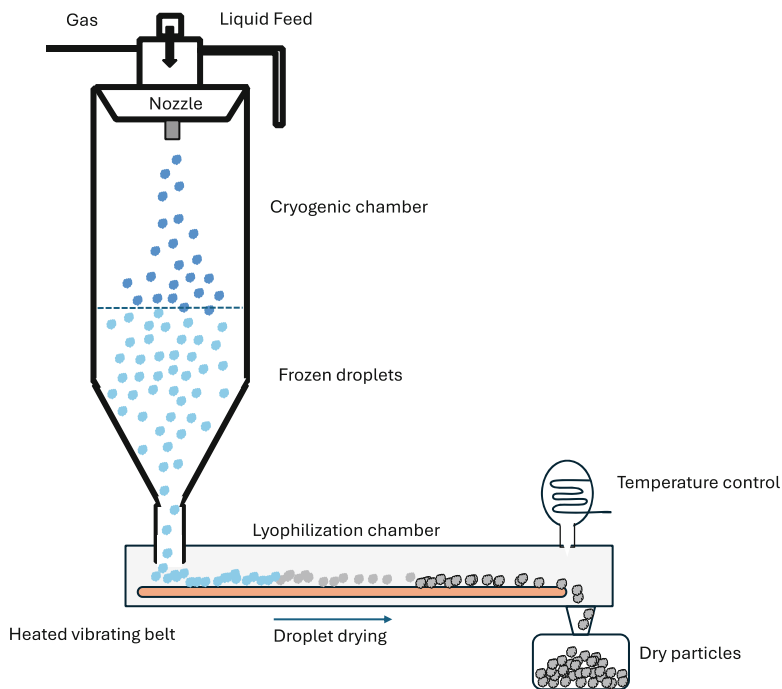


Fig. 10.4 Example of a spray freeze-drying apparatus illustrating the solution source, atomizing nozzle, and cryogenic chamber. The system is equipped with a lyophilization chamber and collecting vessel to which the frozen droplets are transferred and simultaneously dried

between that of a liquid and a gas and behaves accordingly. In this critical region, physicochemical properties such as density, viscosity, diffusion coefficient, and heat conductivity are intermediate between those of the liquid and the gas. In particular, SCFs exhibit a liquid-like density and gas-like viscosity and diffusivity. This enables several possible applications, since the liquid-like properties provide benefits in extraction processes, solubilization of substances, and matrix plasticization, and the gas-like features facilitate mass transfer processes and reaction selectivity.

Several organic solvents and almost all gases can be transformed into SCFs above their specific critical pressure and temperature.

Due to the mild supercritical conditions (31.2 °C, 7.4 MPa), carbon dioxide (CO₂) is by far the most employed SCF in pharmaceutical applications [17]. Additional reasons are the low environmental impact and low costs compared to other solvents or gases.

The SCF technology has become an important industrial process either for food or pharmaceutical applications. In particular, it is useful to process low-soluble drugs and materials to produce fine powders and polymeric micro- and nanocarriers.

This can be done by exploiting the solvent or anti-solvent capacity of the SCF and its subsequent fast removal to specifically induce coacervation phenomena [17].

Table 10.3 Supercritical fluid technique classification

SCF role	Process	
Solvent	Rapid expansion of supercritical solutions	RESS
Anti-solvent	Gas anti-solvent	GAS
	Supercritical anti-solvent process	SAS
	Solution-enhanced dispersion by supercritical process	SEDS
	Aerosol solvent extraction system	ASES
	Supercritical fluid extraction of emulsion	SFEE
Solute	Particle from gas-saturated solution	PGSS

Popular SCF techniques base on CO₂ solvent and anti-solvent effects are listed in Table 10.3.

SCF removal is usually obtained by an abrupt pressure drop below the critical value to trigger fast mass transfer, as the SCF turns into a gas, leading to solvent extraction or fast nucleation of dissolved or dispersed materials.

A schematic representation of popular SCF techniques is displayed in Fig. 10.5.

In some cases, the SCF, as a solvent (RESS, RESOLV), is usually mixed with the materials to be dissolved or with the solution and conveyed into a vessel where it is sprayed through a nozzle, which provokes rapid expansion and pressure drop. As an anti-solvent (SAS, GAS, SEDS, ASES), the pressurized vessel is filled with the SCF, and the organic solution is sprayed into the SCF solution and subsequently expanded, or the two solutions are co-sprayed in the vessel at high pressure.

SAS and SAA techniques have been used to produce micro- and nanoparticles for delivery of insulin, lysozyme, trypsin, methylprednisolone, and hydrocortisone [18].

Beyond the possibility to produce dry powders with controlled size and morphology, an interesting application of the SCF technology is the processing of macromolecules such as proteins, peptides, and nucleic acids. This technique is superior compared to freeze-drying as it can preserve stability of biomolecules by avoiding temperature stress while controlling powder characteristics. In this way, plasmid-DNA powders as well as submicron particles loaded with insulin and catalase have been successfully formulated [18].

SCF technology limitations are mainly linked to high maintenance and management costs due to the need for dedicated facilities and operators with adequate technical skills.

Particle Replication in Non-wetting Templates (PRINT) Technology

The search for exceptional control over particle size and shape has led to the development of cutting-edge techniques based on molding of particles into a pre-modeled matrix. One of these techniques is the particle replication in

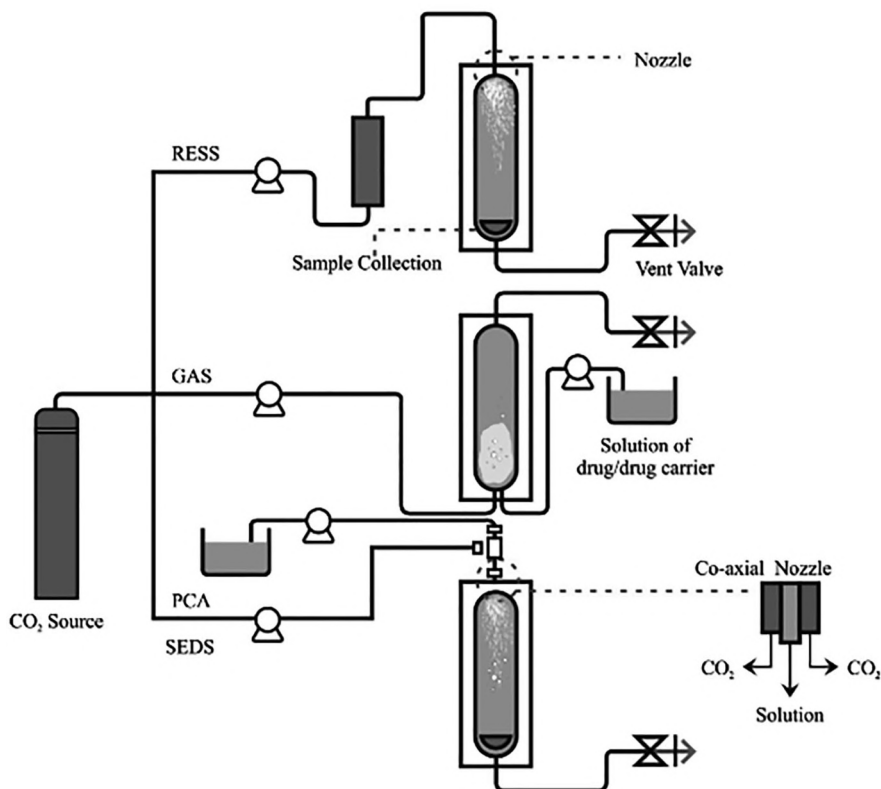


Fig. 10.5 Schematic representation of RESS, GAS, and SEDS supercritical fluid techniques. (Adapted from Ref. [18])

non-wetting templates (PRINT) that is a unique soft lithography-based molding technology.

This technology saw its dawn immediately after the development of a new highly fluorinated perfluoropolyether (PFPE) elastomer, which was conceived to solve the swelling tendency of polydimethylsiloxane used in microfluidics and molding processes. PFPE shows an extremely low surface energy, high gas permeability, low toxicity, high elastic recovery, good mechanical strength, and high chemical and solvent resistance and enables the fabrication of less than 100 nm high-resolution objects, useful for the creation of micro- and nanoparticles [19].

The PRINT technology shows a huge potentiality as it allows precise control on particle size and shape over a wide range of dimensions and morphological attributes. The technique has been designed to work in a GMP environment, which is fundamental to grant industrial development, and it will likely impact the research in drug delivery, electronics, optics, sensing, and imaging in the near future.

A scheme of the PRINT process is reported in Fig. 10.6. After the PFPE mold has been created, the cavities are filled with a liquid precursor via capillary flow, and a

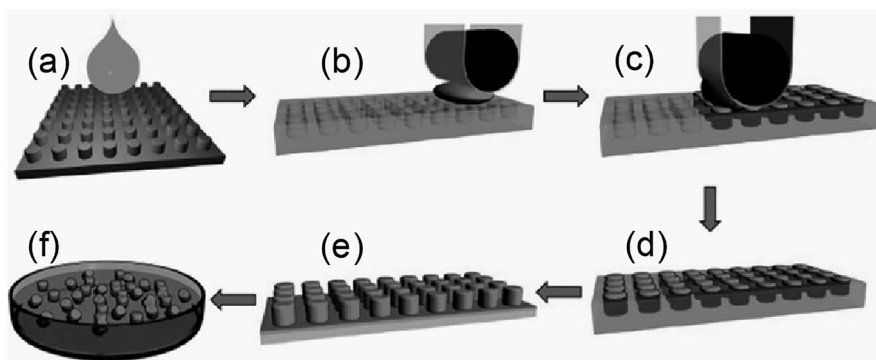


Fig. 10.6 Schematic illustration of a PRINT method for particle fabrication. (a) Liquid PFPE precursor wets the silicon wafer with micro- and nanosized patterns and is photocured to generate an elastomeric PRINT mold; (b, c) a liquid pre-particle material is filled into the cavities using a film-split technique against a high-surface-energy polyethylene terephthalate counter sheet; (d) the liquid in the mold cavities is solidified through photocuring, by temperature quenching, or solvent evaporation; (e) the solid particles are removed by contact with an adhesive layer; (f) the particles are freed by dissolving the adhesive layer. (Adapted from <http://desimone-group.chem.unc.edu/?cat=5>)

high-surface-energy sheet is used to remove the excess material. Then the precursor material in the PFPE mold cavities is solidified by freeze-drying or by other means and the particles are transferred using an adhesive film, which is then dissolved to recover the particles.

In spite of the great potential of the PRINT technology, this technique is still in its infancy, and a huge work is ahead in order to transform this new technology into an established particle engineering and production method.

Nevertheless, the PRINT technology could be powerful and versatile enough to allow customization of delivery systems to achieve cell or tissue selectivity and tailored release profiles, changing the way medicines are developed.

Thin Film Freezing

Thin film freezing is a novel method developed for a range of applications, including parenteral nasal and inhalation products [20].

The approach combines a continuous process with subsequent unit operations. Initially, macroscopic drops are introduced in a controlled fashion to the surface of a rotating drum cooled using liquid nitrogen. The drops instantly freeze to form large porous structures that are removed from the drum by a blade and collected prior to being placed in a lyophilizer. The conditions for lyophilization are optimized to maintain the stability and desired physico-chemical properties, including performance characteristics of solubility and aerosol dispersion. The particles are notably high surface area, high porosity, and low density.

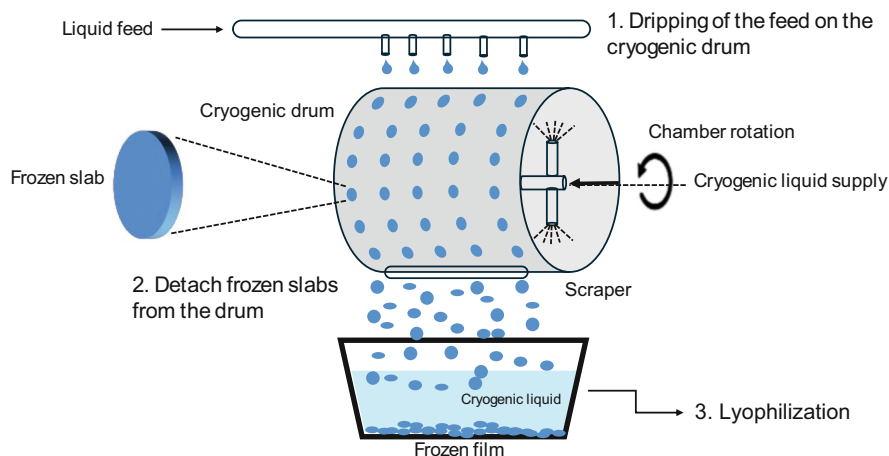


Fig. 10.7 Thin film freezing apparatus showing the major process steps of solution source, feed nozzle array, cryogenic drum, and lyophilizer

Figure 10.7 illustrates the apparatus for preparing thin film frozen particles. The significant elements contributing to the final composition and form of the particles produced are the solution composition, the liquid feed to the nozzle array, the cooling drum temperature and rotation speed, lyophilization conditions (tray temperature, heat, and mass transfer conditions), and post manufacture conditioning.

Conclusion

The need for more performing and efficient powder processing techniques has led to the evolution of novel concepts of pharmaceutical manufacturing. More traditional technologies, such as spray-drying, are nowadays assuming new shape and bringing powder processing and particle engineering approaches to a brand-new level with the introduction of new equipment designs. Moreover, technologies apparently traditionally not suited for solid dosage form manufacturing, such as PRINT or 3D printing, are gaining ground and consideration in pharmaceutical development thanks to tremendous advances in computer-aided design. Thanks to recent new fluidic and hybrid technologies, and in-process controls, fine-tuning is possible on crystallization, powder mixing, and particle formation.

The future will probably see the advent of a new era in pharmaceutical R&D and manufacturing with the possibility to bring the control over particle engineering processes to a level never seen before.

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Chapter 11

Quality by Design for Particulate Systems



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Abstract In earlier chapters, methods of manufacturing and characterization of pharmaceutical particulate systems were described. It is evident that key physico-chemical properties establish the foundation for quality and performance of a dosage form of which they are components. This is fundamental to monitoring and controlling of critical properties of these systems with a view to optimization, ideally establishing quality by design. The use of statistical experimental design during optimization and statistical process control during manufacturing allows the production of the dosage form to meet the target product profile within narrow limits, thereby assuring uniformity. Ultimately, the dosage form quality translates into a safe and efficacious drug product.

Particle systems are an important component of every solid dosage form [1–4]. The previous sections have outlined the important manufacturing and unit operations in addition to the characterization and its link to potential biological endpoint. To ensure the quality and thereby the safety and efficacy of a product, methods of monitoring and controls are needed to guide the manufacturing process to the desired specifications. There are numerous potential sources of error, both in the manufacturing processes properties and the methods of analysis that if compounded may result in poor quality control.

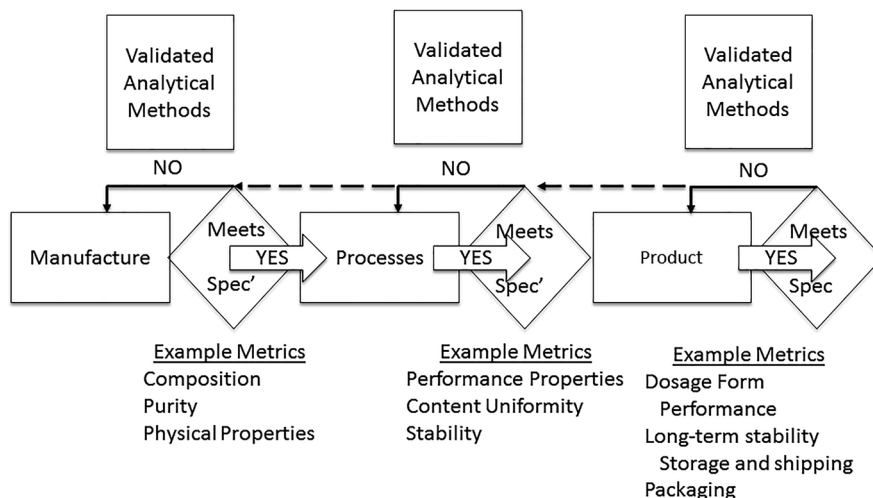


Fig. 11.1 Manufacturing and processing considerations. At each step in manufacturing, the inherent error in either the method itself or the analytical evaluation can be compounded to push a downstream output variable out of specification and impact on the quality of the product. A decision point allows adjustment with respect to analytical specification at each step in development

Broad areas in which control strategies might be applied are given below as a prelude to discussing their application and general methods of mitigating risk associated with quality deviations. Figure 11.1 illustrates the sequence in which events usually occur and the way in which errors can be compounded moving from one stage to the next. Limitations that might exist on analytical capability are identified.

Manufacturing and Characterization

Particles can be manufactured by several different methods, including controlled crystallization from solution, milling, spray drying, and microfluidics. They can then be formulated before incorporation in the final dosage form, e.g., capsules or tablets. In manufacturing processes, particles may exist as powders, blends, granules, or suspensions. Each method has input variables that are subject to control but may be influenced by the physicochemical properties of the drug or excipient being prepared.

Crystallization is influenced by the nature of the solvent (polar, nonpolar) from which the particle is being formed, temperature, and mixing [5–7]. It is frequently the case that to control large-scale manufacturing, a controlled crystallization of nucleating crystals is prepared on a smaller scale, thereby mitigating the influence of inhomogeneities in the larger vessels on crystal system and habit (Chap. 2).

Milling requires that the solid starting material can be supplied, usually from a crystallization process [8, 9]. Milling then depends on conditions which might

include milling media (ball mill) dimensions (pin mill), airflow/pressure applied to the mill (jet mill), internal geometry, and temperature (particularly for unstable substances).

Spray drying is a continuous process where the product properties are dictated by the instrument design and operating conditions of the nozzle design, liquid feed(s), inlet/outlet temperature, and internal geometry [10–14].

Microfluidic systems require consideration of the suspending medium, velocity of fluid and equipment dimensions, and overall scale of the process [15–18].

Blends are complex systems, and consequently their preparation requires control of a number of variables, including component powders and their ratios, loading masses, speed of motion (rotation in drum and ribbon blenders, paddles in planetary mixers), internal dimensions, additional features, and environmental conditions [19–22].

Many solid oral dosage forms require the preparation of granules from starting powders often because this aids with content uniformity, flow, and filling. Input variables include component powders, binding and coating agents, loading masses, coating, roller movement, and positioning in direct compression systems, fluidized bed systems, solution feed, and environmental conditions [23–26].

All variables described must be studied to identify the critical quality attributes (CQAs) and ensure the accuracy and reproducibility of the product performance [27]. In addition, as the product passes from research through various stages of development, scale-up, and technical transfer into pilot and production facilities require further consideration of the impact of the key variable at scale on the product performance to supply clinical or commercial needs.

The nature of the process will significantly impact the degree of difficulty in extrapolating to commercial manufacture. Batch processes such as ball milling or V-blending are unique at each scale and require method validation by designed experiments at each transition. Continuous processes may require less scrutiny, as it may be the case that full-scale production is simply a question of running the same manufacturing method for extended times.

All analytical methods, ranging from characterization of chemical composition and purity through particle properties and performance to the final dosage form behavior, are limited in both sensitivity and quantitation. The ability to control the processes and product is constrained by the accuracy and precision of CQA measurement. Consequently, the analytical method itself must be monitored, and validated, to ensure that it can be relied upon for decision-making with respect to the parameter it is measuring.

Experimental Design

Multivariate Statistics

The application of multivariate statistics to complex systems has been a common practice in engineering and environmental sciences for more than half a

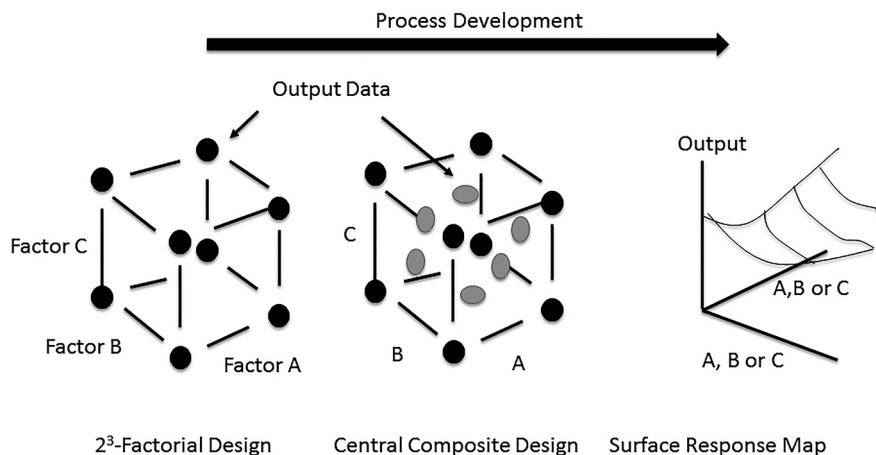


Fig. 11.2 Schematic sequence of designed experiments that leads to definition of property or process space

century [28–32]. The adoption of multivariate statistics methods, referred to as design of experiments (DOE), in pharmaceutical processing has occurred more recently [19, 27, 33, 34]. The ability of DOE methods to efficiently evaluate the critical manufacturing parameters for product quality and performance and the ability to translate the data to define process space by extrapolating from factorial designed experiments to surface response maps are significant advances in understanding the nature of control. Moreover, it can be anticipated that a thorough understanding of the materials composition and process variables with respect to important quality and performance attributes may in the future lead to specifications based on the data collected rather than on arbitrary notions of statistical randomness. The link between experimental design and multivariate statistics was outlined thoroughly by Box et al. [35–38]. For the purposes of this overview, it suffices to say that a sequence of experiments can be conducted that initially follow a factorial design to identify clearly the factors that control CQAs, as illustrated in Fig. 11.2. Once these factors have been identified, a factorial design can be supplemented with a central composite experimental design that leads to definition of the manifold for the surface (property, process) response map [39]. The surface response map in regions of insensitivity of the product quality and performance to the input variable can be considered to define property or process space. In this region, providing input variables do not deviate significantly, the quality of the product can be assured.

Process Analytical Technology

The FDA defines process analytical technology (PAT) as “a system for designing, analyzing, and controlling manufacturing through timely measurements of CQAs

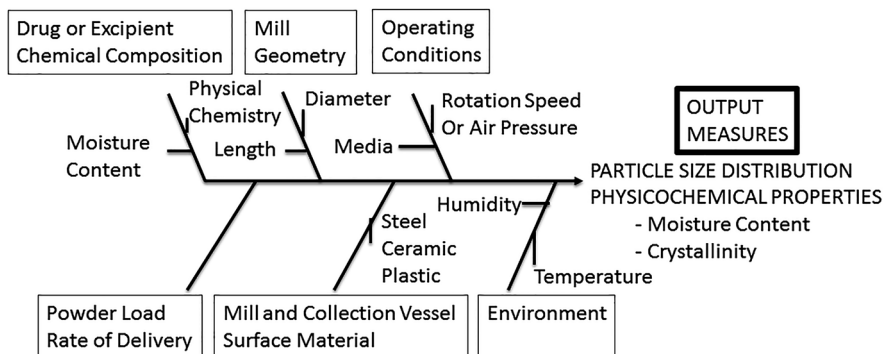


Fig. 11.3 The use of Ishikawa diagrams to capture the input variables with respect to a key property or process variable using milling as an example

and critical performance attributes of raw and in-process materials and processes, with the goal of ensuring final product quality”. PAT is an essential set of tools that includes chemical, physical, microbiological, mathematical, and risk analysis conducted in an integrated manner. Application of validated analytical methods to the process can significantly influence the quality, efficiency, and cost of production [27, 33, 40]. For methods employing offline analytical methods, optimization of the process must be conducted in an iterative manner where each batch of product is evaluated with respect to its output variables. The findings can then be used to focus on the next batch production and so on until sufficient data has been generated to establish the range and control of input variables that would enable predictable and uniform output performance characteristics within acceptable specifications. For online analytical systems, samples are taken and analyzed during the procedure. Rapid feedback allows discrete modifications to the input variables to promote control of the product. Finally, inline analysis measurement conducted without disturbing the manufacturing process allows real-time adjustment to ensure the product conforms with the desired output specifications. Figure 11.3 illustrates the methods whereby analysis can be employed to control the product quality and performance.

Critical Quality Attributes

During the initial evaluation of a process, certain input variables will be identified as critical to the product properties. These CQAs will be the focus for control mechanisms and will be viewed as the greatest source of potential risk of product failure [27, 33, 41]. CQAs may vary based on the physicochemical properties of the drug and the biopharmaceutical requirements for its activity. Identification of the CQAs begins by defining all known variables in the process itself, the environment, and the operation (including the human factors) that might influence the output variables.

These parameters are usually captured through scientific knowledge, technical experience, and brainstorming to access the widest range of opinion. Visualizing data and input variables has long been employed for decision-making [42]. All sources of variability identified may be depicted on an Ishikawa or fishbone diagram [43, 44]. Figure 11.3 illustrates the range of considerations that might be considered for selected processes. These sources of variability have been characterized as the 6 “Ms” (6M) of materials, methods, man, machines, mother nature, and measurement [43]. The time spent on this process and diligence in exploring the influence of each of the factors early in the product and process development will be rewarded in terms of lowering the potential for product failure and increasing efficiency while reducing cost. In other words, minimizing risk and maximizing reward.

Risk Assessment/Management

The risk associated with product failure is primarily one of safety, which can be viewed both directly and indirectly. If the product deviates from specification, delivering more drug than required for therapy, it may become toxic, and there is a direct challenge to patient health. Conversely, if the deviation from specification delivers too little drug, the dosage form may not be efficacious, and for a life-threatening condition, the lack of treatment may also challenge the patient’s health. So the risks are of both direct toxicity and health risks from untreated disease. Figure 11.4 gives a general approach to monitoring and control to mitigate risk during manufacturing, allowing feedback into a process. The traditional approach of

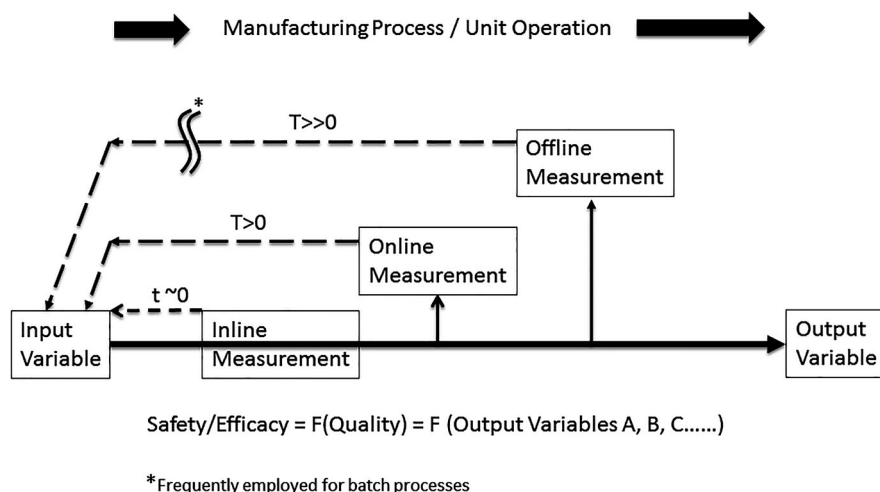


Fig. 11.4 Risk mitigation flow chart indicating measurement and feedback control to the process to ensure output variable meets specification

batch sampling and offline analysis is the least efficient and involves the most risk with respect to the ability to control the product quality by introducing sampling and analytical errors that may be compounded. The sequential nature of sampling and analysis occurs on a timescale that deviations from specification may not be observed quickly enough, if at all, to correct the process. Consequently, the product may need to be discarded, prior to adjusting input variables, a costly and inefficient approach. The greatest control and minimal risk is for continuous processes where instantaneous measurements in product attributes can be monitored and feedback given to input parameters to modulate the process if necessary. This assumes a clear understanding of the process space established during process development.

Other Considerations

Quality by design has been a central tenet of US FDA regulatory strategy since the turn of the millennium [33, 45–47]. Consequently, it is not only a rational approach for good manufacturing practices in product development; it is also an essential prerequisite to regulatory approval. The International Conference on Harmonization of Technical Requirements for Registration of Pharmaceuticals for Human Use (ICH) has several guidance documents that relate to QbD, including those for Pharmaceutical Development (Q8), Quality Risk Management (Q9), Pharmaceutical Quality Systems (Q10), and Development and Manufacture of Drug Substance (Q11) [48].

For research purposes, there are many software packages that allow multivariate statistical approaches to aid the design of experiments, for example, JMP (SAS, Cary, NC) and Design-Expert (Stat-Ease, St. Paul, MN). In addition, project management software allows visualization of process variables (Ishikawa diagrams) and facilitates the development of flow diagrams to address risk mitigation strategies in the context of critical quality attributes, for example, Microsoft Project (Microsoft, Seattle, WA) and iThink (ISEE Systems, Lebanon, NH). The approach to pharmaceutical particulate systems can be viewed in terms of product quality and performance as a form of critical path analysis, which is used in project management [49, 50]. In addition, manufacturing equipment and production lines now include real-time monitoring and control governed by proprietary software. Consequently, once the product leaves the research and pilot scale environments and enters formal development process, engineering systems, equipment, and instrumentation are now fully integrated through software algorithms to allow for computer-aided process monitoring and control.

At the current pace of development against an exponential increase in the collection and sharing of data and information, it is conceivable that at some time in the not-too-distant future, all output variables will be controlled immediately by systems that are capable of instantaneous response, likely through machine learning and artificial intelligence controllers [51–55]. In this ideal situation, processes would be continuous and not batch procedures, and sufficient information would be known

from initial limited but sophisticated experiments to initiate the process with confidence that it will rarely, that is, below the limit for statistically random variation, fail quality, and performance specifications.

Conclusion

Particulate systems are important physical components in solid dosage forms and drug delivery systems containing a solid phase. Consequently, their physical form presents a monitoring and control opportunity to ensure the quality, safety, and efficacy of the pharmaceutical product. The form of the active pharmaceutical ingredient and additive particulates is dictated by the initial method of manufacture and the subsequent processing to achieve the desired properties to support the performance of the product. The approach to controlling quality is to establish methods to assure the uniformity of CQAs of the final particulate system and validated analytical techniques for their monitoring. Manufacturing methods would include crystal system and habit; polymorphism; presence of impurities, including moisture; particle size distribution; shape; and surface rugosity. In assessing the influence of any manufacturing and processing input variables on the product, the first step is to capture all of the factors that might influence the desired properties, including those that might be due to the environment (temperature, humidity, light, mechanical, or electrical phenomena). These factors are usually captured by a thorough review of the specific needs and captured using visual aids such as Ishikawa or fishbone diagrams. Through initial designed experiments, CQAs are identified, those factors that clearly influence the product properties and performance. Establishing properties and processes to which product performance is insensitive will also mitigate risk of deviation. A means of monitoring through process analytical technology and feedback control will significantly reduce the prospect of failure.

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Chapter 12

General Conclusion



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Abstract Complex particulate systems are the basis for all solid dosage forms, starting with the active pharmaceutical ingredient, which has to be produced in the solid state in the first instance, most often by crystallization from solution. The drug is then processed to a form that has utility in the final dosage form, initially contributing to flow and dispersion and finally in terms of release of drug. Characterization of the physicochemical properties of both the starting and processed material is essential to maintaining control of the final product. A variety of methods of defining the properties of the pharmaceutical particulate system and of manufacturing particles and the implications for their performance have been described.

Pharmaceutical particle science is a central element of all solid dosage forms. The underlying principles of particle formation and interaction are well understood. The heterogeneous nature of materials used in dosage forms in terms of both composition and morphology makes behavioral prediction from first principles difficult to achieve. However, sufficient data can be collected and monitoring and controls adopted to guarantee product quality, safety, and efficacy.

Particles

There is a notional concept of the individual particle, the reality of which can only be observed on a macroscopic scale. Individual solid objects exist in the realm of human experience when we consider boulders or pebbles and possibly coarse

sand. As the scale is reduced from meters to centimeters to millimeters, as the scale is reduced further to one at which direct observation is impossible with the naked eye (micrometers), individual particles can be observed by microscopy, but their individuality is an illusion created by sampling and preparation, as they rarely exist outside a population of particles, the interaction of which is central to their behavior and utility. Consequently, methods of analysis and descriptors are mostly aggregate, of average descriptors, whether they encompass the physicochemical or performance properties.

The importance of the concept of individual particles is in identifying assumptions and defining terms by which the particle systems will be described. Ideally, a series of quantitative metrics can be constructed on a foundation of the smallest scale of subdivision (individual particles) to the scale of utility (e.g., bulk powder) in a manner that allows prediction of a variety of performance characteristics of relevance to the dosage form and ultimately to the pharmacological action of the drug.

Particle Systems

Most quantitative measures of the structure and function of particles are based on assembled systems, where interactions are an important feature of any performance attribute.

Figure 12.1 illustrates the approach to characterization that begins with methods of characterizing spatial dimensions of molecular structure from aggregate data, narrows to a scale of scrutiny that is ostensibly that of an individual particle, and then returns to aggregate data when kinetic phenomena are considered. The importance of

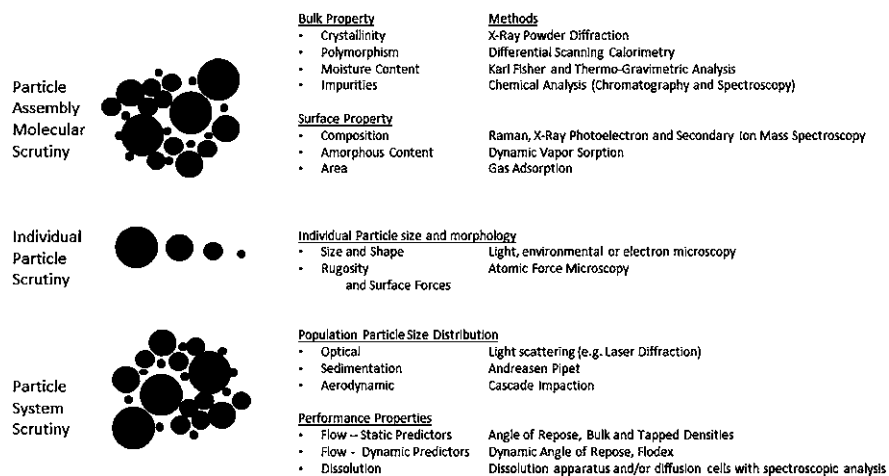


Fig. 12.1 The scale of scrutiny of pharmaceutical particulate systems and methods employed for the purpose

this statement of the sequence of events is in identifying the increasing lack of precision that occurs as observations move from the exactitude of molecular measurements to the approximations of dimensions and behavior that necessarily occur as the reality of the particle system is estimated through a series of assumptions associated with descriptors and sampling. While these assumptions are important to processing of particles and powders, the variability arising from such approximations can be amplified when a pharmaceutical particulate system is incorporated into a dosage form and interacts with the biological processes in therapy, which both have their own sources of variability.

It should be evident from the sequence of considerations covered earlier in the text that important scientific and technological advances with respect to knowledge of powders and particles have been made over more than a century of basic scientific research. These developments have resulted in the adoption of a series of complementary methods that can be selected for their relevance to the specific intended application. This approach achieves the greatest clarity in our understanding, ability to describe and ultimately control pharmaceutical particulate systems. A thorough evaluation of the unknown system should yield sufficient data to allow for adequate control with respect to drug delivery and disposition required to treat disease safely.

In the foregoing sections, we have covered concisely the major physicochemical considerations that underpin any dosage form composed of particles. The sequence, as shown in Fig. 1.1, is chronological, but the steps between functions and the monitoring and control procedures are not equally spaced. The application of general principles of quality by design (QbD) is intended to optimize progress through each of the steps and draws attention to critical attributes that impact progress and may require greater resources (attention, time, expense) to ensure a successful outcome with regard to manufacturing and performance.

Spatial physicochemical properties can be considered with respect to bulk, surface, and extent of subdivision. Bulk properties may be defined in terms of molecular structure (crystallinity), shape (crystal habit, outcome of processing), moisture content, and presence of impurities. Surface properties can be considered in terms of morphology with respect to surface roughness or rugosity (created during crystallization from presence of impurities or through mechanical interactions), charge, and amorphous content. Particle size distribution reflects the degree of subdivision created by a particular process. A variety of processes are available for the production of particles, and as technology advances, more elaborate methods are being devised to prepare multicomponent particles. The monitoring and control of manufacturing methods is central to process engineering and requires the application of statistical methods and/or real-time feedback to the process to meet the desired specifications on product performance. Quality metrics are imposed that allow the final products to meet regulatory expectations.

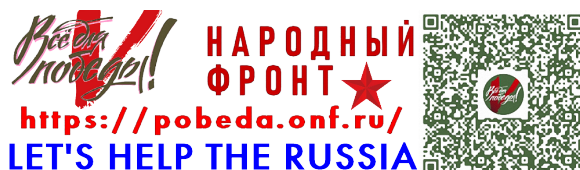
We speculated in the first edition of this volume that the pace of new developments would render monitoring and control of particulate systems and their contribution to drug product performance to assure quality a routine activity in the near term. The evolution of continuous manufacturing processes and developments in machine learning and artificial intelligence mean that the risk of product failure has

been substantially mitigated. Moreover, the prospects of efficiently and reproducibly manufacturing dosage forms with assured quality, safety, efficacy, and at low cost is already a feature of many commercially available drug products.

Conclusion

In conclusion, this short volume gives a succinct overview of pharmaceutical particulate systems and the influence that certain parameters play in their structure and behavior as the foundational components of solid dosage forms. Undoubtedly, readers will need to delve more deeply into each topic if they wish to become thoroughly conversant with methods and techniques, but this volume should be sufficient to understand the key considerations that underpin the role of powders and particles in dosage forms and their importance to ensuring the desired therapeutic outcome.

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